
Variance Reduction for Non-Log-Concave Sampling with Applications to Inverse Problems

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Abstract

Sampling from high-dimensional, non-log-concave distributions with unnormalized densities is a fundamental challenge in machine learning, particularly when the exact gradient of the potential is unavailable and must be approximated via stochastic gradients that exhibit high variance under a fixed budget of gradient computations per iteration. Although variance reduction techniques such as SGD with momentum, STORM, and PAGE have demonstrated improved convergence properties in non-convex optimization, their implications for sampling from non-log-concave distributions remain largely unexplored. In this work, we develop the first unified analysis of these estimators for sampling from non-log-concave distributions. We establish improved non-asymptotic convergence rates in ε -relative Fisher information and, under a Poincaré inequality assumption, in squared total variation distance, and further prove weak convergence to the target distribution. We extend our analysis to solving inverse problems with score-based generative priors. We empirically validate our theory and demonstrate that, under a fixed gradient computations per iteration, variance-reduction techniques consistently improve sample quality in two standard imaging applications.

1 INTRODUCTION

One of the central themes in machine learning is the use of stochasticity to design procedures that are both computationally efficient and statistically consistent. This idea is shared by both Bayesian and frequentist perspectives. From a Bayesian perspective, posterior distributions provide a principled characterization of uncertainty in estimates, and stochasticity is used to represent those distributions

through Monte Carlo (MC) samples. On the frequentist side, stochasticity is used to develop computationally efficient gradient-based optimization methods for obtaining point estimates by replacing intractable or expensive exact gradients with unbiased stochastic estimates. Although they are two different frameworks, there are overlapping methodological challenges. In particular, MC sampling must drive an out-of-equilibrium state towards the steady-state posterior distribution, while achieving fast convergence; consequently, optimization ideas are relevant. On the other hand, frequentist inference often depends on sampling and resampling procedures, which makes MC methods relevant.

Variance reduction has emerged as one of the key tools to address the challenges shared by the two frameworks. In the context of stochastic optimization, stochastic gradient descent (SGD) replaces exact gradients with unbiased stochastic estimates to reduce the computational burden of large-scale learning problems. However, this computational advantage comes at the cost of additional variance, which can slow convergence and lead to a suboptimal stationary point. This raises the question of whether there is a principled way to manage this trade-off. This question has been answered positively in a seminal line of work on stochastic optimization [Johnson and Zhang, 2013, Schmidt et al., 2017, Fang et al., 2018, Cutkosky and Orabona, 2019, Li et al., 2021]. These methods achieve improved convergence rates compared to SGD while relying on stochastic gradients with lower gradient complexity.

In parallel, a related line of research has developed in the Bayesian framework based on MC sampling. Stochastic gradient Langevin diffusion (SGLD) has been proposed to approximate the Langevin dynamics via unbiased stochastic gradients [Welling and Teh, 2011] and several theoretical results have established mixing time bounds for both well behaved (e.g., log-concave) distributions [Cheng and Bartlett, 2018, Dalalyan and Karagulyan, 2019, Crespo Navas et al., 2024] and general non-log-concave settings [Raginsky et al., 2017, Cheng et al., 2020, Balasubramanian et al., 2022]. These developments have in turn motivated variance re-

duction techniques for SGLD [Dubey et al., 2016, Zou et al., 2018, Chatterji et al., 2018, Ding and Li, 2020]. However, these studies rely on the restrictive assumption that the target distribution is strongly log-concave. Although Zou et al. [2019] establish non-asymptotic convergence rates for SAGA- and SVRG-based Langevin diffusion, their analysis relies on a finite-sum structure of the log density together with a dissipativity assumption, which does not hold for non-log-concave distributions in general.

In this work, we develop a unified framework that adapts variance reduction techniques from non-convex optimization to sampling from non-log-concave distributions under relaxed assumptions. In particular, we propose two estimators that provide unified treatments of SGD with momentum and its probabilistic counterpart, and of STORM [Cutkosky and Orabona, 2019] and PAGE [Li et al., 2021]. We refer to these estimators as *mean-Lipschitz variance-reduced Langevin diffusion (ML-VRLD)* and *sample-Lipschitz variance-reduced Langevin diffusion (SL-VRLD)*, respectively. This terminology reflects the underlying regularity conditions: ML-VRLD assumes Lipschitz continuity of the exact gradients, whereas SL-VRLD assumes Lipschitz continuity of unbiased stochastic gradients in the mean-squared sense. The proposed methods improve upon the non-log-concave analysis of SGLD [Balasubramanian et al., 2022], which establishes ε -relative Fisher information convergence with $\mathcal{O}(Ld\sigma^2/\varepsilon)$ stochastic gradient computations per iteration and a total computational complexity of $\mathcal{O}(L^3d^3\sigma^2/\varepsilon^3)$ (see Theorem B.1 in Appendix B.3). Here, d denotes the dimension, L the gradient Lipschitz constant, and σ^2 an upper bound on the variance of the stochastic gradients. In contrast to SGLD, both ML-VRLD and SL-VRLD achieve the same notion of stationarity with $\mathcal{O}(1)$ stochastic gradient computations per iteration, providing memory-efficient algorithms under different assumptions. Moreover, SL-VRLD improves the total computational complexity to $\mathcal{O}(L^2d^2\sigma^2/\varepsilon^3)$. We additionally establish convergence rates in squared total variation distance under a Poincaré inequality assumption and prove weak convergence to the target distribution for both algorithms.

One of the most important practical applications of sampling is solving inverse problems, where score-based generative models serve as implicit priors for modeling complex high-dimensional distributions [Song et al., 2021, Chung et al., 2023]. Recently, several works have explored high-quality image generation as a solution to inverse problems across diverse applications, including image deblurring, MRI and CT reconstruction [Chung and Ye, 2022, Rout et al., 2023, Wu et al., 2024, Chen et al., 2025]. However, most existing posterior sampling methods that employ SGM priors work in a *full-batch* setting, where the entire set of measurements is used at every iteration. This effectively limits their applicability to large datasets [Bottou and Bousquet,

2007] common in three-dimensional imaging (3D) or in dynamic imaging [Kamilov et al., 2016, Ong et al., 2020]. To the best of our knowledge, existing posterior sampling methods with SGM priors do not address this limitation. Related approaches have instead been proposed in other settings [Sun et al., 2019, Tang and Davies, 2020, Liu et al., 2022]. However, these methods rely on mini-batch gradient estimates without variance-reduction mechanisms, resulting in high-variance updates that necessitate larger batch sizes in high-noise regimes. Our second contribution is that we propose two principled, variance-reduced posterior sampling algorithms for inverse problems based on annealed Langevin Monte Carlo (LMC) sampling with an SGM prior. Assuming bounded SGM estimation error, we establish convergence in relative FI, squared TV distance under Poincaré inequality assumption, and prove weak convergence to the target distribution, with $\mathcal{O}(1)$ gradient computations per iteration. We corroborate our theoretical findings with extensive experiments. The code is provided in the [project page](#).

Our contributions are summarized as follows:

- We propose a unified framework built on two estimators ML-VRLD and SL-VRLD for analyzing variance-reduction methods, including SGD with momentum, STORM, and PAGE, in non-log-concave sampling.
- We establish convergence in ε -relative FI with $\mathcal{O}(1)$ gradient computations per iteration, improving upon the $\mathcal{O}(Ld\sigma^2/\varepsilon)$ complexity of SGLD. Moreover, SL-VRLD improves the total computational complexity from $\mathcal{O}(L^3d^3\sigma^2/\varepsilon^3)$ to $\mathcal{O}(L^2d^2\sigma^2/\varepsilon^3)$. We further prove weak convergence to the target distribution and, under a Poincaré inequality assumption, convergence in squared TV distance.
- We extend our framework to posterior sampling via annealed LMC sampling with score-based generative models (SGMs) and establish the same convergence guarantees under bounded SGM estimation error.
- Through synthetic Gaussian mixture and inverse problem experiments, we verify our theoretical results, and obtain more accurate posterior statistics than SGLD. On MRI and sparse-angle CT reconstruction tasks, we achieve superior sample quality compared to the baselines using the same number of gradient computations per iteration.

2 PRELIMINARIES

2.1 SAMPLING ANALOGUE OF STATIONARY POINT ANALYSIS

While optimization and sampling are distinct frameworks, their theoretical analysis share deep structural connections. In particular, tools from non-convex optimization theory can be leveraged to characterize the behavior of sampling algorithms [Balasubramanian et al., 2022]. Consider the

minimization of *Kullback-Leibler (KL)* divergence over the space of probability distributions equipped with the optimal transport geometry:

$$\arg \min_{\mu} \text{KL}(\mu \parallel \pi),$$

where $\text{KL}(\mu \parallel \pi) := \int_{\mathbb{R}^d} \mu(\mathbf{x}) \log \frac{\mu(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$. Here, π denotes the target distribution, and μ denotes the sampling distribution induced by the generated samples. In this setting, Langevin diffusion can be interpreted as the *exact* gradient flow of the KL divergence [Jordan et al., 1998]. For example, the Wasserstein gradient of $\text{KL}(\cdot \parallel \pi)$ at μ is $\nabla \log(\mu/\pi)$, and its expected squared norm is known as the *relative Fisher information (FI)*

$$\text{FI}(\mu \parallel \pi) := \int_{\mathbb{R}^d} \mu(\mathbf{x}) \left\| \nabla \log \frac{\mu(\mathbf{x})}{\pi(\mathbf{x})} \right\|_2^2 d\mathbf{x}.$$

If μ_t denotes the law of the Langevin diffusion (1) at time t , then the calculation rules for gradient flows imply that $\frac{d}{dt} \text{KL}(\mu_t \parallel \pi) = -\text{FI}(\mu_t \parallel \pi)$ [Villani, 2009, Ambrosio et al., 2008, Santambrogio, 2015]. From a non-convex optimization point of view, $\text{FI}(\mu_t \parallel \pi)$ is analogous to the ℓ_2 norm of the gradient in $\mathbb{R}^d$ [Balasubramanian et al., 2022]. Using this analogy, we analyze the convergence of $\text{FI}(\mu_t \parallel \pi)$ under a *linear interpolation* of the distributions obtained by the LMC sampling algorithms. Different than non-convex optimization, if μ and π have positive and smooth densities, $\text{FI}(\mu \parallel \pi) = 0$ implies $\mu = \pi$, which we use to establish weak convergence guarantees. We refer to Balasubramanian et al. [2022] for further discussion on this topic.

2.2 LANGEVIN DIFFUSION

Langevin diffusion is defined as the solution to the stochastic differential equation

$$d\mathbf{x}_t = -\nabla f(\mathbf{x}_t)dt + \sqrt{2}d\mathbf{B}_t, \quad (1)$$

and admits $\pi \propto e^{-f}$ as its unique stationary distribution. Under mild conditions, the process converges to π as $t \rightarrow \infty$. Here, $(\mathbf{B}_t)_{t \geq 0}$ denotes a standard d -dimensional Brownian motion. Discretizing this process with step size $\gamma > 0$ yields the standard Langevin Monte Carlo (LMC) sampling algorithm

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma \nabla f(\mathbf{x}_{k\gamma}) + \sqrt{2\gamma} \mathbf{Z}_k, \quad (2)$$

where $\mathbf{Z}_k \sim \mathcal{N}(0, I)$. In many practical settings, the exact gradient of the potential $\nabla f(\mathbf{x}_{k\gamma})$ is unavailable. Thus, it is replaced by its stochastic estimate $\nabla f(\mathbf{x}_{k\gamma}, \xi_k)$, where ξ_k denotes a random variable capturing the external randomness at iteration k and is independent of all other sources of randomness. However, as shown in prior work [Balasubramanian et al., 2022], these iterates fail to achieve arbitrarily small ε -relative FI. A standard remedy is to replace the

single-sample gradient with a mini-batch estimator, leading to the SGLD iterates

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \frac{\gamma}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i) + \sqrt{2\gamma} \mathbf{Z}_k, \quad (3)$$

where b denotes the batch size and $(\xi_k^i)_{i=1}^b$ are independent and identically distributed. While this reduces the estimator variance, ε -relative FI is guaranteed by $b = \mathcal{O}(Ld\sigma^2/\varepsilon)$ gradient computations per iteration (see Theorem B.1), leading to substantial per-iteration cost in high-dimensional and high-variance settings. Consequently, SGLD requires very large batch sizes, resulting in both high computational cost and increased memory requirements. In contrast, the proposed ML-VRLD and SL-VRLD estimators achieve ε -relative FI error with $\mathcal{O}(1)$ gradient computations per iteration.

2.3 POSTERIOR SAMPLING WITH SGM PRIOR

For the posterior sampling part of our work, we consider a general inverse problem setting modeled as

$$\mathbf{y} = \mathbf{A}(\mathbf{x}) + \mathbf{n}, \quad (4)$$

where the objective is to recover $\mathbf{x} \in \mathbb{R}^d$ from measurements $\mathbf{y} \in \mathbb{R}^m$. The measurement operator $\mathbf{A} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ characterizes the imaging system's response, and $\mathbf{n} \in \mathbb{R}^m$ denotes the measurement noise, generally modeled as additive Gaussian noise (AWGN). The mapping $\mathbf{x} \rightarrow \mathbf{y}$ is not one-to-one, making the inverse problem ill-posed since $\mathbf{x}$ cannot be uniquely recovered from $\mathbf{y}$. In Bayesian framework, uncertainty about $\mathbf{x}$ is modeled through a *prior* $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$, and inference is performed by sampling from the *posterior* $\pi(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ obtained via Bayes' rule, where $p(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ indicates the *likelihood*. Here, f and h denote the potential functions associated with the likelihood and prior, respectively, and are central to our analysis. One standard approach is to sample from the posterior to solve the inverse problem. Using Bayes' rule and the SGLD update rule in (3), we obtain

$$\begin{aligned} \mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma \left(\frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i) + \nabla h(\mathbf{x}_{k\gamma}) \right) \\ + \sqrt{2\gamma} \mathbf{Z}_k. \end{aligned} \quad (5)$$

Since $\nabla h(\mathbf{x}_{k\gamma})$ is generally intractable for complex, high-dimensional priors, score-based generative models (SGMs), parameterized by a neural network, have been proposed to provide computable approximations of this quantity. This approach is commonly referred to as *score matching* [Hyvärinen and Dayan, 2005, Vincent, 2011]. One elegant approximation replaces the intractable score $-\nabla h(\mathbf{x}_{k\gamma})$ by smoothed score $-\nabla h_\sigma(\mathbf{x}_{k\gamma})$, which admits a closed form

expression by Tweedie’s formula [Efron, 2011]. In particular, let $\mathbf{z} \sim p(\mathbf{z})$ be a random sample from the prior, $\mathbf{e} \sim \mathcal{N}(0, \sigma^2 I)$ the AWGN, and $\mathbf{x} = \mathbf{z} + \mathbf{e}$ the noisy image. Then, $\nabla h_\sigma(\mathbf{x}) = (\mathbf{x} - \mathbb{E}[\mathbf{z}|\mathbf{x}])/\sigma^2$, where $p_\sigma(\mathbf{x}) \propto e^{-h_\sigma(\mathbf{x})}$ is the distribution of noisy image with potential function h_σ , and $p_\sigma(\mathbf{x}) := \int_{\mathbb{R}^d} p(\mathbf{z})\mathcal{N}(\mathbf{x}|\mathbf{z}, \sigma^2 I) d\mathbf{z}$. SGMs approximate the smoothed score by minimizing the denoising score matching (DSM) loss derived from Tweedie’s formula [Vincent, 2011].

$$\text{DSM}(\theta) := \mathbb{E} \left[\left\| \frac{\mathbf{z} - \mathbf{x}}{\sigma^2} - \mathcal{S}_\theta(\mathbf{x}, \sigma) \right\|_2^2 \right], \quad (6)$$

where $\mathcal{S}_\theta(\mathbf{x}, \sigma)$ denotes SGM parametrized by θ . Assuming enough data and model capacity, minimization of DSM loss satisfies $\mathcal{S}_{\theta^*}(\mathbf{x}, \sigma) = -\nabla h_\sigma(\mathbf{x})$ almost everywhere [Song and Ermon, 2019]. Thus, for sufficiently small $\sigma > 0$, we have $\mathcal{S}_{\theta^*}(\mathbf{x}_{k\gamma}, \sigma) \approx -\nabla h(\mathbf{x}_{k\gamma})$.

Standard LMC algorithms exhibit slow convergence and mode collapse when targeting high-dimensional, multi-modal distributions. To mitigate these issues, we consider a more general LMC formulation incorporating *weighted annealing* [Sun et al., 2024]. Specifically, we define a sequence of intermediate target distributions

$$\pi_{\sigma_k}^{(\alpha_k)}(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x}) p_{\sigma_k}^{\alpha_k}(\mathbf{x}),$$

where $(\alpha_k)_{k=0}^{N-1}$ and $(\sigma_k)_{k=0}^{N-1}$ denote non-increasing annealing and smoothing schedules, respectively. This formulation corresponds to sampling from a sequence of weighted posterior distributions. Early in the sampling process, the smoothed prior p_{σ_k} dominates, yielding a smoother energy landscape that facilitates exploration. As the iterations progress, the influence of the likelihood increases while the smoothed prior gradually approaches the true prior p , guiding the iterates toward the target posterior. Incorporating this weighted annealing strategy yields the following annealed SGLD iterates with an SGM prior:

$$\begin{aligned} \mathbf{x}_{(k+1)\gamma} = & \mathbf{x}_{k\gamma} - \gamma \left(\frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i) \right. \\ & \left. - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k) \right) + \sqrt{2\gamma} \mathbf{Z}_k. \end{aligned} \quad (7)$$

3 VARIANCE REDUCTION TECHNIQUES

In Balasubramanian et al. [2022], SGLD (3) achieves ε -relative FI with $\mathcal{O}(Ld\sigma^2/\varepsilon)$ gradient computations per iteration. Such a requirement is computationally prohibitive in high-dimensional and high-variance settings. To address this, we propose two unified estimators, ML-VRLD and

Table 1: Comparison of the total and per-iteration computational complexities of Langevin Monte Carlo sampling algorithms for generating samples with distributions that are ε -close to the target distribution in relative Fisher information.

Algorithm	Total Computation	Per-Iteration Computation
SGLD	$\mathcal{O}(L^3 d^3 \sigma^2 / \varepsilon^3)$	$\mathcal{O}(Ld\sigma^2 / \varepsilon)$
ML-VRLD	$\mathcal{O}(L^2 d^2 \sigma^4 / \varepsilon^4)$	$\mathcal{O}(1)$
SL-VRLD	$\mathcal{O}(L^2 d^2 \sigma^2 / \varepsilon^3)$	$\mathcal{O}(1)$

SL-VRLD, that generalize popular variance reduction techniques from optimization, and establish convergence guarantees with $\mathcal{O}(1)$ gradient computations per iteration. Table 1 summarizes the computational complexities.

3.1 ML-VRLD

Our first estimator unifies SGD with momentum and its stochastic counterpart, both of which exhibit lower variance than the SGD estimator. We make the following standard assumptions on the target distribution $\pi \propto e^{-f}$ and the stochastic gradients $\nabla f(\mathbf{x}, \xi)$.

Assumption 1 A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ has *L-Lipschitz continuous gradient*: $\|\nabla f(\mathbf{x}_1) - \nabla f(\mathbf{x}_2)\|_2 \leq L\|\mathbf{x}_1 - \mathbf{x}_2\|_2$, $\forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$ and $\exists L > 0$.

Assumption 2 The stochastic gradient satisfies $\mathbb{E}[\nabla f(\mathbf{x}, \xi) | \mathbf{x}] = \nabla f(\mathbf{x})$ and $\mathbb{E}[\|\nabla f(\mathbf{x}, \xi) - \nabla f(\mathbf{x})\|_2^2 | \mathbf{x}] \leq \sigma^2$, $\forall \mathbf{x} \in \mathbb{R}^d$ and $\exists \sigma > 0$.

We define the *mean-Lipschitz variance-reduced Langevin diffusion (ML-VRLD)* unified estimator as a replacement for the mini-batch SGD estimator in (3):

$$\mathbf{g}_k := \begin{cases} \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i), & \text{with prob. } p \\ \beta \mathbf{g}_{k-1} + (1 - \beta) \nabla f(\mathbf{x}_{k\gamma}, \xi_k), & \text{with prob. } 1 - p \end{cases} \quad (8)$$

where $(\beta, p) \in (0, 1]^2 \setminus \{(1, 1)\}$, $k \geq 1$ is the iteration index, and $\mathbf{g}_0 := \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_0, \xi_0^i)$ is the initial estimate. For the special case of $p = 0$, ML-VRLD reduces to SGD with momentum. When $\beta = 1$, we obtain its stochastic counterpart, which is equivalent to SGD with momentum in expectation and $1 - p$ becomes the momentum parameter. We combine these estimators to develop a unified analysis of both algorithms, while preserving the convergence guarantees obtained by analyzing each algorithm separately. To intuitively see that ML-VRLD estimator has lower variance than the SGLD estimator (3), consider the setting when $p = 0$. The estimator is constructed as a weighted average of a stochastic gradient and past gradient estimates. By

reusing past gradients, stochastic noise is averaged across iterations and, under Assumption 1, this yields an accurate, lower-variance estimate with fixed batch size. To analyze the resulting dynamics, we consider the following interpolation of ML-VRLD:

$$\mathbf{x}_t = \mathbf{x}_{k\gamma} - (t - k\gamma)\mathbf{g}_k + \sqrt{2}(\mathbf{B}_t - \mathbf{B}_{k\gamma}) \quad (9)$$

for $t \in [k\gamma, (k+1)\gamma]$. We are now ready to state our first result.

Theorem 1 (Informal) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 1 and 2. Let $N \geq 1$ denote the total number of iterations, and let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation (9) with estimator (8). Then, for suitable choices of γ, b, p , and β , the time-averaged law $\bar{\mu}_{N\gamma} = \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma} \| \pi) \leq \varepsilon$ with $\mathcal{O}(\sigma^4 L^2 d^2 / \varepsilon^4)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.*

Remark 1. To sample from $\bar{\mu}_{N\gamma}$ in practice, one first draws t uniformly from $[0, N\gamma]$ and identifies the unique integer k such that $t \in [k\gamma, (k+1)\gamma]$. The sample $\mathbf{x}_t$ is then obtained by evaluating the interpolation (9) at time t .

Theorem 1 (see Appendix B.3.1 for the formal statement and proof) establishes the first non-asymptotic convergence guarantee for LMC with stochastic gradients in the non-log-concave setting achieving $\mathcal{O}(1)$ gradient computations per iteration. The dependence on L and d in iteration complexity arises from the discretization error of LMC and matches that of LMC with exact gradients, whose convergence rate scales as $\mathcal{O}(L^2 d^2 / \varepsilon^2)$ [Balasubramanian et al., 2022]. Leveraging the result of Theorem 1, we show that if the target distribution satisfies Poincaré inequality, we obtain stronger convergence guarantees in squared TV distance.

Assumption 3 *A distribution π satisfies a Poincaré inequality if $\exists C_{\text{PI}} > 0$ such that, for all compactly supported functions $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, $\text{Var}_{\mathbf{x} \sim \pi} \phi(\mathbf{x}) \leq C_{\text{PI}} \mathbb{E}_{\mathbf{x} \sim \pi} [\|\nabla \phi(\mathbf{x})\|_2^2]$.*

Corollary 1 (Informal) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 1, 2, and 3. Let $N \geq 1$ denote the total number of iterations, and let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation (9) with estimator (8). Then, for suitable choices of γ, b, p , and β , the time-averaged law $\bar{\mu}_{N\gamma} = \int_0^{N\gamma} \mu_t dt$ satisfies $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $\mathcal{O}(C_{\text{PI}}^4 \sigma^4 L^2 d^2 / \varepsilon^4)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.*

To the best of our knowledge, only two prior works [Balasubramanian et al., 2022, Chewi et al., 2024] establish sampling guarantees for smooth potentials satisfying a Poincaré inequality, both requiring access to exact gradients. In contrast, our results accommodate stochastic gradient estimators and

hold under $\mathcal{O}(1)$ gradient computations per iteration. Moreover, Theorem 1 implies weak convergence to the target distribution under decaying γ_k and p_k , and increasing β_k and b_k .

Theorem 2 (Informal) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 1 and 2. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation (9) with estimator (8), and decreasing step size γ_k and probability p_k , and increasing momentum β_k and batch size b_k . Then, the time-averaged law $\bar{\mu}_{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$, where $\tau_n = \sum_{k=1}^n \gamma_k$ converges weakly to π .*

To our knowledge, existing weak convergence results for LMC either assume Lyapunov-type conditions [Lamberton and Pagès, 2002, Pagès and Panloup, 2012] or having access to exact gradient [Balasubramanian et al., 2022]. In contrast, our result holds when the gradient is replaced by a variance-reduced stochastic gradient estimator, while maintaining constant per-iteration cost. Our analysis builds on the observation that the coefficient of the discrete error difference forms a nonnegative and decreasing sequence, ensuring stability of the resulting error recursion. Combined with the fact that $\text{FI}(\mu \| \pi) = 0$ implies $\mu = \pi$, this establishes weak convergence.

3.2 SL-VRLD

To improve the σ^2 -dependence of the iteration complexity, we propose a gradient estimator that unifies PAGE [Li et al., 2021] and STORM [Cutkosky and Orabona, 2019] under the following assumption, standard in variance-reduced stochastic optimization.

Assumption 4 *A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ has L -Lipschitz continuous stochastic gradients in the mean-squared sense: $\mathbb{E}_{\xi} [\|\nabla f(\mathbf{x}_1, \xi) - \nabla f(\mathbf{x}_2, \xi)\|_2^2] \leq L^2 \|\mathbf{x}_1 - \mathbf{x}_2\|_2^2$, $\forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$ and $\exists L > 0$.*

We define the *sample-Lipschitz variance-reduced Langevin diffusion (SL-VRLD)* unified estimator as a replacement for the mini-batch SGD estimator in (3):

$$\mathbf{g}_k := \begin{cases} \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i), & \text{with prob. } p \\ \nabla f(\mathbf{x}_{k\gamma}, \xi_k) + \beta(\mathbf{g}_{k-1} - \nabla f(\mathbf{x}_{(k-1)\gamma}, \xi_k)), & \text{with prob. } 1 - p \end{cases} \quad (10)$$

where $(\beta, p) \in (0, 1]^2 \setminus \{(1, 1)\}$, $k \geq 1$ is the iteration index, and $\mathbf{g}_0 := \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_0, \xi_0^i)$ is the initial estimate. Setting $\beta = 1$ recovers PAGE, while setting $p = 0$ recovers STORM. Moreover, when $\beta = 1$, taking expectation with respect to p shows that PAGE is equivalent to STORM with momentum parameter $1 - p$, and $p < 1$ yields a lower

per-iteration cost than mini-batch SGD. To highlight the advantage of SL-VRLD over ML-VRLD, consider the case $\beta = 1$. With probability $1 - p$, ML-VRLD relies solely on previous estimates, whereas SL-VRLD incorporates the correction term $\nabla f(\mathbf{x}_{k\gamma}, \xi_k) - \nabla f(\mathbf{x}_{(k-1)\gamma}, \xi_k)$, which, under Assumption 4, improves estimation accuracy and leads to faster convergence with improved FI and squared TV rates.

Theorem 3 (Informal) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 2 and 4. Let $N \geq 1$ denote the total number of iterations, and let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation (9) with estimator (10). Then, for suitable choices of γ, b, p , and β , the time-averaged law $\bar{\mu}_{N\gamma} = \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma} \| \pi) \leq \varepsilon$ with $\mathcal{O}(\sigma^2 L^2 d^2 / \varepsilon^3)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.*

Theorem 3 (with a formal statement and proof provided in Appendix B.3.4) establishes an improved FI convergence rate over that of ML-VRLD, reducing the iteration complexity from $\mathcal{O}(\sigma^4 L^2 d^2 / \varepsilon^4)$ to $\mathcal{O}(\sigma^2 L^2 d^2 / \varepsilon^3)$. A similar improvement holds for the squared TV distance, as stated in Corollary B.1 in Appendix B.3.5, with the complexity reduced from $\mathcal{O}(C_{\text{PI}}^4 \sigma^4 L^2 d^2 / \varepsilon^4)$ to $\mathcal{O}(C_{\text{PI}}^3 \sigma^2 L^2 d^2 / \varepsilon^3)$. In addition, our analysis of FI convergence implies weak convergence of SL-VRLD to the target distribution π ; see Theorem B.1 in Appendix B.3.6.

4 VARIANCE REDUCTION WITH SGM PRIOR

Existing posterior sampling algorithms with SGM priors operate in full-batch regime by using the entire measurement set at each iteration. This limits their applicability to large datasets [Bottou and Bousquet, 2007, Ong et al., 2020]. While mini-batch strategies may appear computationally attractive, accurate sampling typically necessitates large batch sizes. To address this limitation, we develop two principled variance-reduced posterior sampling algorithms by extending our theory in Section 3, and instantiating our unified ML-VRLD and SL-VRLD estimators within the SGM-prior framework. In particular, we replace the mini-batch SGD estimator in (7) by our proposed unified estimators and define the resulting interpolation for posterior sampling

$$\mathbf{x}_t := \mathbf{x}_{k\gamma} - (t - k\gamma) \left(\mathbf{g}_k - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k) \right) + \sqrt{2}(\mathbf{B}_t - \mathbf{B}_{k\gamma}), \quad (11)$$

where $\mathbf{g}_k$ is instantiated as either ML-VRLD (8) or SL-VRLD (10). We provide theoretical guarantees for both choices with the annealing schedule α_k and smoothing schedule σ_k defined as follows

$$\alpha_k := \max\{\alpha_0 \rho_1^k, 1\} \quad \text{and} \quad \sigma_k := \max\{\sigma_0 \rho_2^k, \sigma_{\min}\} \quad (12)$$

for $k \geq 0$, where $\rho_1, \rho_2 \in (0, 1)$ denote decay rates, $\sigma_0 \geq \sigma_{\min}$ and $\alpha_0 \geq 1$ are initial values, and $\sigma_{\min} > 0$ is the minimum noise level. These parameters are selected using the principled techniques of [Song and Ermon, 2020] such that there exists indices $K_\alpha, K_\sigma < N - 1$ satisfying $\alpha_k = 1, \forall k \geq K_\alpha$ and $\sigma_k = \sigma_{\min}, \forall k \geq K_\sigma$ [Sun et al., 2024, Sahin et al., 2026]. Our analysis builds on these schedules together with the interpolation (11). Moreover, since the SGM estimates the true prior score, i.e., $\mathcal{S}_\theta(\mathbf{x}, \sigma_{\min}) \approx -\nabla h(\mathbf{x})$, it is necessary to quantify the estimation error. To this end, we adopt the following assumptions from prior work [Lee et al., 2022a,b, Sun et al., 2024, Sahin et al., 2026].

Assumption 5 *Let $p_{\sigma_k}(\mathbf{x}) := \int p(\mathbf{z}) \mathcal{N}(\mathbf{x} | \mathbf{z}, \sigma_k^2 \mathbf{I}) d\mathbf{z}$, where $h(\mathbf{x})$ and $h_{\sigma_k}(\mathbf{x})$ denote the potential function of $p(\mathbf{x})$ and $p_{\sigma_k}(\mathbf{x})$, respectively. Then, $\exists C_1 > 0$ such that $\|\nabla h_{\sigma_k}(\mathbf{x}) - \nabla h(\mathbf{x})\|_2 \leq C_1 \sigma_k, \forall \mathbf{x} \in \mathbb{R}^d$ and $\forall \sigma_k > 0$.*

Assumption 6 *The score network satisfies $\|\mathcal{S}_\theta(\mathbf{x}, \sigma_k) + \nabla h_{\sigma_k}(\mathbf{x})\|_2 \leq \varepsilon_{\sigma_k}$ and $\|\mathcal{S}_\theta(\mathbf{x}, \sigma_k)\|_2 \leq C_2 \sigma_k^{-1}$ for $\forall \mathbf{x} \in \mathbb{R}^d, \forall \sigma_k > 0$, and $\exists C_2 > 0$.*

Assumption 5 bounds the prior smoothing error. When $p(\mathbf{x})$ is Gaussian, this error admits a closed-form expression but does not hold for general distributions. Assumption 6 bounds the SGM estimation error, thereby relaxing the idealized population setting in which, given sufficient model capacity and data, minimizing the DSM loss (6) recovers the exact score, i.e., $\mathcal{S}_{\theta^*}(\mathbf{x}, \sigma_k) = -\nabla h_{\sigma_k}(\mathbf{x})$ with probability 1. The SGM norm bound is supported by empirical observations in Song and Ermon [2019, 2020]. Unlike prior work [Sun et al., 2024], we do not assume Lipschitz continuity of the SGM. Further discussion of these assumptions is provided in Appendix A. Under these assumptions, we establish the following convergence result.

Theorem 4 (Informal) *Let $\pi(\mathbf{x} | \mathbf{y}) \propto p(\mathbf{y} | \mathbf{x}) p(\mathbf{x})$ be the target posterior, where $p(\mathbf{y} | \mathbf{x}) \propto e^{-f(\mathbf{x})}$ and $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Assume the likelihood potential f satisfies Assumption 2, the prior potential h satisfies Assumptions 1 and 5, and the SGM satisfies Assumption 6 with error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$. Define $L_\pi = L_f + L_h$. Let $N \geq 1$ denote the total number of iterations, and let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation (11) with schedules (12). Then, $\text{FI}(\bar{\mu}_{N\gamma} \| \pi) \leq \varepsilon$ is achieved with $\mathcal{O}(1)$ gradient computations per iteration, and with the following iteration complexities:*

- (a) **(ML-VRLD)** $\mathcal{O}(\sigma^4 L_\pi^2 d^2 / \varepsilon^4)$ if $\mathbf{g}_k$ is defined as in (8) and f satisfies Assumption 1.
- (b) **(SL-VRLD)** $\mathcal{O}(\sigma^2 L_\pi^2 d^2 / \varepsilon^3)$ if $\mathbf{g}_k$ is defined as in (10) and f satisfies Assumption 4.

Formal statements and proofs are provided in Appendices B.4.1 and B.4.2. To the best of our knowledge, this

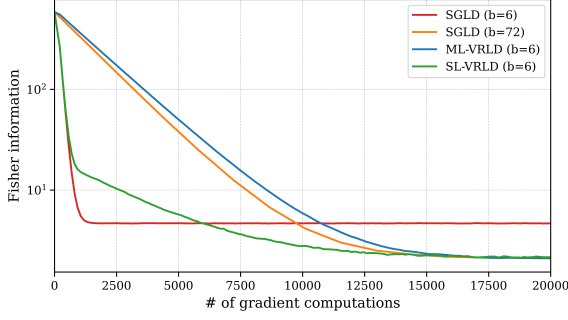


Figure 1: Results on a synthetic 2D Gaussian mixture model (GMM) sampling problem showing the relative FI convergence of each method, where b denotes the batch size. For a fixed batch size, SGLD converges to a suboptimal FI level, whereas ML-VRLD and SL-VRLD achieve substantially lower FI values, consistent with our theoretical results.

result provides the first non-asymptotic ε -accurate convergence guarantee to the target posterior for inverse problems using $\mathcal{O}(1)$ stochastic gradient computations per iteration. Prior work [Sun et al., 2019, Tang and Davies, 2020, Liu et al., 2022] do not establish such guarantees. Our results further imply weak convergence to the target posterior.

Theorem 5 (Informal) *Let $\pi(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ be the target posterior, where $p(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ and $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Assume the likelihood potential f satisfies Assumption 2, the prior potential h satisfies Assumptions 1 and 5, and the SGM satisfies Assumption 6 with error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation (11) with schedules (12), and decreasing step size γ_k and probability p_k , and increasing momentum β_k and batch size b_k . Then, the time-averaged law $\bar{\mu}_{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$, where $\tau_n := \sum_{k=1}^n \gamma_k$, converges weakly to π , provided that one of the following holds:*

- (a) **(ML-VRLD)** $\mathbf{g}_k$ is defined as in (8) and f satisfies Assumption 1.
- (b) **(SL-VRLD)** $\mathbf{g}_k$ is defined as in (10) and f satisfies Assumption 4.

5 EXPERIMENTS

5.1 SYNTHETIC EXPERIMENTS

Numerical Validation. We validate the theoretical results for ML-VRLD and SL-VRLD summarized in Table 1. To this end, we consider two synthetic problems: sampling from a Gaussian mixture model (GMM) and a compressed sensing inverse problem.

For the GMM experiments, we construct an 8-mode target distribution and simulate stochastic gradients by adding Gaussian noise with std. dev. of 30 to the gradient of the

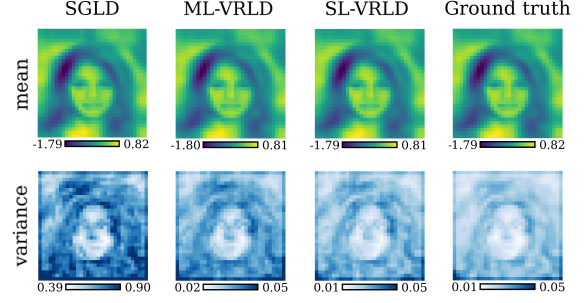


Figure 2: Results on a synthetic inverse problem with known posterior mean and variance. ML-VRLD and SL-VRLD provide more accurate variance estimates than SGLD, which generates samples with significantly higher variance.

potential function. We compare SGLD, ML-VRLD with $\beta = 0.9$, and SL-VRLD with $\beta = 0.999$ over $N = 2000$ iterations using 20 different initializations with $p = 0$. Fig. 1 reports the mean relative FI as a function of the total number of gradient computations. With a fixed batch size of $b = 6$, SGLD converges to a suboptimal relative FI, suggesting that a larger batch size is needed to achieve lower error. Increasing the batch size to $b = 72$ allows SGLD to achieve similar relative FI as ML-VRLD and SL-VRLD, albeit at the cost of substantially more gradient computations per iteration and memory usage. In contrast, SL-VRLD reaches the same error faster than both SGLD and ML-VRLD. These observations are consistent with the theoretical results in Table 1. Visualizations of the generated samples and additional experimental details are provided in Appendix C.1.

For the compressed sensing experiment, we consider a 2D posterior distribution defined by the Gaussian likelihood in (4) and a bimodal Gaussian mixture prior. We simulate stochastic likelihood gradients by adding Gaussian noise with std. dev. of 10 to the likelihood score. We compare SGLD, ML-VRLD, and SL-VRLD with fixed batch size $b = 6$ over 20 test posteriors generated using different realizations of $\mathbf{A}$, and report the mean relative FI in Fig. 5 (Appendix C.1). Consistent with our theory, SL-VRLD converges faster than ML-VRLD, while both achieve lower relative FI than SGLD, which converges to a suboptimal level.

Statistical Validation. We validate ML-VRLD and SL-VRLD for sampling from a posterior distribution over images. We construct a unimodal Gaussian prior obtained from female images in CelebA [Liu et al., 2018]. Each image is rescaled to 32×32 and normalized to $[-1, 1]$. The forward model $\mathbf{A} \in \mathbb{R}^{115 \times 1024}$ is a Gaussian matrix, and the measurements $\mathbf{y}$ are obtained according to (4), with AWGN of std. dev. 0.1. We run SGLD, ML-VRLD, and SL-VRLD with batch size $b = 6$ to generate 1000 samples and compute the sample mean and per-pixel variance of the posterior for each method. We compare these sample statistics to the exact posterior statistics, which are available in closed form,

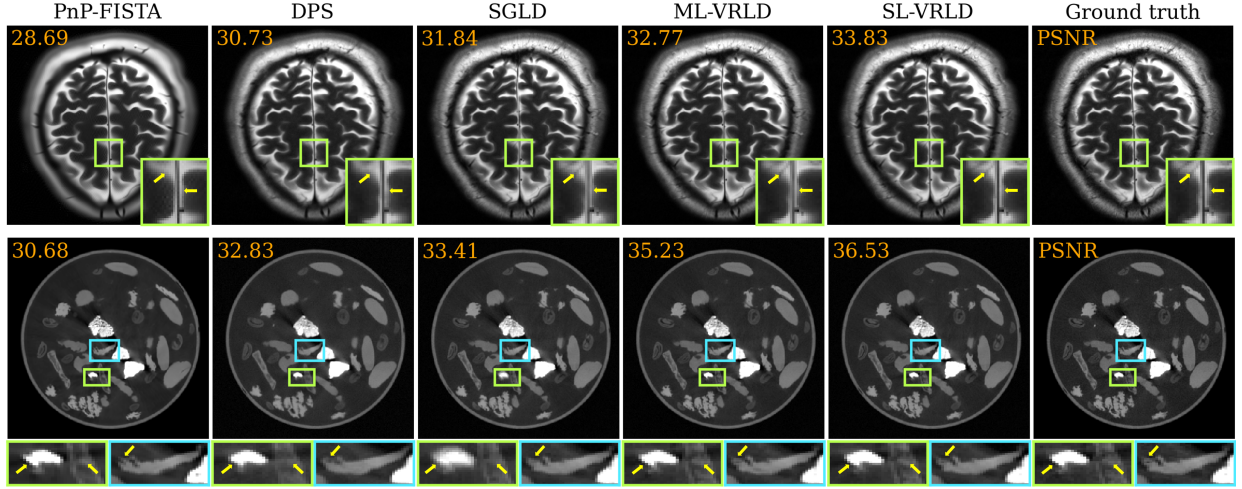


Figure 3: Visual comparison of ML-VRLD, SL-VRLD, and baseline methods for MRI and sparse-angle CT reconstruction. The first row shows MRI results with an acceleration factor of 8 under radial subsampling, using batch size of 32 coils out of 96 coils at each iteration. The second row shows sparse-angle CT reconstructions with a batch size of 30 out of 180 projection views. PSNR values are displayed in the top-left corner of each image. Zoomed-in regions with yellow arrows highlight reconstruction differences. In both tasks, ML-VRLD and SL-VRLD produce sharper structures and fewer artifacts than the baseline methods.

as shown in Fig. 2. SGLD exhibits substantially larger sample variance than ML-VRLD, SL-VRLD, and the ground truth statistics. Both ML-VRLD and SL-VRLD significantly reduce this variance, with SL-VRLD achieving the closest match to the ground truth variance. These results support our theoretical findings. Under a fixed batch size, SGLD fails to recover the correct posterior variance and therefore does not converge to the target posterior, whereas ML-VRLD and SL-VRLD achieve substantially lower estimation error in posterior statistics. Additional results and experimental details are provided in Appendix C.1.

5.2 ACCELERATED PARALLEL MRI RECONSTRUCTION

We simulate a multi-coil MRI acquisition with a radial subsampling pattern [Lustig et al., 2008]. The measurement operator $\mathbf{A}$ is constructed from K receiver coil sensitivity maps $\{\mathbf{S}_i\}_{i=1}^K$ [Pruessmann et al., 1999], where the measurement operator for the i -th coil is given by $\mathbf{A}_i = \mathbf{P}\mathbf{F}\mathbf{S}_i$. Here, $\mathbf{F}$ denotes the Fourier transform and $\mathbf{P}$ is a diagonal matrix with radial subsampling mask. We consider two acceleration factors $R \in \{8, 12\}$, representing highly undersampled regimes used in fast and volumetric MRI acquisitions [Zhang et al., 2020]. To simulate a multi-coil MRI system [Lustig et al., 2008], we synthesize $K = 96$ coil sensitivity maps using SigPy [Ong and Lustig, 2019].

We compare SGLD, ML-VRLD, and SL-VRLD, implemented as annealed LMC sampling with likelihood score estimates defined as in (7), (8), and (10), respectively. The prior score is approximated using the pretrained SGM

from [Sun et al., 2024]. As diffusion-based baselines, we evaluate DPS [Chung et al., 2023] and PnP-DM [Wu et al., 2024], which adopt different sampling strategies for solving inverse problems, and additionally include denoiser-based methods, namely PnP-FISTA [Kamilov et al., 2018, Sun et al., 2019] and RED [Romano et al., 2017]. We use U-Net architecture [Dhariwal and Nichol, 2021] as a denoiser. We further include classical baselines, including TV regularization [Rudin et al., 1992] and zero-filled reconstruction, which is obtained by setting unsampled points to zero followed by inverse Fourier transform. We use brain MRI data from the fastMRI dataset [Zbontar et al., 2019, Knoll et al., 2020] to train the denoiser and diffusion models used in the baselines, and to construct the test set. The training and test sets consist of 70,684 and 100 slices, respectively, where each slice has a shape of 256×256 . For the annealed LMC methods, we adopt the annealing schedule defined in (12) and set $p=0$ for both ML-VRLD and SL-VRLD. The joint use of $\beta < 1$ and $p > 0$ is introduced for theoretical unification. In practice, one may instead set either $p = 0$ or $\beta = 1$. We tune the step size γ and momentum parameter β for ML-VRLD and SL-VRLD using Optuna [Akiba et al., 2019]. All methods use a fixed batch size of $b = 32$ randomly selected coils at each iteration. Additional experimental details are provided in Appendix C.2.

The first row in Fig. 3 shows representative reconstructions at an acceleration factor of $R = 8$. SL-VRLD and ML-VRLD more effectively preserve fine anatomical details, particularly in the zoomed-in regions, whereas the baseline methods tend to oversmooth or introduce artifacts. Table 2 reports quantitative comparison of all methods for two dif-

Method	MRI						CT					
	Accel. = 8×			Accel. = 12×			360 views			180 views		
	PSNR	SSIM	MSE	PSNR	SSIM	MSE	PSNR	SSIM	MSE	PSNR	SSIM	MSE
Zero-filled/FBP	20.01	0.709	10.5e-3	17.67	0.664	17.7e-3	25.05	0.727	31.25e-4	22.31	0.598	58.94e-4
TV	25.12	0.916	32.4e-4	22.12	0.861	65.5e-4	28.58	0.938	15.33e-4	28.25	0.932	16.35e-4
RED	30.36	0.915	9.3e-4	29.09	0.908	12.5e-4	29.07	0.944	13.56e-4	28.93	0.943	13.96e-4
PnP-FISTA	30.56	0.942	9.0e-4	29.04	0.935	12.8e-4	29.97	0.954	10.81e-4	29.39	0.951	12.29e-4
DPS	33.22	0.968	5.0e-4	31.27	0.959	7.8e-4	31.8	0.961	7.17e-4	31.64	0.961	7.43e-4
PnP-DM	33.46	0.969	4.6e-4	<u>31.53</u>	0.961	<u>7.3e-4</u>	33.59	0.971	4.53e-4	31.66	0.959	7.02e-4
SGLD	33.84	0.975	4.4e-4	30.81	0.966	9.1e-4	33.12	0.976	5.36e-4	32.6	0.974	6.06e-4
ML-VRLD	<u>34.63</u>	<u>0.977</u>	<u>3.66e-4</u>	31.42	<u>0.968</u>	7.99e-4	<u>35.02</u>	<u>0.98</u>	<u>3.41e-4</u>	<u>34.35</u>	<u>0.978</u>	<u>4.06e-4</u>
SL-VRLD	35.34	0.978	3.02e-4	33.11	0.971	5.06e-4	35.52	0.981	2.97e-4	34.54	0.979	3.72e-4

Table 2: Quantitative comparison of ML-VRLD and SL-VRLD against baseline methods for MRI reconstruction on the fastMRI brain dataset [Zbontar et al., 2019] and sparse-angle CT reconstruction on the 2DeteCT dataset [Kiss et al., 2023]. The **best** and second-best results are highlighted. For MRI reconstruction, acceleration factors of 8 and 12 are used with a batch size of 32 coils (out of 96 total coils) across all methods. For sparse-angle CT reconstruction, batch sizes of 60 and 30 (out of 360 and 180 projection views, respectively) are used for all methods. See Table 3 (Appendix C) for memory usage.

ferent acceleration factors. SL-VRLD consistently achieves the best performance across all methods, with ML-VRLD yielding the second-best results. These findings demonstrate the advantage of incorporating variance reduction into annealed LMC posterior sampling.

5.3 SPARSE-ANGLE CT RECONSTRUCTION

We consider a sparse-angle CT reconstruction task, which aims to recover high-quality images from a limited number of projection views, resulting in an ill-posed inverse problem due to incomplete angular sampling. We simulate parallel-beam CT measurements with 360 and 180 projection views uniformly distributed over a half circle using the implementation of the measurement operator A in `DeepInverse` [Tachella et al., 2025]. We adopt the same baselines as in the MRI experiments, except that zero-filled reconstruction is replaced by filtered back-projection (FBP), which is more appropriate for CT reconstruction. For all methods, we use the same batch sizes of 60 and 30 out of 360 and 180 projection views, respectively, to estimate the likelihood score. We use abdominal CT scans of size 256×256 from the 2DeteCT dataset [Kiss et al., 2023] to construct the training and test sets. The training set comprises 4,500 slices and is used to train the denoiser, diffusion model, and SGM for their corresponding methods described in the previous section, while the test set consists of 100 slices and is used for evaluation. Additional details are provided in Appendix C.3.

The second row in Fig. 3 shows a visual comparison of reconstructions using a batch size of 30 out of 180 projection views. As highlighted in the zoomed-in regions, SGLD

and the baseline methods exhibit a loss of structural sharpness, whereas ML-VRLD and SL-VRLD better preserve fine anatomical details. We provide quantitative comparison in Table 2. ML-VRLD and SL-VRLD consistently outperforms the baseline methods at two different projection views, demonstrating the benefit of variance reduction.

6 CONCLUSION

We proposed ML-VRLD and SL-VRLD, two unified estimators for non-log-concave Langevin sampling under different regularity conditions. These estimators encompass variance reduction techniques from non-convex optimization, including SGD with momentum, STORM, and PAGE. We showed that both methods achieve ε -relative Fisher information convergence with $\mathcal{O}(1)$ gradient computations per iteration, enabling convergence under fixed memory requirements. In contrast, SGLD requires a batch size scaling as $\mathcal{O}(Ld\sigma^2/\varepsilon)$, making it increasingly expensive in high-dimensional and high-variance settings. Moreover, SL-VRLD reduces the total computational complexity compared to ML-VRLD and SGLD. We further established weak convergence to the target distribution and, under a Poincaré inequality assumption, convergence in squared total variation distance. Motivated by the limitations of existing full-batch posterior sampling methods in large-scale inverse problems, we extended our framework to score-based generative priors and developed principled posterior sampling algorithms with the same convergence guarantees. Experiments on synthetic problems validated our theoretical findings, while MRI and sparse-angle CT reconstruction experiments demonstrated improved sample quality using the proposed estimators.

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Variance Reduction for Non-Log-Concave Sampling with Applications to Inverse Problems (Supplementary Material)

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A DISCUSSION ON ASSUMPTIONS

In this section, we explicitly list and discuss all the assumptions used for our theoretical results in Section 3 and 4. We detail in which case each assumption is verified, especially in solving inverse problems with SGM priors.

- Assumption 1 is standard in both the sampling [Dalalyan, 2017, Cheng et al., 2020, Balasubramanian et al., 2022] and optimization [Nesterov et al., 2018] literature.
- Assumptions 2 and 4 are standard in the analysis of stochastic gradient methods with variance reduction [Fang et al., 2018, Cutkosky and Orabona, 2019, Liu et al., 2020, Li et al., 2021], and are also commonly imposed in the Langevin sampling with stochastic gradients [Chatterji et al., 2018, Dalalyan and Karagulyan, 2019, Cheng et al., 2020, Balasubramanian et al., 2022].
- Assumption 3 is satisfied by all strongly log-concave measures, but it does not hold in general for non-log-concave distributions. This assumption enables convergence in squared total variation distance with constant per-iteration computational cost $\mathcal{O}(1)$, a guarantee not established for strongly log-concave distributions in prior work on variance-reduced Langevin sampling [Chatterji et al., 2018].
- Assumption 5 bounds the score mismatch between the true score $-\nabla h(\mathbf{x})$ and the smoothed score $-\nabla h_{\sigma_k}(\mathbf{x})$. This assumption follows prior work on annealed Langevin sampling with SGM prior [Sun et al., 2024]. In special cases, such as when the prior $p(\mathbf{x})$ is Gaussian, the upper bound can be derived analytically. However, obtaining a closed-form expression for the mismatch is generally intractable for arbitrary distributions.
- Assumption 6 imposes two conditions on the learned score network $\mathcal{S}_\theta(\mathbf{x}, \sigma_k)$: a bounded norm and a bounded approximation error. The bounded-norm condition is consistent with the scaling behavior of diffusion processes on manifolds, where the score scales as $1/\sigma_k$ to provide the restoring force [Song and Ermon, 2019, 2020]. The approximation error bound is justified by interpreting the target prior p_{σ_k} as the population distribution rather than the empirical distribution. Under this interpretation, the assumption enforces a generalization condition on the score network, namely that it approximates the smooth population score with arbitrary precision ($\varepsilon_{\sigma_k} \rightarrow 0$) [Pidstrigach, 2022]. These assumptions align with prior theoretical analyses in [Lee et al., 2022b, Sun et al., 2024].

B THEOREMS AND LEMMAS

B.1 NOTATION

Basic Notation. We denote by $\mathbb{R}^d$ the d -dimensional Euclidean space. For $\mathbf{x} \in \mathbb{R}^d$, $\|\mathbf{x}\|$ denotes the Euclidean norm $\|\mathbf{x}\|_2$. Vectors and matrices are denoted by **boldface** letters.

Langevin Diffusion. We consider the Langevin diffusion defined as the solution to the stochastic differential equation (SDE)

$$d\mathbf{x}_t = -\nabla f(\mathbf{x}_t) dt + \sqrt{2} d\mathbf{B}_t,$$

where $(\mathbf{B}_t)_{t \geq 0}$ is a standard d -dimensional Brownian motion. The target distribution is denoted by π , and is given by

$$\pi(\mathbf{x}) \propto \exp(-f(\mathbf{x})).$$

Under mild conditions, this SDE admits π as its unique stationary distribution. Throughout, we assume that f is continuously differentiable and that its gradient ∇f is Lipschitz continuous (see Assumption 1). We assume access to stochastic first-order oracles $\nabla f(\mathbf{x}, \xi)$, where ξ denotes a random variable capturing external randomness. Note that the stochastic first-order oracles are unbiased and have bounded variance (see Assumption 2). Discretizing the above SDE with step size $\gamma > 0$ yields the Langevin Monte Carlo (LMC) update

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma \nabla f(\mathbf{x}_{k\gamma}) + \sqrt{2} (\mathbf{B}_{(k+1)\gamma} - \mathbf{B}_{k\gamma}),$$

where $\mathbf{B}_{k\gamma} \in \mathbb{R}^d$ is Brownian motion. We further define $\mathbf{g}_k$ as estimator of the gradient $\nabla f(\mathbf{x}_{k\gamma})$ at iteration k . Replacing the exact gradient by $\mathbf{g}_k$, the LMC update becomes

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma \mathbf{g}_k + \sqrt{2} (\mathbf{B}_{(k+1)\gamma} - \mathbf{B}_{k\gamma}).$$

The corresponding continuous-time interpolation is given by

$$\mathbf{x}_t = \mathbf{x}_{k\gamma} - (t - k\gamma)\mathbf{g}_k + \sqrt{2} (\mathbf{B}_t - \mathbf{B}_{k\gamma}), \quad t \in [k\gamma, (k+1)\gamma],$$

and we denote by μ_t the law of $\mathbf{x}_t$.

Posterior Sampling. In the posterior sampling setting for ill-posed inverse problems of the form $\mathbf{y} = \mathbf{A}(\mathbf{x}) + \mathbf{n}$, we model the prior as $p(\mathbf{x}) \propto \exp(-h(\mathbf{x}))$, where $h(\mathbf{x})$ is the prior potential. We model the likelihood as $p(\mathbf{y}|\mathbf{x}) \propto \exp(-f(\mathbf{x}))$, where $f(\mathbf{x})$ is the likelihood potential. By Bayes' rule, the posterior satisfies

$$p(\mathbf{x}|\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{x})p(\mathbf{x})}{\int p(\mathbf{y}|\mathbf{u})p(\mathbf{u})d\mathbf{u}} \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \propto \exp(-(f(\mathbf{x}) + h(\mathbf{x}))).$$

Sampling from the posterior can be performed via Langevin dynamics

$$d\mathbf{x}_t = -(\nabla f(\mathbf{x}_t) + \nabla h(\mathbf{x}_t))dt + \sqrt{2}d\mathbf{B}_t,$$

which admits $\pi(\mathbf{x}|\mathbf{y})$ as its stationary distribution. For ease of notation, we omit the explicit dependence on $\mathbf{y}$ and write $\pi(\mathbf{x}|\mathbf{y})$ simply as $\pi(\mathbf{x})$ in the proofs. Discretizing this SDE with step size $\gamma > 0$ yields

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma(\nabla f(\mathbf{x}_{k\gamma}) + \nabla h(\mathbf{x}_{k\gamma})) + \sqrt{2}(\mathbf{B}_{(k+1)\gamma} - \mathbf{B}_{k\gamma}).$$

We use an estimator $\mathbf{g}_k$ (instantiated as ML-VRLD (8) or SL-VRLD (10)) to approximate $\nabla f(\mathbf{x}_{k\gamma})$. Moreover, we use a score network $\mathcal{S}_\theta(\mathbf{x}, \sigma_k)$ to approximate the smoothed prior score, i.e., $\mathcal{S}_\theta(\mathbf{x}, \sigma_k) \approx -\nabla h_{\sigma_k}(\mathbf{x})$. With annealing and smoothing schedules $(\alpha_k)_{k=0}^N$ and $(\sigma_k)_{k=0}^N$, the annealed posterior update is

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma(\mathbf{g}_k - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)) + \sqrt{2}(\mathbf{B}_{(k+1)\gamma} - \mathbf{B}_{k\gamma}).$$

The corresponding continuous-time interpolation on $t \in [k\gamma, (k+1)\gamma]$ is

$$\mathbf{x}_t = \mathbf{x}_{k\gamma} - (t - k\gamma)(\mathbf{g}_k - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)) + \sqrt{2}(\mathbf{B}_t - \mathbf{B}_{k\gamma}), \quad t \in [k\gamma, (k+1)\gamma].$$

KL Divergence and Fisher Information. The Kullback–Leibler divergence between two probability measures μ and π is defined as

$$\text{KL}(\mu\|\pi) := \int_{\mathbb{R}^d} \mu(\mathbf{x}) \log \frac{\mu(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}.$$

The relative Fisher information is

$$\text{FI}(\mu\|\pi) := \int_{\mathbb{R}^d} \mu(\mathbf{x}) \left\| \nabla \log \frac{\mu(\mathbf{x})}{\pi(\mathbf{x})} \right\|_2^2 d\mathbf{x}.$$

Probability Measures. Expectation with respect to a distribution μ is denoted by $\mathbb{E}_{\mathbf{x} \sim \mu}[\cdot]$. When clear from context, we write $\mathbb{E}[\cdot]$.

Big-O Notation. For two positive sequences $f(n)$ and $g(n)$, we write $f(n) = \mathcal{O}(g(n))$ if there exists $C > 0$ such that $f(n) \leq Cg(n)$ for sufficiently large n . In particular, $\mathcal{O}(1)$ denotes a positive constant.

B.2 LEMMAS

We begin by presenting several technical lemmas that are used throughout the proofs of the main theorems. The following lemma provides the starting point for our analysis.

Lemma 1 (Lemma 23 in Balasubramanian et al. [2022]) *Consider the stochastic process defined by*

$$\mathbf{x}_t := \mathbf{x}_0 - t\mathbf{g}_0 + \sqrt{2}\mathbf{B}_t, \quad \text{with } \mathbf{g}_0 = g(\mathbf{x}_0), \quad \mathbf{x}_0 \sim \mu_0 \quad (13)$$

where $\mathbf{g}_0$ is integrable and $(\mathbf{B}_t)_{t \geq 0}$ is a standard Brownian motion in $\mathbb{R}^d$ independent of $(\mathbf{x}_0, \mathbf{g}_0)$. Then, writing μ_t for the law of $\mathbf{x}_t$, we have

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t\|\pi) &\leq -\frac{3}{4} \text{FI}(\mu_t\|\pi) + \mathbb{E}[\|\nabla f(\mathbf{x}_t) - \mathbb{E}[\mathbf{g}_0|\mathbf{x}_t]\|^2] \\ &\leq -\frac{3}{4} \text{FI}(\mu_t\|\pi) + \mathbb{E}[\|\nabla f(\mathbf{x}_t) - \mathbf{g}_0\|^2], \end{aligned} \quad (14)$$

where $\pi \propto e^{-f}$.

Lemma 2 (Lemma 20 in Chewi et al. [2024]) Assume that $f(\mathbf{x})$ has an L -Lipschitz gradient, and let $\pi \propto e^{-f}$. Then, for any probability measure μ , it holds that

$$\mathbb{E}_{\mathbf{x} \sim \mu} [\|\nabla f(\mathbf{x})\|^2] \leq \text{FI}(\mu \|\pi) + 2dL. \quad (15)$$

We use the transportation inequality to analyze the convergence in squared total variation (TV) distance under a Poincaré inequality assumption.

Lemma 3 (Theorem 3.1 in Guillin et al. [2009]) If $\pi \propto e^{-f}$ satisfies a Poincaré inequality, i.e. for every smooth, compactly supported $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\text{Var}_\pi(f) \leq C_{\text{PI}} \mathbb{E}_\pi[\|\nabla f\|^2],$$

then for any probability measure μ ,

$$\|\mu - \pi\|_{\text{TV}}^2 \leq 4C_{\text{PI}} \text{FI}(\mu \|\pi).$$

B.3 THEOREMS

We first state a theorem characterizing the sufficient number of gradient computations per iteration to achieve ε -relative Fisher information for SGLD (3) algorithm.

Theorem B.1 (Theorem 15 in Balasubramanian et al. [2022]) Let $(\mu_t)_{t \geq 0}$ denote the law of the interpolation generated by (9) with the estimator defined as $\mathbf{g}_k = \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i)$, where $b \geq 1$. Assume that the potential f defining the target distribution $\pi \propto e^{-f}$ satisfies Assumptions 1 and 2. Then, for any step size $\gamma \in (0, \frac{1}{14L})$, and for any $N \geq 1$, it holds that

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2\text{KL}(\mu_0 \|\pi)}{N\gamma} + 16L^2 d\gamma + \frac{8\sigma^2}{b}$$

Furthermore, choose $\gamma = \frac{1}{2L} \sqrt{\frac{\text{KL}(\mu_0 \|\pi)}{dN}}$ and $b = \lceil \sigma^2 \sqrt{N} \rceil$. Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \varepsilon$ with $\mathcal{O}(\frac{L^2 d^2}{\varepsilon^2})$ iterations and $\mathcal{O}(\frac{L d \sigma^2}{\varepsilon})$ gradient computations per iteration.

Proof. We consider the following interpolation for SGLD with mini-batch estimates

$$\mathbf{x}_t := \mathbf{x}_{k\gamma} - (t - k\gamma) \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{k\gamma}, \xi_k^i) + \sqrt{2}(\mathbf{B}_t - \mathbf{B}_{k\gamma})$$

for $t \in [k\gamma, (k+1)\gamma]$. Under Assumptions 1 and 2, we follow the proof of Theorem 15 in Balasubramanian et al. [2022] to obtain

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2\text{KL}(\mu_0 \|\pi)}{N\gamma} + 16L^2 d\gamma + \frac{8\sigma^2}{b} \quad (16)$$

for all $\gamma \in (0, \frac{1}{14L})$. Here, $\text{KL}(\mu_0 \|\pi)$ denotes the initial Kullback-Leibler divergence and it scales on the order of d (see, e.g., Lemma 1 in Vempala and Wibisono [2019] or Appendix A in Chewi et al. [2025]). We then choose

$$\gamma = \frac{1}{2L} \sqrt{\frac{\text{KL}(\mu_0 \|\pi)}{dN}}, \quad \text{and} \quad b = \lceil \sigma^2 \sqrt{N} \rceil.$$

Plugging these definitions into (16), we get

$$\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{(12L \sqrt{\text{KL}(\mu_0 \|\pi)d} + 8)}{\sqrt{N}}, \quad (17)$$

where the first inequality follows from the convexity of the Fisher information combined with Jensen's inequality. Since $\text{KL}(\mu_0 \|\pi) = \mathcal{O}(d)$, we have $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \varepsilon$ if

$$N = \mathcal{O}\left(\frac{L^2 d^2}{\varepsilon^2}\right), \quad (18)$$

which implies $b = \mathcal{O}(\frac{Ld\sigma^2}{\varepsilon})$. This concludes the proof. $\square$

In the remainder of this section, we present our theoretical results and their proofs. First, we give results for ML-VRLD and SL-VRLD sampling algorithms to sample from a target distribution $\pi \propto e^{-f(\mathbf{x})}$. Then, we present the results for the application of these algorithms to inverse problems with score-based generative prior, where the goal is to sample from the posterior $p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ with prior $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$ and likelihood $p(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$. In practice, the settings of interest are often high-variance, where σ^2 , the upper bound on the variance of stochastic gradients in Assumption 2, satisfies $\sigma^2 > 1$. Nevertheless, to also cover the case $\sigma^2 < 1$, we state our theoretical results in terms of $\sigma_m := \max\{\sigma, 1\}$. For ease of exposition, we use σ throughout the main manuscript.

B.3.1 Proof of Theorem 1 (ML-VRLD: FI convergence)

Theorem 1 (ML-VRLD: FI convergence) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 1 and 2. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation generated by (9) with estimator (8). Then, for any step size $\gamma \in \left(0, \frac{1}{L\sqrt{27\phi(p,\beta)}}\right]$, where $\phi(p,\beta) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$, and for any $N \geq 1$, it holds that*

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t|\pi) dt \leq \frac{2C_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1 - (1-p)\beta} \right] + \frac{72\gamma L^2 d}{(1 - (1-p)\beta)^2}, \quad (19)$$

where

$$C_0 = \text{KL}(\mu_0|\pi) + \frac{9\gamma}{(1 - (1-p)\beta)^2} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2]. \quad (20)$$

Furthermore, choose

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L}}, \quad p = 1 - \beta = \frac{\sqrt{Ld}}{\sigma_m N^{1/4}}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\}.$$

Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma}|\pi) \leq \varepsilon$ with $\mathcal{O}(\frac{\sigma_m^4 L^2 d^2}{\varepsilon^4})$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. Using the interpolation argument in (9), we begin the proof by recalling Lemma 1

$$\frac{d}{dt} \text{KL}(\mu_t|\pi) \leq -\frac{3}{4} \text{FI}(\mu_t|\pi) + \mathbb{E}[\|\nabla f(\mathbf{x}_t) - \mathbf{g}_k\|^2].$$

We decompose the error term by introducing $\nabla f(\mathbf{x}_{k\gamma})$ and then apply Young's inequality (i.e., $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$), obtaining

$$\frac{d}{dt} \text{KL}(\mu_t|\pi) \leq -\frac{3}{4} \text{FI}(\mu_t|\pi) + 2\mathbb{E}[\|\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{k\gamma})\|^2] + 2\mathbb{E}[\|\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k\|^2].$$

Invoking Assumption 1, we upper bound $\mathbb{E}[\|\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{k\gamma})\|^2]$ using Lipschitz continuity of the potential ∇f , which gives

$$\frac{d}{dt} \text{KL}(\mu_t|\pi) \leq -\frac{3}{4} \text{FI}(\mu_t|\pi) + 2L^2 \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2] + 2e_k^2, \quad (21)$$

where we define $e_k^2 := \mathbb{E}[\|\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k\|^2]$. The next step is to bound e_k^2 . Using the definition of the ML-VRLD estimator (8), e_{k+1}^2 can be expressed as follows. With probability p , a mini-batch estimator of size b is used, while with probability $1 - p$ a momentum-based update is applied. This yields

$$\begin{aligned} e_{k+1}^2 &= p \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1}^i) \right\|^2 \right] \\ &\quad + (1-p) \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - (\beta \mathbf{g}_k + (1-\beta) \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1})) \right\|^2 \right]. \end{aligned}$$

Using Assumption 2, we upper bound the variance of the mini-batch estimator, which gives

$$e_{k+1}^2 \leq \frac{p\sigma^2}{b} + (1-p) \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - (\beta \mathbf{g}_k + (1-\beta) \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1})) \right\|^2 \right].$$

We decompose the second term by adding and subtracting $\nabla f(\mathbf{x}_{k\gamma})$ inside expectation, regrouping the terms and using Assumption 2

$$\begin{aligned} e_{k+1}^2 &\leq \frac{p\sigma^2}{b} + (1-p) \mathbb{E} \left[\left\| \beta(\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k) + \beta(\nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{k\gamma})) \right. \right. \\ &\quad \left. \left. + (1-\beta)(\nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1})) \right\|^2 \right] \\ &\leq \frac{p\sigma^2}{b} + (1-p)\beta^2 \mathbb{E} \left[\left\| (\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k) + (\nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{k\gamma})) \right\|^2 \right] + (1-\beta)^2(1-p)\sigma^2. \end{aligned}$$

Applying Young's inequality with parameter $\rho_0 > 0$ (i.e., $\|a+b\|^2 \leq (1+\rho_0)\|a\|^2 + (1+1/\rho_0)\|b\|^2$), together with Assumption 1 to bound $\mathbb{E} \left[\left\| (\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k) + (\nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{k\gamma})) \right\|^2 \right]$, we obtain

$$\begin{aligned} e_{k+1}^2 &\leq \frac{p\sigma^2}{b} + (1-p)\beta^2 L^2 \left(1 + \frac{1}{\rho_0} \right) \Delta_k + (1-p)\beta^2(1+\rho_0)e_k^2 + (1-p)(1-\beta)^2\sigma^2 \\ &= (1-p)\beta^2 L^2 \left(1 + \frac{1}{\rho_0} \right) \Delta_k + (1-p)\beta^2(1+\rho_0)e_k^2 + \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] \sigma^2, \end{aligned}$$

where $\Delta_k := \mathbb{E} \left[\left\| \mathbf{x}_{(k+1)\gamma} - \mathbf{x}_{k\gamma} \right\|^2 \right]$. Furthermore, we choose $\rho_0 = \frac{1-(1-p)\beta}{2(1-p)\beta}$. Rearranging the terms then yields

$$\begin{aligned} e_k^2 &\leq -\frac{2}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{\left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] 2\sigma^2}{1-(1-p)\beta} + \frac{2(1-p)\beta^2 L^2 (1+(1-p)\beta)}{(1-(1-p)\beta)^2} \Delta_k \\ &\leq -\frac{2}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{\left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] 2\sigma^2}{1-(1-p)\beta} + \frac{4(1-p)\beta^2 L^2}{(1-(1-p)\beta)^2} \Delta_k. \end{aligned} \quad (22)$$

Since $p \in (0, 1]$ and $\beta \in (0, 1]$, it follows that $(1-p)\beta \in [0, 1]$, and hence $1 + (1-p)\beta \leq 2$. We have used this bound to simplify the last term. We now bound Δ_k by using the continuous interpolation (9) of LMC. For $t \in [k\gamma, (k+1)\gamma]$,

$$\mathbf{x}_t - \mathbf{x}_{k\gamma} = -(t - k\gamma)\mathbf{g}_k + \sqrt{2}(\mathbf{B}_t - \mathbf{B}_{k\gamma}).$$

Taking squared norms on both sides and then taking expectations, and using the independence of $\mathbf{g}_k$ and the Brownian increment, we obtain

$$\begin{aligned} \mathbb{E} \left[\left\| \mathbf{x}_t - \mathbf{x}_{k\gamma} \right\|^2 \right] &= (t - k\gamma)^2 \mathbb{E} \left[\left\| \mathbf{g}_k \right\|^2 \right] + 2\mathbb{E} \left[\left\| \mathbf{B}_t - \mathbf{B}_{k\gamma} \right\|^2 \right] \\ &= (t - k\gamma)^2 \mathbb{E} \left[\left\| \mathbf{g}_k \right\|^2 \right] + 2(t - k\gamma)d \\ &\leq \gamma^2 \mathbb{E} \left[\left\| \mathbf{g}_k \right\|^2 \right] + 2(t - k\gamma)d \\ &= \Delta_k, \end{aligned}$$

where we have used the identity $\mathbb{E} \left[\left\| \mathbf{B}_t - \mathbf{B}_{k\gamma} \right\|^2 \right] = (t - k\gamma)d$. Writing

$$\mathbf{g}_k = (\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma})) + (\nabla f(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_t)) + \nabla f(\mathbf{x}_t),$$

and applying the inequality $\|a+b+c\|^2 \leq 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$, we obtain

$$\begin{aligned} \Delta_k &= \gamma^2 \mathbb{E} \left[\left\| \mathbf{g}_k \right\|^2 \right] + 2\gamma d \\ &\leq 3\gamma^2 \mathbb{E} \left[\left\| \mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma}) \right\|^2 \right] + 3\gamma^2 \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_t) \right\|^2 \right] + 3\gamma^2 \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_t) \right\|^2 \right] + 2\gamma d. \end{aligned}$$

Using Lipschitz continuity of ∇f (Assumption 1) and the bound $\mathbb{E} \left[\left\| \mathbf{x}_t - \mathbf{x}_{k\gamma} \right\|^2 \right] \leq \Delta_k$ yields

$$\Delta_k \leq 3\gamma^2 e_k^2 + 3\gamma^2 L^2 \Delta_k + 3\gamma^2 \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_t) \right\|^2 \right] + 2\gamma d.$$

Substituting the upper bound (22) for e_k^2 and rearranging the terms yields

$$\begin{aligned}\Delta_k \leq & -\frac{6\gamma^2}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{6\gamma^2\sigma^2}{1-(1-p)\beta} \left[\frac{p}{b} + (1-p)(1-\beta^2) \right] \\ & + 3\gamma^2 L^2 \left[1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2} \right] \Delta_k + 3\gamma^2 \mathbb{E}[\|\nabla f(x_t)\|^2] + 2\gamma d.\end{aligned}$$

To simplify notation, we define $\phi(p, \beta) := 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$. With this definition and rearranging the terms, the inequality becomes

$$\begin{aligned}[1 - 3\gamma^2 L^2 \phi(p, \beta)] \Delta_k \leq & -\frac{6\gamma^2}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{6\gamma^2\sigma^2}{1-(1-p)\beta} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] \\ & + 3\gamma^2 \mathbb{E}[\|\nabla f(x_t)\|^2] + 2\gamma d.\end{aligned}$$

Assuming the step size satisfies $\gamma \leq \frac{1}{27L^2\phi(p, \beta)}$, the coefficient of Δ_k on the left-hand side satisfies $1 - 3\gamma^2 L^2 \phi(p, \beta) \geq \frac{8}{9}$. Applying this bound and dividing both sides by $\frac{8}{9}$ yields

$$\Delta_k \leq -\frac{27\gamma^2}{4(1-(1-p)\beta)} (e_{k+1}^2 - e_k^2) + \frac{27\gamma^2\sigma^2}{4(1-(1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + \frac{27\gamma^2}{8} \mathbb{E}[\|\nabla f(x_t)\|^2] + \frac{9}{4} \gamma d. \quad (23)$$

Now, recall from (21), we have

$$\frac{d}{dt} \text{KL}(\mu_t \| \pi) \leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + 2L^2 \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2] + 2e_k^2,$$

Applying (22) and (23) together with $\mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2] \leq \Delta_k$, we obtain

$$\begin{aligned}\frac{d}{dt} \text{KL}(\mu_t \| \pi) \leq & -\frac{3}{4} \text{FI}(\mu_t \| \pi) - \frac{1}{1-(1-p)\beta} \left[4 + \frac{27}{2} \gamma^2 L^2 \phi(p, \beta) \right] (e_{k+1}^2 - e_k^2) \\ & + \frac{\sigma^2}{1-(1-p)\beta} \left[4 + \frac{27}{2} \gamma^2 L^2 \phi(p, \beta) \right] \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] \\ & + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta) \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{9}{2} \gamma L^2 \phi(p, \beta) d.\end{aligned}$$

Applying Lemma 2 to bound $\mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2]$, and using the step size condition $\gamma \leq \frac{1}{27L^2\phi(p, \beta)}$, we simplify the above inequality as

$$\begin{aligned}\frac{d}{dt} \text{KL}(\mu_t \| \pi) \leq & -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{1}{1-(1-p)\beta} \left[4 + \frac{27}{2} \gamma^2 L^2 \phi(p, \beta) \right] (e_{k+1}^2 - e_k^2) \\ & + \frac{9\sigma^2}{2(1-(1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + 9\gamma L^2 \phi(p, \beta) d.\end{aligned} \quad (24)$$

We now integrate both sides over $t \in [k\gamma, (k+1)\gamma]$ and define $\mathcal{L}_k := \text{KL}(\mu_{k\gamma} \| \pi) + \frac{\gamma}{1-(1-p)\beta} \left[4 + \frac{27}{2} \gamma^2 L^2 \phi(p, \beta) \right] e_k^2$. This yields

$$\mathcal{L}_{k+1} - \mathcal{L}_k \leq -\frac{1}{2} \int_{k\gamma}^{(k+1)\gamma} \text{FI}(\mu_t \| \pi) dt + \frac{9\gamma\sigma^2}{2(1-(1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + 9\gamma^2 L^2 \phi(p, \beta) d.$$

Iterating the inequality for $k = 0, 1, \dots, N-1$, we obtain

$$\mathcal{L}_N - \mathcal{L}_0 \leq -\frac{1}{2} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt + \frac{9N\gamma\sigma^2}{2(1-(1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + 9N\gamma^2 L^2 \phi(p, \beta) d.$$

Rearranging the terms, we get

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \frac{2\mathcal{L}_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1-(1-p)\beta} \right] + 18\gamma L^2 \phi(p, \beta) d.$$

Note that $\mathcal{L}_0 = \text{KL}(\mu_0 \|\pi) + \frac{\gamma}{1-(1-p)\beta} \left(4 + \frac{27}{2}\gamma^2 L^2 \phi(p, \beta)\right) e_0^2 \leq \text{KL}(\mu_0 \|\pi) + \frac{9\gamma e_0^2}{2(1-(1-p)\beta)} =: C_0$, then

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2C}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1-(1-p)\beta} \right] + 18\gamma L^2 \phi(p, \beta) d.$$

Also $\phi(p, \beta) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2} \leq 1 + \frac{4(1-p)\beta}{(1-(1-p)\beta)^2} = \left(\frac{1+(1-p)\beta}{1-(1-p)\beta} \right)^2 \leq \frac{4}{(1-(1-p)\beta)^2}$ since $1 + (1-p)\beta \leq 2$, then the inequality can be rewritten as

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2C_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1-(1-p)\beta} \right] + \frac{72\gamma L^2 d}{(1-(1-p)\beta)^2}. \quad (25)$$

This concludes the first part of the proof. Next we choose

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L}}, \quad p = (1-\beta) = \frac{\sqrt{Ld}}{N^{1/4} \sigma_m} \quad \text{and} \quad b = \left\lceil \frac{1}{p} \right\rceil \quad \text{where} \quad \sigma_m := \max\{\sigma, 1\}$$

and define the time-averaged law $\bar{\mu}_{N\gamma} := \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$. Since $\text{FI}(\cdot \|\pi)$ is convex, using Jensen's inequality, we obtain the upper bound for $\text{FI}(\bar{\mu}_{N\gamma} \|\pi)$ as

$$\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \underbrace{\frac{2C_0}{N\gamma}}_{(I)} + \underbrace{9\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1-(1-p)\beta} \right]}_{(II)} + \underbrace{\frac{72\gamma L^2 d}{(1-(1-p)\beta)^2}}_{(III)}.$$

Substituting the definition of γ into (I), we have

$$(I) = \frac{2C_0 \sigma_m \sqrt{L}}{N^{1/4} \sqrt{d}}. \quad (26)$$

To bound (II), we proceed as

$$(II) \leq 9\sigma_m^2 \left[\frac{p^2 + (1-p)p^2}{1-(1-p)^2} \right] \quad (27)$$

$$\leq 18\sigma_m^2 p. \quad (28)$$

We use $\sigma \leq \sigma_m$, $b \geq 1/p$, and $1-\beta = p$ to get (27). Following that, we use $p \in (0, 1]$ to get (28). Substituting the definition of p , we have

$$(II) \leq \frac{18\sigma_m \sqrt{Ld}}{N^{1/4}}. \quad (29)$$

Similarly, to bound (III),

$$(III) \leq \frac{72\gamma L^2 d}{p^2} \quad (30)$$

$$\leq \frac{72\sigma_m \sqrt{Ld}}{N^{1/4}}, \quad (31)$$

where we use $p = 1-\beta$ and $p \in (0, 1]$ to obtain (30), and substitute the definitions of γ and p to get (31). Combining (26, 29, 31), we have

$$\begin{aligned} \text{FI}(\bar{\mu}_{N\gamma} \|\pi) &\leq \frac{2C_0 \sigma_m \sqrt{L}}{N^{1/4} \sqrt{d}} + \frac{90\sigma_m \sqrt{Ld}}{N^{1/4}} \\ &\leq \left(\frac{2C_0}{d} + 90 \right) \frac{\sigma_m \sqrt{Ld}}{N^{1/4}}, \end{aligned}$$

where C_0 is usually taken to be of order d , see e.g. Lemma 1 in Vempala and Wibisono [2019] or Appendix A in Chewi et al. [2025]. Thus, the above bound implies that, to ensure $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \varepsilon$,

$$N = \mathcal{O}\left(\frac{\sigma_m^4 L^2 d^2}{\varepsilon^4}\right)$$

iterations is sufficient. And the average number of gradient computations per iteration is

$$pb + (1 - p) = \mathcal{O}(1). \quad (32)$$

We next verify that the parameter constraints imposed throughout the analysis are compatible with this choice of N . First, the requirement $p = (1 - \beta) = \frac{\sqrt{dL}}{N^{1/4}\sigma_m} \leq 1$ imposes the lower bound $N \geq \frac{d^2 L^2}{\sigma_m^4}$, which is satisfied by the above choice of N . Second, we show that the step-size condition $\gamma \leq \frac{1}{L\sqrt{27\phi(p, \beta)}}$ is satisfied. Since $\phi(p, \beta) \leq \frac{4}{(1 - (1 - p)\beta)^2}$, we have

$$\gamma \leq \frac{1 - (1 - p)\beta}{L\sqrt{108}} \leq \frac{1}{L\sqrt{27\phi(p, \beta)}}.$$

Substituting the definitions of $\beta = 1 - p$ and $p = \frac{\sqrt{dL}}{\sigma_m N^{1/4}}$, we get

$$\begin{aligned} \frac{1 - (1 - p)\beta}{L\sqrt{108}} &= \frac{1 - (1 - p)^2}{L\sqrt{108}} \\ &\geq \frac{p}{L\sqrt{108}} \\ &= \frac{\sqrt{d}}{\sigma_m N^{1/4} \sqrt{108L}}. \end{aligned}$$

Thus, the step-size condition holds provided that

$$\gamma \leq \frac{\sqrt{d}}{\sigma_m N^{1/4} \sqrt{108L}}.$$

Using the definition of step size

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L}},$$

this condition reduces to the following lower bound $N \geq 108$, which is satisfied by $N = \mathcal{O}\left(\frac{\sigma_m^4 L^2 d^2}{\varepsilon^4}\right)$. $\square$

B.3.2 Proof of Corrolary 1 (ML-VRLD: squared TV distance convergence)

Corrolary 1 (ML-VRLD: squared TV distance convergence) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 1, 2, and 3. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation generated by (9) with estimator (8). Then, for any step size $\gamma \in \left(0, \frac{1}{L\sqrt{27\phi(p, \beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{4(1-p)\beta^2}{(1 - (1-p)\beta)^2}$, and for any $N \geq 1$, it holds that*

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0 C_{\text{PI}}}{N\gamma} + 36C_{\text{PI}}\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1 - (1-p)\beta} \right] + \frac{288C_{\text{PI}}\gamma L^2 d}{(1 - (1-p)\beta)^2}, \quad (33)$$

where

$$C_0 = \text{KL}(\mu_0 \|\pi) + \frac{9\gamma}{(1 - (1-p)\beta)^2} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2]. \quad (34)$$

Furthermore, choose

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L}}, \quad p = 1 - \beta = \frac{\sqrt{dL}}{N^{1/4} \sigma_m}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\}.$$

Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $\mathcal{O}\left(\frac{C_{\text{PI}}^4 \sigma_m^4 L^2 d^2}{\varepsilon^4}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. Recall from the proof of Theorem 1 (Appendix B.3.1) that the Fisher information of the time-averaged law $\bar{\mu}_{N\gamma}$ satisfies (25)

$$\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2C_0}{N\gamma} + 9\sigma_m^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1 - (1-p)\beta} \right] + \frac{72\gamma L^2 d}{(1 - (1-p)\beta)^2},$$

where we use $\sigma \leq \sigma_m$. Using Lemma 3 and Assumption 3, we get

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0 C_{\text{PI}}}{N\gamma} + 36C_{\text{PI}} \sigma_m^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1 - (1-p)\beta} \right] + \frac{288C_{\text{PI}} \gamma L^2 d}{(1 - (1-p)\beta)^2}.$$

Choose

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L}}, \quad p = 1 - \beta = \frac{\sqrt{Ld}}{N^{1/4} \sigma_m}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m := \max\{\sigma, 1\}.$$

Following the same steps in (26, 29, 31), we obtain

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \left(\frac{2C_0}{d} + 90 \right) \frac{4C_{\text{PI}} \sigma_m \sqrt{Ld}}{N^{1/4}},$$

where C_0 is in the order of d . This implies that, to ensure $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$, it suffices to choose N such that $N^{1/4} \geq \frac{\tilde{C} C_{\text{PI}} \sigma_m \sqrt{Ld}}{\varepsilon}$, where $\tilde{C} := \frac{8C_0}{d} + 360 > 0$. Equivalently, this yields iteration complexity of $\mathcal{O}\left(\frac{C_{\text{PI}}^4 \sigma_m^4 L^2 d^2}{\varepsilon^4}\right)$ with $\mathcal{O}(1)$ gradient computations per iteration. Since the constraints on N in Theorem 1 remain unchanged, they continue to hold. $\square$

B.3.3 Proof of Theorem 2 (ML-VRLD: weak convergence)

Theorem 2 (ML-VRLD: weak convergence) Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 1 and 2. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation generated by (9) with estimator (8). Define the time-varying parameters as follows

$$\gamma_k = \frac{C_\gamma}{k^{3/2}}, \quad p_k = 1 - \beta_k = \frac{C_p}{k^{1/2}}, \quad b_k = \left\lceil \frac{1}{p_k} \right\rceil, \quad \forall k \geq 1,$$

where $0 < C_p < 1$ and $C_\gamma > 0$ are numerical constants. Then, the time-averaged law $\bar{\mu}_{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$, where $\tau_n = \sum_{k=1}^n \gamma_k$, converges weakly to π .

Proof. Given time-varying parameters γ_k, b_k, p_k at iteration k , define the cumulative time τ_n and averaged law $\bar{\mu}_{\tau_n}$ at iteration n as

$$\tau_n := \sum_{k=1}^n \gamma_k, \quad \bar{\mu}_{\tau_n} := \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt,$$

where μ_t denotes the law of the process $\mathbf{x}_t$ under the following continuous-time interpolation

$$\mathbf{x}_t := \mathbf{x}_{\tau_n} - (t - \tau_n) \mathbf{g}_n + \sqrt{2} (\mathbf{B}_t - \mathbf{B}_{\tau_n}), \quad t \in [\tau_n, \tau_{n+1}]. \quad (35)$$

$\mathbf{g}_n$ is defined as follows

$$\mathbf{g}_n := \begin{cases} \frac{1}{b_n} \sum_{i=1}^{b_n} \nabla f(\mathbf{x}_{\tau_n}, \xi_n^i) & \text{w.p. } p_n, \\ \beta_n \mathbf{g}_{n-1} + (1 - \beta_n) \nabla f(\mathbf{x}_{\tau_n}, \xi_n) & \text{w.p. } 1 - p_n, \end{cases} \quad (36)$$

for all $n \geq 1$. At initialization, we choose $\gamma_0, b_0 > 0$ and $(\beta_0, p_0) \in (0, 1]^2 \setminus \{(1, 1)\}$, and define $\mathbf{g}_0$ as

$$\mathbf{g}_0 := \frac{1}{b_0} \sum_{i=1}^{b_0} \nabla f(\mathbf{x}_0, \xi_0^i).$$

We establish weak convergence by adapting the proof of Theorem 1. To this end, we first verify the step sizes $(\gamma_k)_{k \geq 1}$ satisfy the step-size condition of Theorem 1, namely,

$$\gamma_k \in \left(0, \frac{1}{L\sqrt{27\phi(p_k, \beta_k)}}\right], \quad \text{where} \quad \phi(p_k, \beta_k) = 1 + \frac{4(1-p_k)\beta_k^2}{(1-(1-p_k)\beta_k)^2} \quad (37)$$

for all $k \geq 1$. Using the definitions of p_k and β_k , we have

$$\begin{aligned} \phi(p_k, \beta_k) &= 1 + \frac{4(1-p_k)\beta_k^2}{(1-(1-p_k)\beta_k)^2} \\ &\leq 1 + \frac{4}{1-(1-p_k)^2} \\ &\leq 1 + \frac{4}{p_k} \\ &= 1 + \frac{4k^{1/2}}{C_p} \\ &\leq \frac{8k^{1/2}}{C_p}, \end{aligned} \quad (38)$$

where we use the fact that $0 < C_p < 1$ to obtain (38). Then, we have

$$\frac{1}{L\sqrt{27\phi(p_k, \beta_k)}} \geq \frac{\sqrt{C_p}}{L\sqrt{216}k^{1/4}}.$$

Choosing $C_\gamma = \frac{\sqrt{C_p}}{L\sqrt{216}}$, we have

$$\gamma_k = \frac{C_\gamma}{k^{3/2}} \leq \frac{\sqrt{C_p}}{L\sqrt{216}k^{1/4}} \leq \frac{1}{L\sqrt{27\phi(p_k, \beta_k)}}$$

for $k \geq 1$. This verifies (37). In addition, by definition of p_k and β_k , we have $p_k < 1$ and $\beta_k < 1$. Therefore, we can follow the same steps in Theorem 1 (Appendix B.3.1) up to (24) since the the same arguments remain valid for $t \in [\tau_{n-1}, \tau_n]$ with time-varying parameters. We obtain

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{1}{1-(1-p_n)\beta_n} \left[4 + \frac{27}{2} \gamma_n^2 L^2 \phi(p_n, \beta_n) \right] (e_n^2 - e_{n-1}^2) \\ &\quad + \frac{9\sigma^2}{2(1-(1-p_n)\beta_n)} \left[\frac{p_n}{b_n} + (1-p_n)(1-\beta_n)^2 \right] + 9\gamma_n L^2 \phi(p_n, \beta_n) d \end{aligned} \quad (39)$$

for $t \in [\tau_{n-1}, \tau_n]$. Integrating both sides of the inequality over the interval $[\tau_{n-1}, \tau_n]$ yields

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \| \pi) - \text{KL}(\mu_{\tau_{n-1}} \| \pi) &\leq -\frac{1}{2} \int_{\tau_{n-1}}^{\tau_n} \text{FI}(\mu_t \| \pi) dt - \frac{\gamma_n}{1-(1-p_n)\beta_n} \left[4 + \frac{27}{2} \gamma_n^2 L^2 \phi(p_n, \beta_n) \right] (e_n^2 - e_{n-1}^2) \\ &\quad + \frac{9\gamma_n \sigma^2}{2(1-(1-p_n)\beta_n)} [p_n^2 + (1-p_n)(1-\beta_n)^2] + 9\gamma_n^2 L^2 \phi(p_n, \beta_n) d, \end{aligned}$$

where we use $b_n \geq \frac{1}{p_n}$. Iterating the above bound, we obtain

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \| \pi) - \text{KL}(\mu_0 \| \pi) &\leq -\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \| \pi) dt - \underbrace{\sum_{k=1}^n \frac{\gamma_k}{1-(1-p_k)\beta_k} \left[4 + \frac{27}{2} \gamma_k^2 L^2 \phi(p_k, \beta_k) \right] (e_k^2 - e_{k-1}^2)}_{\text{(I)}} \\ &\quad + \underbrace{\frac{9\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + (1-p_k)(1-\beta_k)^2}{1-(1-p_k)\beta_k} \right)}_{\text{(II)}} + \underbrace{9L^2 d \sum_{k=1}^n \gamma_k^2 \phi(p_k, \beta_k)}_{\text{(III)}}. \end{aligned} \quad (40)$$

It remains to control the terms labeled (I)–(III) and show that their contribution remains uniformly bounded as $k \rightarrow \infty$. This will imply that the upper bound (40) stays bounded, which is a key step toward establishing weak convergence.

First, analyze the term (I). Define

$$c_k := \frac{\gamma_k}{1 - (1 - p_k)\beta_k} \left(4 + \frac{27}{2} \gamma_k^2 L^2 \phi(p_k, \beta_k) \right).$$

If $(c_k)_{k \geq 1}$ is nonnegative and decreasing sequence, we can bound (I) as follows

$$(I) = - \sum_{k=1}^n c_k (e_k^2 - e_{k-1}^2) = c_1 e_0^2 + \sum_{k=1}^{n-1} (c_{k+1} - c_k) e_k^2 - c_n e_n^2 \leq c_1 e_0^2.$$

Thus, it remains to show that $(c_k)_{k \geq 1}$ is nonnegative and decreasing sequence. Nonnegativity is obvious as all the terms are positive and $p_k < 1$ and $\beta_k < 1$. To prove that the sequence is decreasing, we substitute γ_k , β_k , and p_k . We then have

$$\begin{aligned} \frac{\gamma_k}{1 - (1 - p_k)\beta_k} &= \frac{\gamma_k}{1 - \beta_k^2} \\ &= \frac{C_\gamma}{2C_p k - C_p^2 k^{1/2}}, \end{aligned}$$

which is decreasing for $k \geq 1$. In addition,

$$\frac{27}{2} \gamma_k^2 L^2 \phi(p_k, \beta_k) = \frac{27C_\gamma^2 L^2}{2k^3} \left(1 + \frac{4 \left(1 - \frac{C_p}{k^{1/2}} \right)^3}{\left(\frac{2C_p}{k^{1/2}} - \frac{C_p^2}{k} \right)^2} \right).$$

We can write this sequence as a continuous function

$$c(x) = \frac{27C_\gamma^2 L^2}{2x^3} \left(1 + \frac{4 \left(1 - \frac{C_p}{x^{1/2}} \right)^3}{\left(\frac{2C_p}{x^{1/2}} - \frac{C_p^2}{x} \right)^2} \right)$$

for $x \geq 1$. We apply change of variables $r = \frac{C_p}{x^{1/2}}$ and get

$$c(x) = \frac{27C_\gamma^2 L^2}{2C_p^6} r^6 \underbrace{\left(1 + \frac{4(1-r)^3}{r^2(2-r)^2} \right)}_{=: h(r)}. \quad (41)$$

Differentiating $h(r)$ with respect to r , we have

$$\begin{aligned} h'(r) &= \frac{2r^3}{(2-r)^3} [16 - 64r + 114r^2 - 88r^3 + 28r^4 - 3r^5] \\ &= \frac{2r^3}{(2-r)^3} [16(1-r)^5 + 16r(1-r)^4 + 18r^2(1-r)^3 + 30r^3(1-r)^2 + 30r^3(1-r)^2 + 18r^4(1-r) + 3r^5] \\ &> 0 \end{aligned}$$

since $0 < r < 1$. Hence, differentiating $h(r(x))$ with respect to x , we get

$$\begin{aligned} c'(x) &= \frac{27C_\gamma^2 L^2}{2C_p^6} h'(r(x)) r'(x) \\ &= - \frac{27C_\gamma^2 L^2}{4C_p^5 x^{3/2}} h'(r(x)) < 0 \end{aligned}$$

for $x \geq 1$. Therefore, $c(x)$ is decreasing on $[1, \infty)$, which implies $(c_k)_{k \geq 1}$ is decreasing sequence. This implies

$$\limsup_{n \rightarrow \infty} (I) \leq c_1 e_0^2. \quad (42)$$

We now analyze the term (II)

$$\begin{aligned}
\frac{9\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + (1-p_k)(1-\beta_k)^2}{1 - (1-p_k)\beta_k} \right) &\leq \frac{9\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + (1-p_k)(1-\beta_k)^2}{p_k} \right) \\
&\leq \frac{9\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + (1-\beta_k)^2}{p_k} \right) \\
&= \frac{9\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + p_k^2}{p_k} \right) \\
&= 9\sigma^2 \sum_{k=1}^n \gamma_k p_k.
\end{aligned}$$

Using the definitions of γ_k and p_k , we obtain the following upper bound for (II):

$$\frac{9\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + (1-p_k)(1-\beta_k)^2}{1 - (1-p_k)\beta_k} \right) \leq 9C_\gamma C_p \sigma^2 \sum_{k=1}^n \frac{1}{k^2}.$$

Taking the limit as $n \rightarrow \infty$ yields

$$\lim_{n \rightarrow \infty} (\text{II}) \leq 9C_\gamma C_p \sigma^2 \sum_{k=1}^{\infty} \frac{1}{k^2} =: S_2 < \infty. \quad (43)$$

Finally, we analyze (III)

$$9L^2 d \sum_{k=1}^n \gamma_k^2 \left(1 + \frac{4(1-p_k)\beta_k^2}{(1 - (1-p_k)\beta_k)^2} \right) \leq 72L^2 d \sum_{k=1}^n \frac{\gamma_k^2}{p_k^2} = \frac{72C_\gamma^2 L^2 d}{C_p^2} \sum_{k=1}^n \frac{1}{k^2}.$$

Taking the limit as $n \rightarrow \infty$ yields

$$\lim_{n \rightarrow \infty} (\text{III}) \leq \frac{36C_\gamma^2 L^2 d}{C_p^2} \sum_{n=1}^{\infty} \frac{1}{n^2} =: S_3 < \infty. \quad (44)$$

Now that we have shown the terms (I), (II) and (III) are uniformly bounded, we bound (40) as

$$\text{KL}(\mu_{\tau_n} \parallel \pi) - \text{KL}(\mu_0 \parallel \pi) \leq -\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt + c_1 e_0^2 + S_2 + S_3. \quad (45)$$

Since KL divergence is nonnegative, we may discard $\text{KL}(\mu_{\tau_n} \parallel \pi)$ and rearrange the terms to get

$$\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt \leq \text{KL}(\mu_0 \parallel \pi) + c_1 e_0^2 + S_2 + S_3.$$

Dividing both sides by $\frac{\tau_n}{2}$, we obtain

$$\frac{1}{\tau_n} \int_0^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt \leq \frac{2\text{KL}(\mu_0 \parallel \pi)}{\tau_n} + \frac{2c_1 e_0^2}{\tau_n} + \frac{2S_2}{\tau_n} + \frac{2S_3}{\tau_n}.$$

Since $\text{FI}(\cdot \parallel \pi)$ is convex, we use Jensen's inequality to obtain

$$\text{FI}(\bar{\mu}_{\tau_n} \parallel \pi) \leq \frac{1}{\tau_n} \int_0^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt \leq \frac{2\text{KL}(\mu_0 \parallel \pi)}{\tau_n} + \frac{2c_1 e_0^2}{\tau_n} + \frac{2S_2}{\tau_n} + \frac{2S_3}{\tau_n}, \quad (46)$$

where $\mu_{\tau_n} := \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$. Given that $\text{KL}(\mu_0 \parallel \pi) < \infty$, we have $\text{FI}(\bar{\mu}_{\tau_n} \parallel \pi) < \infty$. On the other hand, if $t \in [\tau_n, \tau_{n+1}]$, considering (39) for the interval $[\tau_n, \tau_{n+1}]$, integrating both sides from τ_n to t , and dropping the negative integral over the

Fisher information gives us

$$\begin{aligned}
\text{KL}(\mu_t \|\pi) &\leq \text{KL}(\mu_{\tau_n} \|\pi) - \frac{(t - \tau_n)}{1 - (1 - p_{n+1})\beta_{n+1}} \left[4 + \frac{27}{2} \gamma_{n+1}^2 L^2 \phi(p_{n+1}, \beta_{n+1}) \right] (e_{n+1}^2 - e_n^2) \\
&\quad + \frac{9\sigma^2(t - \tau_n)}{2(1 - (1 - p_{n+1})\beta_{n+1})} \left[\frac{p_{n+1}}{b_{n+1}} + (1 - p_{n+1})(1 - \beta_{n+1})^2 \right] + 9(t - \tau_n) \gamma_{n+1} L^2 \phi(p_{n+1}, \beta_{n+1}) d \\
&\leq \text{KL}(\mu_0 \|\pi) + \frac{9\sigma^2(t - \tau_n)}{2(1 - (1 - p_{n+1})\beta_{n+1})} \left[\frac{p_{n+1}}{b_{n+1}} + (1 - p_{n+1})(1 - \beta_{n+1})^2 \right] \\
&\quad + 9(t - \tau_{n+1}) \gamma_{n+1} L^2 \phi(p_{n+1}, \beta_{n+1}) d + c_1 e_0^2 + S_2 + S_3 \\
&\leq \text{KL}(\mu_0 \|\pi) + c_1 e_0^2 + 2S_2 + 2S_3 < \infty,
\end{aligned}$$

where the second inequality follows from (40), the bounds on (II) and (III), and the fact that

$$c_{n+1} \geq \frac{(t - \tau_n)}{1 - (1 - p_{n+1})\beta_{n+1}} \left[4 + \frac{27}{2} \gamma_{n+1}^2 L^2 \phi(p_{n+1}, \beta_{n+1}) \right] \quad (47)$$

for $t \in [\tau_n, \tau_{n+1}]$. To obtain the last inequality, we again use the bounds on (II) and (III).

This proves $\text{KL}(\mu_t \|\pi)$ is uniformly bounded for all $t \geq 0$. By the convexity of the KL divergence, it implies that $\{\text{KL}(\bar{\mu}_{\tau_n} \|\pi) \mid n \in \mathbb{N}\}$ is uniformly bounded as well. Since the sub-level sets of $\text{KL}(\cdot \|\pi)$ are weakly compact, $(\bar{\mu}_{\tau_n})_{n \in \mathbb{N}}$ is tight. To show that $\bar{\mu}_{\tau_n} \rightarrow \pi$ weakly, it is sufficient to show that every cluster point of $(\bar{\mu}_{\tau_n})_{n \in \mathbb{N}}$ is equal to π .

Consider a subsequence of $(\bar{\mu}_{\tau_n})_{n \in \mathbb{N}}$ converging to some limit $\bar{\mu}$. As $n \rightarrow \infty$, $\tau_n \rightarrow \infty$, and by inequality (46) we also have $\text{FI}(\bar{\mu}_{\tau_n} \|\pi) \rightarrow 0$. Therefore, this is still true along the subsequence. By the weak lower semi-continuity of the Fisher information along the subsequence, $\text{FI}(\bar{\mu}_{\tau_n} \|\pi) = 0$. That implies; for $\psi := \frac{d\mu}{d\pi}$, we have $\sqrt{\psi} \in \text{dom}(\mathcal{E})$ and $\mathcal{E}(\sqrt{\psi}) = 0$, where $\mathcal{E}$ denotes the Dirichlet energy (i.e. the squared $L^2(\pi)$ -norm of the gradient; see Section 3 in Balasubramanian et al. [2022]). Since ∇f is Lipschitz by Assumption 1, then π has a continuous and strictly positive density on $\mathbb{R}^d$, so $\mathcal{E}(\sqrt{\psi}) = 0$ implies that ψ is a constant π with probability 1. Hence, $\bar{\mu} = \pi$ as $n \rightarrow \infty$. $\square$

B.3.4 Proof of Theorem 3 (SL-VRLD: FI convergence)

Theorem 3 (SL-VRLD: FI convergence) *Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 2 and 4. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation generated by (9) with estimator (10).*

Then, for any step size $\gamma \in \left(0, \frac{1}{L\sqrt{27\phi(p, \beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{2(1-p)}{1-(1-p)\beta}$, and for any $N \geq 1$, it holds that

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2C_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{2(1-(1-p)\beta)} \right] + \frac{30\gamma L^2 d}{1-(1-p)\beta}, \quad (48)$$

where

$$C_0 = \text{KL}(\mu_0 \|\pi) + \frac{9\gamma}{4(1-(1-p)\beta)} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2]. \quad (49)$$

Furthermore, choose

$$\gamma = \left(\frac{d}{\sigma_m^2 N^2 L^2} \right)^{1/3}, \quad p = 1 - \beta = \left(\frac{d^2 L^2}{\sigma_m^4 N} \right)^{1/3}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\}.$$

Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \varepsilon$ with $\mathcal{O}\left(\frac{\sigma_m^2 L^2 d^2}{\varepsilon^3}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. As in the proof of Theorem 1, we start by invoking Lemma 1

$$\begin{aligned}
\frac{d}{dt} \text{KL}(\mu_t \|\pi) &\leq -\frac{3}{4} \text{FI}(\mu_t \|\pi) + \mathbb{E}[\|\nabla f(\mathbf{x}_t) - \mathbf{g}_k\|^2] \\
&\leq -\frac{3}{4} \text{FI}(\mu_t \|\pi) + 2\mathbb{E}[\|\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{k\gamma})\|^2] + 2\mathbb{E}[\|\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k\|^2] \\
&\leq -\frac{3}{4} \text{FI}(\mu_t \|\pi) + 2L^2 \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2] + 2e_k^2.
\end{aligned} \quad (50)$$

where $e_k^2 := \mathbb{E}[\|\nabla f(\mathbf{x}_{k\gamma}) - \mathbf{g}_k\|^2]$. We decompose the expectation by adding and subtracting $\nabla f(\mathbf{x}_{k\gamma})$, apply Young's inequality, and use Assumption 1 to obtain (50). The next step is to bound e_k^2 . Using the definition of the SL-VRLD estimator (10), e_{k+1}^2 can be expressed as follows.

$$\begin{aligned} e_{k+1}^2 &= p \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - \frac{1}{b} \sum_{i=1}^b \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1}^i) \right\|^2 \right] \\ &\quad + (1-p) \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1}) - \beta (\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma}, \xi_{k+1})) \right\|^2 \right] \\ &\leq \frac{p\sigma^2}{b} + (1-p) \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1}) - \beta (\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma}, \xi_{k+1})) \right\|^2 \right] \end{aligned} \quad (51)$$

$$\begin{aligned} &= \frac{p\sigma^2}{b} + (1-p) \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1}) - \beta (\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma})) - \beta (\nabla f(\mathbf{x}_{k\gamma}) \right. \right. \\ &\quad \left. \left. - \nabla f(\mathbf{x}_{k\gamma}, \xi_{k+1})) \right\|^2 \right] \end{aligned} \quad (52)$$

$$\begin{aligned} &= \frac{p\sigma^2}{b} + (1-p) \mathbb{E} \left[\underbrace{\left\| \nabla f(\mathbf{x}_{(k+1)\gamma}) - \nabla f(\mathbf{x}_{k\gamma}) - (\nabla f(\mathbf{x}_{(k+1)\gamma}, \xi_{k+1}) - \nabla f(\mathbf{x}_{k\gamma}, \xi_{k+1})) \right\|}_{=: \mathbf{A}_k} - \beta (\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma})) \right. \\ &\quad \left. + (1-\beta) \underbrace{(\nabla f(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_{k\gamma}, \xi_{k+1}))}_{=: \mathbf{B}_k} \right]^2 \end{aligned} \quad (53)$$

$$\begin{aligned} &= \frac{p\sigma^2}{b} + (1-p) \mathbb{E} [\|\mathbf{A}_k + (1-\beta)\mathbf{B}_k - \beta(\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma}))\|^2] \\ &= \frac{p\sigma^2}{b} + (1-p)\beta^2 e_k^2 + (1-p) \mathbb{E} [\|\mathbf{A}_k + (1-\beta)\mathbf{B}_k\|^2] \end{aligned} \quad (54)$$

$$\leq \frac{p\sigma^2}{b} + (1-p)\beta^2 e_k^2 + 2(1-p) \mathbb{E} [\|\mathbf{A}_k\|^2] + 2(1-p)(1-\beta)^2 \mathbb{E} [\|\mathbf{B}_k\|^2] \quad (55)$$

$$\leq \frac{p\sigma^2}{b} + (1-p)\beta e_k^2 + 2(1-p)L^2 \Delta_k + 2(1-p)(1-\beta)^2 \sigma^2, \quad (56)$$

where $\Delta_k := \mathbb{E}[\|\mathbf{x}_{(k+1)\gamma} - \mathbf{x}_{k\gamma}\|^2]$. We first use Assumption 2 to bound the first term to get (51). Next, we add and subtract $\beta \nabla f(\mathbf{x}_{k\gamma})$ inside the expected squared norm to obtain (52). We then apply the decomposition $-\beta a = (1-\beta)a - a$ for any $a \in \mathbb{R}$ and $\beta \in [0, 1]$ inside the expected squared norm to derive (53). Defining $\mathbf{A}_k$ and $\mathbf{B}_k$, and using the fact that $\mathbb{E}[\mathbf{A}_k - (1-\beta)\mathbf{B}_k \mid \mathcal{F}_k] = 0$, we obtain (54). Here, $\mathcal{F}_k$ denotes the filtration defined as

$$\mathcal{F}_k := \sigma(\mathbf{x}_0, \mathbf{Z}_0, \xi_0, p_0, \dots, \mathbf{Z}_k, \xi_k, p_k),$$

where ξ_k denotes all ξ_k^i with $i \in \{1, \dots, b\}$ if the batch is calculated, and $\mathbf{Z}_k := \mathbf{B}_{k+1} - \mathbf{B}_k$. Following this, we use Young's inequality $((a+b)^2 \leq 2a^2 + 2b^2)$ to obtain (55). Lastly, we use the fact that $\beta \leq 1$ together with the Assumptions 2 and 4, and the second-moment variance bound, to get (56). Rearranging the terms yields

$$\begin{aligned} e_k^2 &\leq \frac{p\sigma^2}{b(1-(1-p)\beta)} - \frac{1}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{2(1-p)L^2 \Delta_k}{1-(1-p)\beta} + \frac{2(1-p)(1-\beta)^2 \sigma^2}{1-(1-p)\beta} \\ &= -\frac{1}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{2(1-p)L^2 \Delta_k}{1-(1-p)\beta} + \left[\frac{p}{b} + 2(1-p)(1-\beta)^2 \right] \frac{\sigma^2}{1-(1-p)\beta}. \end{aligned} \quad (57)$$

We next bound Δ_k by exploiting the interpolation in (9). For any $t \in [k\gamma, (k+1)\gamma]$, the interpolation satisfies

$$\mathbf{x}_t - \mathbf{x}_{k\gamma} = -(t - k\gamma)\mathbf{g}_k + \sqrt{2}(\mathbf{B}_t - \mathbf{B}_{k\gamma}).$$

Taking squared norms and expectations, and using the independence of $\mathbf{g}_k$ and the Brownian increment, we obtain

$$\begin{aligned} \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2] &= (t - k\gamma)^2 \mathbb{E}[\|\mathbf{g}_k\|^2] + 2 \mathbb{E}[\|\mathbf{B}_t - \mathbf{B}_{k\gamma}\|^2] \\ &= (t - k\gamma)^2 \mathbb{E}[\|\mathbf{g}_k\|^2] + 2(t - k\gamma)d, \end{aligned}$$

where we use the fact that $\mathbb{E}[\|\mathbf{B}_t - \mathbf{B}_{k\gamma}\|^2] = (t - k\gamma)d$. Since $t \in [k\gamma, (k+1)\gamma]$, it follows that

$$\mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2] \leq \gamma^2 \mathbb{E}[\|\mathbf{g}_k\|^2] + 2\gamma d = \Delta_k.$$

Then writing $\mathbf{g}_k = [\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma})] + [\nabla f(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_t)] + \nabla f(\mathbf{x}_t)$, and using Assumption 2 with Young's inequality, we obtain

$$\begin{aligned} \Delta_k &= \gamma^2 \mathbb{E}[\|\mathbf{g}_k\|^2] + 2\gamma d \\ &\leq 3\gamma^2 \mathbb{E}[\|\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma})\|^2] + 3\gamma^2 \mathbb{E}[\|\nabla f(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_t)\|^2] + 3\gamma^2 \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + 2\gamma d. \\ &\leq 3\gamma^2 e_k^2 + 3\gamma^2 L^2 \Delta_k + 3\gamma^2 \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + 2\gamma d. \end{aligned}$$

We now substitute the upper bound (57) on e_k^2 to obtain

$$\begin{aligned} \Delta_k &\leq -\frac{3\gamma^2}{1 - (1-p)\beta} (e_{k+1}^2 - e_k^2) + 3\gamma^2 L^2 \left(1 + \frac{2(1-p)}{1 - (1-p)\beta}\right) \Delta_k \\ &\quad + \frac{3\gamma^2 \sigma^2}{1 - (1-p)\beta} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] + 3\gamma^2 \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + 2\gamma d. \end{aligned}$$

Defining $\phi(p, \beta) := 1 + \frac{2(1-p)}{1 - (1-p)\beta}$ and rearranging the terms, we obtain

$$\begin{aligned} (1 - 3\gamma^2 L^2 \phi(p, \beta)) \Delta_k &\leq -\frac{3\gamma^2}{1 - (1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{3\gamma^2 \sigma^2}{1 - (1-p)\beta} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] \\ &\quad + 3\gamma^2 \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + 2\gamma d. \end{aligned}$$

Setting γ such that $\gamma \leq \frac{1}{L\sqrt{27\phi(p, \beta)}}$ and dividing both sides by $\frac{8}{9}$, we have

$$\Delta_k \leq -\frac{27\gamma^2}{8(1 - (1-p)\beta)} (e_{k+1}^2 - e_k^2) + \frac{27\gamma^2 \sigma^2}{8(1 - (1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] + \frac{27\gamma^2}{8} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{9}{4} \gamma d. \quad (58)$$

Given the upper bounds (57, 58) on e_k^2 and Δ_k , respectively, we return back to the main inequality (50). Recall that

$$\frac{d}{dt} \text{KL}(\mu_t \| \pi) \leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + 2L^2 \Delta_k + 2e_k^2.$$

Plugging in the bounds (57, 58) for e_k^2 and Δ_k yields

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) - \frac{1}{1 - (1-p)\beta} \left(2 + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta)\right) (e_{k+1}^2 - e_k^2) \\ &\quad + \frac{\sigma^2}{1 - (1-p)\beta} \left(2 + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta)\right) \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] \\ &\quad + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta) \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{9}{2} \gamma L^2 \phi(p, \beta) d. \end{aligned}$$

Applying Lemma 2 with the step-size condition $\gamma \leq \frac{1}{L\sqrt{27\phi(p, \beta)}}$, we obtain

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{1}{1 - (1-p)\beta} \left(2 + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta)\right) (e_{k+1}^2 - e_k^2) \\ &\quad + \frac{9\sigma^2}{4(1 - (1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] + \left[\frac{27}{2} \gamma L + \frac{9}{2}\right] \gamma L^2 d \phi(p, \beta). \end{aligned}$$

To simplify the last term, we note that since $\phi(p, \beta) \geq 1$, $\gamma L \leq \frac{1}{\sqrt{27}}$, which implies

$$\frac{27}{2} \gamma L + \frac{9}{2} \leq \frac{15}{2}.$$

Then, we have

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{1}{1 - (1-p)\beta} \left(2 + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta) \right) (e_{k+1}^2 - e_k^2) \\ &\quad + \frac{9\sigma^2}{4(1 - (1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2 \right] + \frac{15}{2} \gamma L^2 d \phi(p, \beta). \end{aligned} \quad (59)$$

We now integrate both sides over $t \in [k\gamma, (k+1)\gamma]$ and define $\mathcal{L}_k := \text{KL}(\mu_{k\gamma} \| \pi) + \frac{\gamma}{1 - (1-p)\beta} \left[2 + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta) \right] e_k^2$. This yields,

$$\mathcal{L}_{k+1} - \mathcal{L}_k \leq -\frac{1}{2} \int_{k\gamma}^{(k+1)\gamma} \text{FI}(\mu_t \| \pi) dt + \frac{9\gamma\sigma^2}{4(1 - (1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2 \right] + \frac{15}{2} \gamma^2 L^2 \phi(p, \beta) d.$$

Iterating the inequality for $k = 0, 1, \dots, N-1$ we obtain

$$\mathcal{L}_N - \mathcal{L}_0 \leq -\frac{1}{2} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt + \frac{9N\gamma\sigma^2}{4(1 - (1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2 \right] + \frac{15}{2} N\gamma^2 L^2 \phi(p, \beta) d.$$

Rearranging the terms and multiplying both sides by $\frac{2}{N\gamma}$, we get

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \frac{2\mathcal{L}_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{2(1 - (1-p)\beta)} \right] + 15\gamma L^2 d \left[1 + \frac{2(1-p)}{1 - (1-p)\beta} \right].$$

Note that $\mathcal{L}_N > 0$ and $\mathcal{L}_0 = \text{KL}(\mu_0 \| \pi) + \frac{\gamma}{1 - (1-p)\beta} \left(2 + \frac{27}{4} \gamma^2 L^2 \phi(p, \beta) \right) e_0^2 \leq \text{KL}(\mu_0 \| \pi) + \frac{9\gamma e_0^2}{4(1 - (1-p)\beta)} =: C_0$, then

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \frac{2C_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{2(1 - (1-p)\beta)} \right] + \frac{30\gamma L^2 d}{1 - (1-p)\beta} \quad (60)$$

where we additionally used the fact that $1 + \frac{2(1-p)}{1 - (1-p)\beta} = \frac{1 + (1-p)\beta}{1 - (1-p)\beta} \leq \frac{2}{1 - (1-p)\beta}$ to rewrite the last term. This completes the first part of the proof. Furthermore, choose

$$\gamma = \left(\frac{d}{\sigma_m^2 N^2 L^2} \right)^{1/3}, \quad p = 1 - \beta = \left(\frac{d^2 L^2}{\sigma_m^4 N} \right)^{1/3}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m := \max\{\sigma, 1\}.$$

and define the time-averaged law $\bar{\mu}_{N\gamma} := \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$. Since $\text{FI}(\cdot \| \pi)$ is convex, using Jensen's inequality, we obtain the upper bound for $\text{FI}(\bar{\mu}_{N\gamma} \| \pi)$

$$\text{FI}(\bar{\mu}_{N\gamma} \| \pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \underbrace{\frac{2C_0}{N\gamma}}_{\text{(I)}} + \underbrace{9\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{2(1 - (1-p)\beta)} \right]}_{\text{(II)}} + \underbrace{\frac{30\gamma L^2 d}{1 - (1-p)\beta}}_{\text{(III)}}.$$

Substituting the definition of γ into (I), we have

$$\text{(I)} = \frac{2C_0 \sigma_m^{2/3} L^{2/3}}{N^{1/3} d^{1/3}}. \quad (61)$$

To bound (II), we

$$\begin{aligned} \text{(II)} &\leq 9\sigma_m^2 \left[\frac{p^2 + (1-p)p^2}{1 - (1-p)^2} \right] \\ &\leq 18\sigma_m^2 p, \end{aligned} \quad (62)$$

where we use $\sigma \leq \sigma_m$, $b \geq 1/p$, and $1 - \beta = p \leq 1$ to get (63). Substituting the definition of p , we get

$$\text{(II)} \leq \frac{18\sigma_m^{2/3} L^{2/3} d^{2/3}}{N^{1/3}}. \quad (63)$$

Similarly, to bound (III),

$$\begin{aligned}
\text{(III)} &= \frac{30\gamma L^2 d}{1 - (1-p)\beta} \\
&\leq \frac{30\gamma L^2 d}{p} \\
&= \frac{30\sigma_m^{2/3} L^{2/3} d^{2/3}}{N^{1/3}}.
\end{aligned} \tag{64}$$

Combining the bounds for (I), (II), and (III), we have

$$\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \left(\frac{2C_0}{d} + 48 \right) \frac{\sigma_m^{2/3} L^{2/3} d^{2/3}}{N^{1/3}},$$

where C_0 is usually taken to be of order d , see e.g. Lemma 1 in Vempala and Wibisono [2019] or Appendix A in Chewi et al. [2025]. Thus, the above bound implies that, to ensure $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \varepsilon$,

$$N = \mathcal{O} \left(\frac{\sigma_m^2 L^2 d^2}{\varepsilon^3} \right)$$

iterations is sufficient. And the average number of gradient computations per iteration is

$$pb + (1-p) = \mathcal{O}(1).$$

We next verify that the constraints on N imposed throughout the analysis are consistent with the stated iteration complexity.

First, the requirement $p = 1 - \beta = \frac{d^{2/3} L^{2/3}}{\sigma_m^{4/3} N^{1/3}} \leq 1$ imposes the lower bound $N \geq \frac{d^2 L^2}{\sigma_m^4}$, which is satisfied by the above choice of N .

Second, we show that the step-size condition $\gamma \leq \frac{1}{L\sqrt{27\phi(p,\beta)}}$ is satisfied. We note that

$$\begin{aligned}
\phi(p, \beta) &= 1 + \frac{2(1-p)}{1 - (1-p)\beta} \\
&= \frac{1 + (1-p)\beta}{1 - (1-p)\beta} \\
&\leq \frac{2}{1 - (1-p)\beta}.
\end{aligned}$$

Then, γ can be chosen as

$$\gamma \leq \frac{\sqrt{1 - (1-p)\beta}}{L\sqrt{54}} \leq \frac{1}{L\sqrt{27\phi(p,\beta)}}.$$

Substituting the definitions of $\beta = 1 - p$ and $p = \frac{d^{2/3} L^{2/3}}{\sigma_m^{4/3} N^{1/3}}$, we get

$$\begin{aligned}
\frac{\sqrt{1 - (1-p)\beta}}{L\sqrt{54}} &= \frac{\sqrt{1 - (1-p)^2}}{L\sqrt{54}} \\
&\geq \frac{\sqrt{p}}{L\sqrt{54}} \\
&= \frac{d^{1/3}}{\sigma_m^{2/3} L^{2/3} N^{1/6} \sqrt{54}}.
\end{aligned}$$

Thus, the step-size condition holds provided that

$$\gamma \leq \frac{d^{1/3}}{\sigma_m^{2/3} L^{2/3} N^{1/6} \sqrt{54}}.$$

Using the definition of step size

$$\gamma = \frac{d^{1/3}}{\sigma_m^{2/3} L^{2/3} N^{2/3}},$$

this condition reduces to the following lower bound $N \geq 54$, which is satisfied by $N = \mathcal{O} \left(\frac{\sigma_m^2 L^2 d^2}{\varepsilon^3} \right)$. $\square$

B.3.5 Proof of Corollary B.1 (SL-VRLD: squared TV distance convergence)

Corollary B.1 (SL-VRLD: squared TV distance convergence) Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 2, 3, and 4. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation generated by (9) with estimator (10). Then, for any step size $\gamma \in \left(0, \frac{1}{L\sqrt{27\phi(p,\beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{2(1-p)}{1-(1-p)\beta}$, and for any $N \geq 1$, it holds that

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0C_{\text{PI}}}{N\gamma} + 18C_{\text{PI}}\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{1-(1-p)\beta} \right] + \frac{80C_{\text{PI}}\gamma L^2 d}{1-(1-p)\beta}, \quad (65)$$

where

$$C_0 = \text{KL}(\mu_0 \parallel \pi) + \frac{9\gamma}{4(1-(1-p)\beta)} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2]. \quad (66)$$

Furthermore, choose

$$\gamma = \left(\frac{d}{\sigma_m^2 N^2 L^2} \right)^{1/3}, \quad p = 1 - \beta = \left(\frac{d^2 L^2}{\sigma_m^4 N} \right)^{1/3}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\}.$$

Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $\mathcal{O}\left(\frac{C_{\text{PI}}^3 \sigma_m^2 L^2 d^2}{\varepsilon^3}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. Recall from the proof (Appendix B.3.4) that the Fisher information of the time-averaged law $\bar{\mu}_{N\gamma}$ satisfies (60),

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \parallel \pi) dt \leq \frac{2C_0}{N\gamma} + 9\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{2(1-(1-p)\beta)} \right] + \frac{30\gamma L^2 d}{1-(1-p)\beta}.$$

Using Lemma 3 and Assumption 3 we get

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0C_{\text{PI}}}{N\gamma} + 36C_{\text{PI}} \frac{(p^2 + 2(1-p)(1-\beta)^2)}{(1-(1-p)\beta)} + \frac{120C_{\text{PI}}\gamma L^2 d}{1-(1-p)\beta}.$$

Following the same steps in (61,63,64), we obtain

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \left(\frac{2C_0}{d} + 48 \right) \frac{4C_{\text{PI}}\sigma_m^{2/3} L^{2/3} d^{2/3}}{N^{1/3}},$$

where C_0 is in the order of d . This implies it suffices to choose N such that $N^{1/3} \geq \frac{\tilde{C}C_{\text{PI}}\sigma_m^{2/3} L^{2/3} d^{2/3}}{\varepsilon}$, where $\tilde{C} := \frac{8C_0}{d} + 192 > 0$. Equivalently, this yields iteration complexity of $\mathcal{O}\left(\frac{C_{\text{PI}}^3 \sigma_m^2 L^2 d^2}{\varepsilon^3}\right)$ with $\mathcal{O}(1)$ gradient computations per iteration. Since the constraints on N in Theorem 3 remain unchanged, they continue to hold. $\square$

B.3.6 Proof of Theorem B.1 (SL-VRLD: weak convergence)

Theorem B.1 (SL-VRLD: weak convergence) Let $\pi \propto e^{-f}$ be the target distribution, where the potential function f satisfies Assumptions 2 and 4. Let $(\mu_t)_{t \geq 0}$ denote the law of continuous-time interpolation generated by (9) with estimator (10). Define the time-varying parameters as follows

$$\gamma_k = \frac{C_\gamma}{k^{3/2}}, \quad p_k = \frac{C_p}{k^{1/2}}, \quad b_k = \left\lceil \frac{1}{p_k} \right\rceil, \quad \forall k \geq 1,$$

where $0 < C_p < 1$ and $C_\gamma > 0$ are numerical constants. Then, the time-averaged law $\bar{\mu}_{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$, where $\tau_n = \sum_{k=1}^n \gamma_k$, converges weakly to π .

Proof. Given time-varying parameters γ_k, b_k, p_k at iteration k , define the cumulative time τ_n and averaged law $\bar{\mu}_{\tau_n}$ at iteration n as

$$\tau_n := \sum_{k=1}^n \gamma_k, \quad \bar{\mu}_{\tau_n} := \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt,$$

where μ_t denotes the law of the process $\mathbf{x}_t$ under the following continuous-time interpolation

$$\mathbf{x}_t := \mathbf{x}_{\tau_n} - (t - \tau_n) \mathbf{g}_n + \sqrt{2} (\mathbf{B}_t - \mathbf{B}_{\tau_n}), \quad t \in [\tau_n, \tau_{n+1}]. \quad (67)$$

$\mathbf{g}_n$ is defined as follows

$$\mathbf{g}_n := \begin{cases} \frac{1}{b_n} \sum_{i=1}^{b_n} \nabla f(\mathbf{x}_{\tau_n}, \xi_n^i) & \text{w.p. } p_n, \\ \nabla f(\mathbf{x}_{\tau_n}, \xi_n) + \beta_n (\mathbf{g}_{n-1} - \nabla f(\mathbf{x}_{\tau_{n-1}}, \xi_n)) & \text{w.p. } 1 - p_n, \end{cases} \quad (68)$$

for all $n \geq 1$. At initialization, we choose $\gamma_0, b_0 > 0$ and $(\beta_0, p_0) \in (0, 1]^2 \setminus \{(1, 1)\}$, and define $\mathbf{g}_0$ as

$$\mathbf{g}_0 := \frac{1}{b_0} \sum_{i=1}^{b_0} \nabla f(\mathbf{x}_0, \xi_0^i).$$

We establish weak convergence by adapting the proof of Theorem 3. To this end, we first verify the step sizes $(\gamma_k)_{k \geq 1}$ satisfy the step-size condition of Theorem 3, namely

$$\gamma_k \in \left(0, \frac{1}{L\sqrt{\phi(p_k, \beta_k)}}\right], \quad \text{where} \quad \phi(p_k, \beta_k) = 1 + \frac{2(1-p)}{1-(1-p)\beta} \quad (69)$$

for all $k \geq 1$. Using the definitions of p_k and β_k , we have

$$\begin{aligned} \phi(p_k, \beta_k) &= 1 + \frac{2(1-p_k)}{1-(1-p_k)\beta_k} \\ &\leq 1 + \frac{2}{1-(1-p_k)^2} \\ &\leq 1 + \frac{2}{p_k} \\ &\leq 1 + \frac{2k^{1/2}}{C_p} \\ &\leq \frac{4k^{1/2}}{C_p}, \end{aligned}$$

where we use the fact that $0 < C_p < 1$. Then, we have

$$\frac{1}{L\sqrt{27\phi(p_k, \beta_k)}} \geq \frac{\sqrt{C_p}}{L\sqrt{108}k^{1/4}}.$$

Choosing $C_\gamma = \frac{\sqrt{C_p}}{L\sqrt{108}}$, we have

$$\gamma_k = \frac{C_\gamma}{k^{3/2}} \leq \frac{\sqrt{C_p}}{L\sqrt{108}k^{1/4}} \leq \frac{1}{L\sqrt{27\phi(p_k, \beta_k)}}$$

for $k \geq 1$. This verifies (69). Therefore, we can follow the same steps in Theorem 3 (Appendix B.3.4) up to (59) since the same arguments remain valid for $t \in [\tau_{n-1}, \tau_n]$ with time-varying parameters. We obtain

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{1}{1-(1-p_n)\beta_n} \left(2 + \frac{27}{4} \gamma_n^2 L^2 \phi(p_n, \beta_n) \right) (e_n^2 - e_{n-1}^2) \\ &\quad + \frac{9\sigma^2}{4(1-(1-p_n)\beta_n)} \left[\frac{p_n}{b_n} + 2(1-p_n)(1-\beta_n)^2 \right] + \frac{15}{2} \gamma_n L^2 \phi(p_n, \beta_n) d \end{aligned} \quad (70)$$

for $t \in [\tau_{n-1}, \tau_n]$, where $\phi(p_n, \beta_n) := 1 + \frac{2(1-p_n)}{1-(1-p_n)\beta_n}$. Integrating both sides of the inequality over the interval $[\tau_{n-1}, \tau_n]$ yields

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \|\pi) - \text{KL}(\mu_{\tau_{n-1}} \|\pi) &\leq -\frac{1}{2} \int_{\tau_{n-1}}^{\tau_n} \text{FI}(\mu_t \|\pi) dt - \frac{\gamma_n}{1 - (1-p_n)\beta_n} \left[2 + \frac{27}{4} \gamma_n^2 L^2 \phi(p_n, \beta_n) \right] (e_n^2 - e_{n-1}^2) \\ &\quad + \frac{9\gamma_n \sigma^2}{4(1 - (1-p_n)\beta_n)} [p_n^2 + 2(1-p_n)(1-\beta_n)^2] + \frac{15}{2} \gamma_n^2 L^2 \phi(p_n, \beta_n) d, \end{aligned}$$

where we use $b_n \geq \frac{1}{p_n}$. Iterating the above bound, we obtain

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \|\pi) - \text{KL}(\mu_0 \|\pi) &\leq -\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \|\pi) dt - \underbrace{\sum_{k=1}^n \frac{\gamma_k}{1 - (1-p_k)\beta_k} \left[2 + \frac{27}{4} \gamma_k^2 L^2 \phi(p_k, \beta_k) \right] (e_k^2 - e_{k-1}^2)}_{\text{(I)}} \\ &\quad + \underbrace{\frac{9\sigma^2}{4} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + 2(1-p_k)(1-\beta_k)^2}{1 - (1-p_k)\beta_k} \right)}_{\text{(II)}} + \underbrace{\frac{15L^2 d}{2} \sum_{k=1}^n \gamma_k^2 \phi(p_k, \beta_k)}_{\text{(III)}}. \end{aligned} \quad (71)$$

It remains to control the terms labeled (I)–(III) and show that their contribution remains uniformly bounded as $k \rightarrow \infty$. This will imply that the upper bound (71) stays bounded, which is a key step toward establishing weak convergence.

First, analyze the term (I). Define

$$c_k := \frac{\gamma_k}{1 - (1-p_k)\beta_k} \left(4 + \frac{27}{2} \gamma_k^2 L^2 \phi(p_k, \beta_k) \right).$$

If $(c_k)_{k \geq 1}$ is nonnegative and decreasing sequence, we have

$$\text{(I)} = -\sum_{k=1}^n c_k (e_k^2 - e_{k-1}^2) = c_1 e_0^2 + \sum_{k=1}^{n-1} (c_{k+1} - c_k) e_k^2 - c_n e_n^2 \leq c_1 e_0^2$$

Thus, it remains to show that $(c_k)_{k \geq 1}$ is nonnegative and decreasing sequence. Nonnegativity is obvious as all the terms are positive and $p_k < 1$ and $\beta_k < 1$. To prove that the sequence is decreasing, we substitute γ_k, β_k , and p_k . As shown in the proof of Theorem 2, we have

$$\frac{\gamma_k}{1 - (1-p_k)\beta_k} = \frac{C_\gamma}{2C_p k - C_p^2 k^{1/2}},$$

which is decreasing for $k \geq 1$. In addition,

$$\frac{27}{2} \gamma_k^2 L^2 \phi(p_k, \beta_k) = \frac{27C_\gamma^2 L^2}{2k^3} \left(1 + \frac{2 \left(1 - \frac{C_p}{k^{1/2}} \right)}{1 - \left(1 - \frac{C_p}{k^{1/2}} \right)^2} \right).$$

We can write this sequence as a continuous function

$$c(x) = \frac{27C_\gamma^2 L^2}{2x^3} \left(1 + \frac{2 \left(1 - \frac{C_p}{x^{1/2}} \right)}{1 - \left(1 - \frac{C_p}{x^{1/2}} \right)^2} \right)$$

for $x \geq 1$. We use change of variables $r = \frac{C_p}{x^{1/2}}$ and get

$$c(x) = \frac{27C_\gamma^2 L^2}{2C_p^6} \underbrace{\left(\frac{r^5(2-r^2)}{2-r} \right)}_{:=h(r)}.$$

Differentiating $h(r)$ with respect to r , we have

$$h'(r) = \frac{2r^4(10 - 4r - 7r^2 + 3r^3)}{(2 - r)^2}.$$

Defining $q(r) := 10 - 4r - 7r^2 + 3r^3$, we note that, for $r \in (0, 1)$, $q'(r) = -4 - 14r + 9r^2 < -4 - 5r < 0$. Therefore,

$$q(r) > q(1) = 2 > 0,$$

which implies $h'(r) > 0$ for $r \in (0, 1)$. Hence,

$$\begin{aligned} c'(x) &\leq \frac{27C_\gamma^2 L^2}{2C_p^6} h'(r(x)) r'(x) \\ &\leq -\frac{27C_\gamma^2 L^2}{4C_p^5 x^{3/2}} h'(r(x)) < 0 \end{aligned}$$

for $x \geq 1$. Therefore, $c(x)$ is decreasing on $[1, \infty)$, which implies $(c_k)_{k \geq 1}$ is decreasing sequence. This implies

$$\limsup_{n \rightarrow \infty} (\text{I}) \leq c_1 e_0^2. \quad (72)$$

We now analyze the term (II)

$$\begin{aligned} (\text{II}) &= \frac{9\sigma^2}{4} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + 2(1 - p_k)(1 - \beta_k)^2}{p_k} \right) \\ &\leq \frac{9\sigma^2}{4} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + 2(1 - \beta_k)^2}{p_k} \right) \\ &= \frac{9\sigma^2}{4} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + 2p_k^2}{p_k} \right) \\ &= \frac{27}{4} \sigma^2 \sum_{k=1}^n \gamma_k p_k \end{aligned}$$

Using the definitions of γ_k and p_k , we obtain the following upper bound for (II):

$$\frac{9\sigma^2}{4} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + 2(1 - p_k)(1 - \beta_k)^2}{1 - (1 - p_k)\beta_k} \right) \leq \frac{27C_\gamma C_p \sigma^2}{4} \sum_{k=1}^n \frac{1}{k^2}.$$

Taking the limit as $n \rightarrow \infty$ yields

$$\lim_{k \rightarrow \infty} (\text{II}) \leq \frac{27C_\gamma C_p \sigma^2}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} =: S_2 < \infty. \quad (73)$$

Finally, we analyze (III)

$$(\text{III}) = \frac{15L^2 d}{2} \sum_{k=1}^n \gamma_k^2 \left(1 + \frac{2(1 - p_k)}{1 - (1 - p_k)\beta_k} \right) \leq \frac{15L^2 d}{2} \sum_{k=1}^n \gamma_k^2 \left(1 + \frac{2(1 - p_k)}{p_k} \right) \leq \frac{15L^2 d}{2} \sum_{k=1}^n \frac{2\gamma_k^2}{p_k}.$$

Using the definitions of γ_k and p_k , and taking the limit as $n \rightarrow \infty$ yields

$$\lim_{k \rightarrow \infty} (\text{III}) \leq \frac{15C_\gamma^2 L^2 d}{C_p} \sum_{k=1}^{\infty} \frac{1}{k^{\frac{5}{2}}} =: S_3 < \infty. \quad (74)$$

Now that we have shown the terms (I), (II) and (III) are uniformly bounded, we bound (71) as

$$\text{KL}(\mu_{\tau_n} \parallel \pi) - \text{KL}(\mu_0 \parallel \pi) \leq -\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt + c_1 e_0^2 + S_2 + S_3. \quad (75)$$

Since KL divergence is nonnegative, we may discard $\text{KL}(\mu_{\mu_{\tau_n}} \|\pi)$ and rearrange the terms to get

$$\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \|\pi) dt \leq \text{KL}(\mu_0 \|\pi) + c_1 e_0^2 + S_2 + S_3.$$

Dividing both sides by $\frac{\tau_n}{2}$, we obtain

$$\frac{1}{\tau_n} \int_0^{\tau_n} \text{FI}(\mu_t \|\pi) dt \leq \frac{2\text{KL}(\mu_0 \|\pi)}{\tau_n} + \frac{2c_1 e_0^2}{\tau_n} + \frac{2S_2}{\tau_n} + \frac{2S_3}{\tau_n}.$$

Since $\text{FI}(\cdot \|\pi)$ is convex, we use Jensen's inequality to obtain

$$\text{FI}(\bar{\mu}_{\tau_n} \|\pi) \leq \frac{1}{\tau_n} \int_0^{\tau_n} \text{FI}(\mu_t \|\pi) dt \leq \frac{2\text{KL}(\mu_0 \|\pi)}{\tau_n} + \frac{2c_1 e_0^2}{\tau_n} + \frac{2S_2}{\tau_n} + \frac{2S_3}{\tau_n}, \quad (76)$$

where $\mu_{\tau_n} := \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$. Given that $\text{KL}(\mu_0 \|\pi) < \infty$, we have $\text{FI}(\bar{\mu}_{\tau_n} \|\pi) < \infty$. On the other hand, if $t \in [\tau_n, \tau_{n+1}]$, considering (70) for the interval $[\tau_n, \tau_{n+1}]$, integrating both sides from τ_n to t , and dropping the negative integral over the Fisher information gives us

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \|\pi) &\leq -\frac{1}{1 - (1 - p_{n+1})\beta_{n+1}} \left(2 + \frac{27}{4} \gamma_{n+1}^2 L^2 \phi(p_{n+1}, \beta_{n+1}) \right) (e_{n+1}^2 - e_n^2) \\ &\quad + \frac{9\sigma^2}{4(1 - (1 - p_{n+1})\beta_{n+1})} [p_{n+1}^2 + 2(1 - p_{n+1})(1 - \beta_{n+1})^2] + \frac{15}{2} \gamma_{n+1} L^2 d \phi(p_{n+1}, \beta_{n+1}). \end{aligned}$$

Furthermore, integrating both sides over the interval $[\tau_n, t]$, we obtain

$$\begin{aligned} \text{KL}(\mu_t \|\pi) &\leq \text{KL}(\mu_{\tau_n} \|\pi) - \frac{(t - \tau_n)}{1 - (1 - p_{n+1})\beta_{n+1}} \left[2 + \frac{27}{4} \gamma_{n+1}^2 L^2 \phi(p_{n+1}, \beta_{n+1}) \right] (e_{n+1}^2 - e_n^2) \\ &\quad + \frac{9\sigma^2(t - \tau_n)}{4(1 - (1 - p_{n+1})\beta_{n+1})} \left[\frac{p_{n+1}}{b_{n+1}} + 2(1 - p_{n+1})(1 - \beta_{n+1})^2 \right] + \frac{15}{2} (t - \tau_n) \gamma_{n+1} L^2 d \phi(p_{n+1}, \beta_{n+1}) \\ &\leq \text{KL}(\mu_0 \|\pi) + \frac{9\sigma^2(t - \tau_n)}{4(1 - (1 - p_{n+1})\beta_{n+1})} \left[\frac{p_{n+1}}{b_{n+1}} + 2(1 - p_{n+1})(1 - \beta_{n+1})^2 \right] \\ &\quad + \frac{15}{2} (t - \tau_n) \gamma_{n+1} L^2 d \phi(p_{n+1}, \beta_{n+1}) + c_1 e_0^2 + S_2 + S_3 \\ &\leq \text{KL}(\mu_0 \|\pi) + c_1 e_0^2 + 2S_2 + 2S_3 < \infty. \end{aligned}$$

where the second inequality follows from (71), the bounds on (II) and (III), and the fact that

$$c_{n+1} \geq \frac{(t - \tau_n)}{1 - (1 - p_{n+1})\beta_{n+1}}$$

for $t \in [\tau_n, \tau_{n+1}]$. To obtain the last inequality, we again use the bounds on (II) and (III). Thus $\{\text{KL}(\mu_t \|\pi) \mid t \geq 0\}$ is bounded. The weak convergence of $\bar{\mu}$ to π then follows by the same argument as in the proof of Theorem 2. $\square$

B.4 POSTERIOR SAMPLING WITH SGM PRIOR PROOFS

B.4.1 Proof of Theorem 4a (ML-VRLD with SGM prior: FI convergence)

Theorem 4a (ML-VRLD with SGM prior: FI convergence) *Let $\pi(\mathbf{x}|\mathbf{y}) \propto \ell(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ be the posterior with the likelihood $\ell(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ and the prior $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Suppose the likelihood potential f satisfies Assumptions 1 and 2 with Lipschitz constant L_f , the prior potential h satisfies Assumption 1 and 5 with Lipschitz constant L_h , and the SGM satisfies Assumption 6 with decreasing error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$ for $k \geq 1$. Define $L_\pi = L_f + L_h$. Let $(\mu_t)_{t \geq 0}$ denote*

the law of the interpolation generated by (11) with the estimator (8) and the schedules defined in (12). For any step size $\gamma \in \left(0, \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$, and for any $N \geq 1$, it holds that

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \frac{2C_0}{N\gamma} + 13\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1-(1-p)\beta} \right] + \frac{84\gamma L_\pi^2 d}{(1-(1-p)\beta)^2} + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2, \quad (77)$$

where

$$\bar{\sigma}^2 = \frac{39C_1^2}{2N} \sum_{k=0}^{N-1} \sigma_k^2, \quad \bar{\varepsilon}_\sigma^2 = \frac{39}{2N} \sum_{k=0}^{N-1} \varepsilon_{\sigma_k}^2, \quad \bar{\alpha}^2 = \frac{39C_2^2}{2N} \sum_{k=0}^{N-1} \frac{(\alpha_k - 1)^2}{\sigma_k^2}, \quad (78)$$

$$C_0 = \text{KL}(\mu_0 \| \pi) + \frac{13\gamma}{2(1-(1-p)\beta)^2} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2], \quad (79)$$

and C_0, C_1, C_2 are positive constants. Furthermore, let

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L_\pi}}, \quad p = 1 - \beta = \frac{\sqrt{L_\pi d}}{\sigma_m N^{1/4}}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\}, \quad (80)$$

and let the initial parameters satisfy $\sigma_0 \geq \sigma_{\min}$, $\varepsilon_{\sigma_0} > 0$, $\alpha_0 \geq 1$, and $\sigma_{\min} > 0$ is the minimum noise level. Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma} \| \pi) \leq \varepsilon$ with $\mathcal{O}\left(\frac{\sigma_m^4 L_\pi^2 d^2}{\varepsilon^4}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. In the case of posterior sampling with SGM priors recall that the SDE in interest is:

$$d\mathbf{x}_t = -(\nabla f(\mathbf{x}_t) + \nabla h(\mathbf{x}_t)) dt + \sqrt{2} d\mathbf{B}_t.$$

We again begin by invoking Lemma 1,

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + \mathbb{E}[\|\nabla \log \pi(\mathbf{x}_t | \mathbf{y}) + \mathbf{g}_k - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)\|^2] \\ &\leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + \mathbb{E}[\|-\nabla h(\mathbf{x}_t) - \nabla f(\mathbf{x}_t) + \mathbf{g}_k - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)\|^2] \\ &\leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + \mathbb{E}[\|\nabla f(\mathbf{x}_{k\gamma}) + \nabla h(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_t) - \nabla h(\mathbf{x}_t) - \nabla f(\mathbf{x}_{k\gamma}) + \mathbf{g}_k \\ &\quad - \nabla h(\mathbf{x}_{k\gamma}) + \nabla h_{\sigma_k}(\mathbf{x}_{k\gamma}) - \nabla h_{\sigma_k}(\mathbf{x}_{k\gamma}) - \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k) - (\alpha_k - 1) \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)\|^2]. \end{aligned}$$

Applying Young's inequality together with Assumptions 1, 5, and 6, we get

$$\frac{d}{dt} \text{KL}(\mu_t \| \pi) \leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + 3L_\pi \Delta_k + 3e_k^2 + 3(\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2 \sigma_k^{-1})^2 \quad (81)$$

where $\Delta_k := \mathbb{E}[\|\mathbf{x}_{(k+1)\gamma} - \mathbf{x}_{k\gamma}\|^2] \geq \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}_{k\gamma}\|^2]$, $L_\pi := L_f + L_h$ and $e_k := \mathbb{E}[\|\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma})\|^2]$. We use the recursive upper bound on the estimation error e_k^2 in (22):

$$e_k^2 \leq -\frac{2}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{2\sigma^2}{1-(1-p)\beta} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + \frac{4(1-p)\beta^2 L_f^2}{(1-(1-p)\beta)^2} \Delta_k. \quad (82)$$

Plugging this into (81), we obtain

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{3}{4} \text{FI}(\mu_t \| \pi) + 3L_\pi^2 \phi(p, \beta) \Delta_k + \frac{6\sigma^2}{1-(1-p)\beta} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] \\ &\quad - \frac{6}{1-(1-p)\beta} (e_{k+1}^2 - e_k^2) + 3(\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2 \sigma_k^{-1})^2, \end{aligned} \quad (83)$$

where $\phi(p, \beta) := 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$. We bound Δ_k using the interpolation (11):

$$\begin{aligned} \Delta_k &= \mathbb{E}[\|\mathbf{x}_{(k+1)\gamma} - \mathbf{x}_{k\gamma}\|^2] = \gamma^2 \mathbb{E}[\|\mathbf{g}_k - \alpha_k \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)\|^2] + 2\gamma d \\ &= \gamma^2 \mathbb{E}[\|\mathbf{g}_k - \nabla f(\mathbf{x}_{k\gamma}) + \nabla f(\mathbf{x}_t) + \nabla h(\mathbf{x}_t) + \nabla f(\mathbf{x}_{k\gamma}) - \nabla f(\mathbf{x}_t) + \nabla h(\mathbf{x}_{k\gamma}) \\ &\quad - \nabla h(\mathbf{x}_t) - \nabla h(\mathbf{x}_{k\gamma}) + \nabla h_{\sigma_k}(\mathbf{x}_{k\gamma}) - \nabla h_{\sigma_k}(\mathbf{x}_{k\gamma}) - \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k) - (\alpha_k - 1) \mathcal{S}_\theta(\mathbf{x}_{k\gamma}, \sigma_k)\|^2] + 2\gamma d. \end{aligned}$$

Applying Young's inequality together with Assumptions 1, 5 and 6 yields

$$\Delta_k \leq 4\gamma^2 e_k^2 + 4\gamma^2 \mathbb{E} [\|\nabla \log \pi(\mathbf{x}_t | \mathbf{y})\|^2] + 4\gamma^2 L_\pi^2 \Delta_k + 4\gamma^2 (\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2 \sigma_k^{-1})^2 + 2\gamma d. \quad (84)$$

We now substitute the bound (82) on e_k^2 and rearrange the terms as follows

$$(1 - 4\gamma^2 L_\pi^2 \phi(p, \beta)) \Delta_k \leq -\frac{8\gamma^2}{1 - (1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{8\gamma^2 \sigma^2}{1 - (1-p)\beta} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + 4\gamma^2 \mathbb{E} [\|\nabla \log \pi(\mathbf{x}_t | \mathbf{y})\|^2] + 4\gamma^2 (\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2 \sigma_k^{-1})^2 + 2\gamma d.$$

Setting $\gamma \leq \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}$ and dividing both sides by $\frac{13}{13}$, we obtain

$$\Delta_k \leq -\frac{26\gamma^2}{3(1 - (1-p)\beta)} (e_{k+1}^2 - e_k^2) + \frac{26\gamma^2 \sigma^2}{3(1 - (1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + \frac{13}{3} \gamma^2 \mathbb{E} [\|\nabla \log \pi(\mathbf{x}_t | \mathbf{y})\|^2] + \frac{13}{3} \gamma^2 (\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2 \sigma_k^{-1})^2 + \frac{13}{6} \gamma d. \quad (85)$$

Substituting the above bound into (83), applying Lemma 2, using the step-size condition $\gamma \leq \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}$, and using Young's inequality for $(\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2 \sigma_k^{-1})^2$ yields

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{6 + 26\gamma^2 L_\pi^2 \phi(p, \beta)}{1 - (1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{13\sigma^2}{2(1 - (1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] \\ &\quad + \left(26\gamma L_\pi + \frac{13}{2} \right) \gamma L_\pi^2 d\phi(p, \beta) + \frac{39}{4} \sigma_k^2 C_1^2 + \frac{39}{4} \varepsilon_{\sigma_k}^2 + \frac{39C_2^2 (\alpha_k - 1)^2}{4\sigma_k^2}. \end{aligned} \quad (86)$$

Since $\phi(p, \beta) \geq 1$, we have $\gamma L_\pi \geq \frac{1}{\sqrt{52}}$, which implies

$$26\gamma L_\pi + \frac{13}{2} \leq \frac{21}{2}.$$

Then, we have

$$\begin{aligned} \frac{d}{dt} \text{KL}(\mu_t \| \pi) &\leq -\frac{1}{2} \text{FI}(\mu_t \| \pi) - \frac{6 + 26\gamma^2 L_\pi^2 \phi(p, \beta)}{1 - (1-p)\beta} (e_{k+1}^2 - e_k^2) + \frac{13\sigma^2}{2(1 - (1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] \\ &\quad + \frac{21}{2} \gamma L_\pi^2 d\phi(p, \beta) + \frac{39}{4} \sigma_k^2 C_1^2 + \frac{39}{4} \varepsilon_{\sigma_k}^2 + \frac{39C_2^2 (\alpha_k - 1)^2}{4\sigma_k^2}. \end{aligned} \quad (87)$$

Next, we define $\mathcal{L}_k := \text{KL}(\mu_{k\gamma} \| \pi) + \frac{\gamma(6+26\gamma^2 L_\pi^2 \phi(p, \beta))}{1-(1-p)\beta} e_k^2$ and integrate both sides over $t \in [k\gamma, (k+1)\gamma]$ to get

$$\begin{aligned} \mathcal{L}_{k+1} - \mathcal{L}_k &\leq -\frac{1}{2} \int_{k\gamma}^{(k+1)\gamma} \text{FI}(\mu_t \| \pi) dt + \frac{13\gamma\sigma^2}{2(1 - (1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2 \right] + \frac{21}{2} \gamma L_\pi^2 d\phi(p, \beta) \\ &\quad + \frac{39}{4} \gamma \sigma_k^2 C_1^2 + \frac{39}{4} \gamma \varepsilon_{\sigma_k}^2 + \frac{39\gamma C_2^2 (\alpha_k - 1)^2}{4\sigma_k^2} \end{aligned}$$

Note that $\mathcal{L}_0 = \text{KL}(\mu_0 \| \pi) + \frac{\gamma(6+26\gamma^2 L_\pi^2 \phi(p, \beta))}{1-(1-p)\beta} e_0^2 \leq \text{KL}(\mu_0 \| \pi) + \frac{13\gamma e_0^2}{2(1-(1-p)\beta)} =: C_0$. Iterating the above inequality for $k = 0, 1, \dots, N-1$ and rearranging the terms, we obtain

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \frac{2C_0}{N\gamma} + 13\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{(1 - (1-p)\beta)} \right] + \frac{84\gamma L_\pi^2 d}{(1 - (1-p)\beta)^2} + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2,$$

where $\bar{\sigma}^2$, $\bar{\varepsilon}_\sigma^2$, and $\bar{\alpha}^2$ are defined in (78). Note that we use $\phi(p, \beta) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2} \leq 1 + \frac{4(1-p)\beta}{(1-(1-p)\beta)^2} = \frac{(1+(1-p)\beta)^2}{(1-(1-p)\beta)^2} \leq \frac{4}{(1-(1-p)\beta)^2}$ to bound $\phi(p, \beta)$ in the above inequality. This completes the first part of the proof.

To establish the iteration complexity, we note that, replacing L by L_π , all the terms in the upper bound of Theorem 1 (except the bias terms $\bar{\varepsilon}_\sigma^2$, $\bar{\alpha}^2$, $\bar{\sigma}^2$) are the same with different numerical coefficients. Therefore, choosing the same parameters (replacing L with L_π) as in Theorem 1

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{\frac{3}{4}} \sqrt{L_\pi}}, \quad p = 1 - \beta = \frac{\sqrt{L_\pi d}}{\sigma_m N^{\frac{1}{4}}}, \quad \text{where } \sigma_m := \max\{\sigma, 1\},$$

we have

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \mathcal{O} \left(\frac{\sigma_m L_\pi^{1/2} d^{1/2}}{N^{1/4}} \right) + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2 + \bar{\sigma}^2.$$

We recall that the noise $(\sigma_k)_{k=0}^{N-1}$ and $(\alpha_k)_{k=0}^{N-1}$ schedules are defined as

$$\alpha_k = \max\{\alpha_0 \rho_1^k, 1\} \quad \text{and} \quad \sigma_k = \max\{\sigma_0 \rho_2^k, \sigma_{\min}\},$$

where $\rho_1, \rho_2 \in (0, 1)$ denote decay rates, $\sigma_0 > 0$, $\alpha_0 > 0$, and $\sigma_{\min} > 0$ is the minimum noise level. By definition in (12), there exists indices $K_\alpha < N - 1$ and $K_\sigma < N - 1$ independent of N such that $\alpha_k = 1$ for $\forall k \geq K_\alpha$ and $\sigma_k = \sigma_{\min}$ for $\forall k \geq K_\sigma$. Using this definition, we next analyze the convergence behavior of the bias contributions due to the noise schedule, annealing schedule, and SGM estimation error separately.

To bound $\bar{\sigma}^2$, we proceed as

$$\begin{aligned} \bar{\sigma}^2 &= \frac{39C_1^2}{2N} \sum_{k=0}^{N-1} \max\{\sigma_0^2 \rho_2^{2k}, \sigma_{\min}^2\} \\ &\stackrel{(*)}{=} \frac{39C_1^2}{2N} \sum_{k=0}^{K_\sigma-1} \sigma_0^2 \rho_2^{2k} + \frac{39C_1^2}{2N} \sum_{k=K_\sigma}^{N-1} \sigma_{\min}^2 \\ &= \underbrace{\frac{39C_1^2 \sigma_0^2 (1 - \rho_2^{2K_\sigma})}{2N(1 - \rho_2^2)}}_{\mathcal{O}(N^{-1})} + \frac{39C_1^2 (N - K_\sigma) \sigma_{\min}^2}{2N}, \end{aligned}$$

where $(*)$ follows from the definition of the annealing schedule (12). Setting $\sigma_{\min} = \mathcal{O}(N^{-1/8})$, we obtain

$$\bar{\sigma}^2 = \mathcal{O}(N^{-1}) + \mathcal{O}(N^{-1/4}) = \mathcal{O}(N^{-1/4}). \quad (88)$$

To bound $\bar{\alpha}^2$, we use the fact that $(\alpha_k)_{k=0}^{N-1}$ is nonincreasing sequence and that there exists a constant $K_\alpha < N - 1$, independent of N , such that $\alpha_k = 1$ for all $k \geq K_\alpha$. Thus, we obtain

$$\begin{aligned} \bar{\alpha}^2 &= \frac{39C_2^2}{2N} \sum_{k=0}^{N-1} \frac{(\alpha_k - 1)^2}{\sigma_k^2} \\ &= \frac{39C_2^2}{2N} \sum_{k=0}^{K_\alpha-1} \frac{(\alpha_0 \rho_1^k - 1)^2}{\sigma_k^2} \\ &= \frac{39C_2^2}{2N\sigma_{\min}^2} \sum_{k=0}^{K_\alpha-1} (\alpha_0 \rho_1^k - 1)^2 \\ &= \frac{39C_2^2}{2N\sigma_{\min}^2} \sum_{k=0}^{K_\alpha-1} (\alpha_0^2 \rho_1^{2k} - 2\alpha_0 \rho_1^k + 1) \\ &\leq \frac{39C_2^2 \alpha_0^2}{2N\sigma_{\min}^2} \sum_{k=0}^{K_\alpha-1} \rho_1^{2k} + \frac{39C_2^2 K_\alpha}{2N\sigma_{\min}^2} \\ &\leq \underbrace{\frac{39C_2^2 \alpha_0^2 (1 - \rho_1^{2K_\alpha})}{2N\sigma_{\min}^2 (1 - \rho_1^2)}}_{\mathcal{O}(N^{-3/4})} + \underbrace{\frac{39C_2^2 K_\alpha}{2N\sigma_{\min}^2}}_{\mathcal{O}(N^{-3/4})} \\ &= \mathcal{O}(N^{-3/4}). \end{aligned} \quad (89)$$

Finally, to bound $\bar{\varepsilon}_\sigma^2$, we use the assumption that the SGM estimation error decays as $\varepsilon_{\sigma_k}^2 \leq \frac{C'}{k}$ for some constant $C' > 0$. We proceed as

$$\begin{aligned}\bar{\varepsilon}_\sigma^2 &= \frac{39}{2N} \sum_{k=0}^{N-1} \varepsilon_{\sigma_k}^2 \\ &= \frac{39\varepsilon_{\sigma_0}^2}{2N} + \frac{39C'}{2N} \sum_{k=1}^{N-1} \frac{1}{k} \\ &= \mathcal{O}\left(\frac{\log N}{N}\right).\end{aligned}\tag{90}$$

Combining the upper bounds (88), (89), and (90), and using the Jensen's inequality with the convexity of $\text{FI}(\cdot\|\pi)$, we obtain

$$\begin{aligned}\text{FI}(\bar{\mu}_{N\gamma}\|\pi) &= \mathcal{O}\left(\frac{\sigma_m L_\pi^{1/2} d^{1/2}}{N^{1/4}}\right) + \mathcal{O}\left(\frac{1}{N^{1/4}}\right) + \mathcal{O}\left(\frac{1}{N^{3/4}}\right) + \mathcal{O}\left(\frac{\log N}{N}\right) \\ &= \mathcal{O}\left(\frac{\sigma_m L_\pi^{1/2} d^{1/2}}{N^{1/4}}\right).\end{aligned}$$

This implies that $\text{FI}(\bar{\mu}_{N\gamma}\|\pi) \leq \varepsilon$ is achieved with $\mathcal{O}\left(\frac{\sigma_m^4 L_\pi^2 d^2}{\varepsilon^4}\right)$ iterations with $pb + (1-p) = \mathcal{O}(1)$ gradient computations per iteration. $\square$

B.4.2 Proof of Theorem 4b (SL-VRLD with SGM prior: FI convergence)

Theorem 4 (SL-VRLD with SGM prior: FI convergence) *Let $\pi(\mathbf{x}|\mathbf{y}) \propto \ell(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ be the posterior with the likelihood $\ell(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ and the prior $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Suppose the likelihood potential f satisfies Assumptions 2 and 4 with Lipschitz constant L_f , the prior potential h satisfies Assumptions 1 and 5 with Lipschitz constant L_h , and the SGM satisfies Assumptions 6 with decreasing error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$ for $k \geq 1$. Define $L_\pi = L_f + L_h$. Let $(\mu_t)_{t \geq 0}$ denote the law of the interpolation generated by (11) with the estimator (10) and the schedules defined in (12). For any step size $\gamma \in \left(0, \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{2(1-p)}{1-(1-p)\beta}$, and for any $N \geq 1$, it holds that*

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t\|\pi) dt \leq \frac{2C_0}{N\gamma} + \frac{13\sigma^2}{2} \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{(1-(1-p)\beta)} \right] + \frac{63\gamma L_\pi^2 d}{1-(1-p)\beta} + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2, \tag{91}$$

where

$$\bar{\sigma}^2 = \frac{39C_1^2}{2N} \sum_{k=0}^{N-1} \sigma_k^2, \quad \bar{\varepsilon}_\sigma^2 = \frac{39}{2N} \sum_{k=0}^{N-1} \varepsilon_{\sigma_k}^2, \quad \bar{\alpha}^2 = \frac{39C_2^2}{2N} \sum_{k=0}^{N-1} \frac{(\alpha_k - 1)^2}{\sigma_k^2}, \tag{92}$$

$$C_0 = \text{KL}(\mu_0\|\pi) + \frac{13\gamma}{4(1-(1-p)\beta)} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2], \tag{93}$$

and C_0, C_1, C_2 are positive constants. Furthermore, let

$$\gamma = \left(\frac{d}{\sigma_m^2 N^2 L_\pi^2}\right)^{1/3}, \quad p = 1 - \beta = \left(\frac{d^2 L_\pi^2}{\sigma_m^4 N}\right)^{1/3}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\}, \tag{94}$$

and let the initial parameters satisfy $\sigma_0 \geq \sigma_{\min}$, $\varepsilon_{\sigma_0} > 0$, $\alpha_0 \geq 1$, and $\sigma_{\min} > 0$ is the minimum noise level. Then, the time-averaged law $\bar{\mu}_{N\gamma} := \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ satisfies $\text{FI}(\bar{\mu}_{N\gamma}\|\pi) \leq \varepsilon$ with $\mathcal{O}\left(\frac{\sigma_m^2 L_\pi^2 d^2}{\varepsilon^3}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. Similar to the proof in Appendix B.4.1, we start by using Lemma 1,

$$\frac{d}{dt} \text{KL}(\mu_t\|\pi) \leq -\frac{3}{4} \text{FI}(\mu_t\|\pi) + 3L_\pi^2 \Delta_k + 3e_k^2 + 3(\sigma_k C + \varepsilon_{\sigma_k} + (\alpha_k - 1)C\sigma_k^{-1})^2. \tag{95}$$

From the analysis of Theorem 3 in Appendix B.3.4, we have the upper bound (57) on e_k^2

$$e_k^2 \leq -\frac{1}{1-(1-p)\beta}(e_{k+1}^2 - e_k^2) + \frac{2(1-p)L_\pi^2}{1-(1-p)\beta}\Delta_k + \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] \frac{\sigma^2}{1-(1-p)\beta}, \quad (96)$$

where we use the fact that $L_\pi \geq L_f$. Plugging this bound into (95), we obtain

$$\begin{aligned} \frac{d}{dt}\text{KL}(\mu_t|\pi) &\leq -\frac{3}{4}\text{FI}(\mu_t|\pi) + 3L_\pi^2\phi(p, \beta)\Delta_k - \frac{3}{1-(1-p)\beta}(e_{k+1}^2 - e_k^2) \\ &\quad + \frac{3\sigma^2}{1-(1-p)\beta} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] + 3(\sigma_k C + \varepsilon_{\sigma_k} + (\alpha_k - 1)C\sigma_k^{-1})^2, \end{aligned} \quad (97)$$

where $\phi(p, \beta) := 1 + \frac{2(1-p)}{1-(1-p)\beta}$. Following the analysis of Theorem 4a, from the inequality (84), we have the following upper bound on Δ_k :

$$\Delta_k \leq 4\gamma^2 e_k^2 + 4\gamma^2 \mathbb{E} [\|\nabla \log \pi(\mathbf{x}_t|\mathbf{y})\|^2] + 4\gamma^2 L_\pi^2 \Delta_k + 4\gamma^2 (\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2\sigma_k^{-1})^2 + 2\gamma d.$$

Plugging in the upper (96) on e_k^2 and rearranging the terms yields

$$\begin{aligned} (1 - 4\gamma^2 L_\pi^2 \phi(p, \gamma)) \Delta_k &\leq -\frac{4\gamma^2}{1-(1-p)\beta}(e_{k+1}^2 - e_k^2) + \frac{4\gamma^2 \sigma^2}{1-(1-p)\beta} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] \\ &\quad + 4\gamma^2 \mathbb{E} [\|\nabla \log \pi(\mathbf{x}_t|\mathbf{y})\|^2] + 4\gamma^2 (\sigma_k C + \varepsilon_{\sigma_k} + (\alpha_k - 1)C\sigma_k^{-1})^2 + 2\gamma d. \end{aligned}$$

Setting γ such that $\gamma \leq \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}$ and dividing both sides by $\frac{13}{13}$, we obtain

$$\begin{aligned} \Delta_k &\leq -\frac{13\gamma^2}{3(1-(1-p)\beta)}(e_{k+1}^2 - e_k^2) + \frac{13\gamma^2 \sigma^2}{3(1-(1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] \\ &\quad + \frac{13}{3}\gamma^2 \mathbb{E} [\|\nabla \log \pi(\mathbf{x}_t|\mathbf{y})\|^2] + \frac{13}{3}\gamma^2 (\sigma_k C + \varepsilon_{\sigma_k} + (\alpha_k - 1)C\sigma_k^{-1})^2 + \frac{13}{6}\gamma d. \end{aligned} \quad (98)$$

Substituting the above bound into (97), applying Lemma 2, using the step-size condition $\gamma \leq \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}$, and using Young's inequality for $(\sigma_k C_1 + \varepsilon_{\sigma_k} + (\alpha_k - 1)C_2\sigma_k^{-1})^2$ yields

$$\begin{aligned} \frac{d}{dt}\text{KL}(\mu_t|\pi) &\leq -\frac{1}{2}\text{FI}(\mu_t|\pi) - \frac{3 + 13\gamma^2 L_\pi^2 \phi(p, \beta)}{1-(1-p)\beta}(e_{k+1}^2 - e_k^2) + \frac{13\sigma^2}{4(1-(1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] \\ &\quad + \frac{21}{2}\gamma L_\pi^2 d\phi(p, \beta) + \frac{39}{4}\sigma_k^2 C_1^2 + \frac{39}{4}\varepsilon_{\sigma_k}^2 + \frac{39C_2^2(\alpha_k - 1)^2}{4\sigma_k^2} \end{aligned} \quad (99)$$

Next, we define $\mathcal{L}_k := \text{KL}(\mu_{k\gamma}|\pi) + \frac{\gamma(3+13\gamma^2 L_\pi^2 \phi(p, \beta))}{1-(1-p)\beta}e_k^2$ and integrate both sides over $t \in [k\gamma, (k+1)\gamma]$.

$$\begin{aligned} \mathcal{L}_{k+1} - \mathcal{L}_k &\leq -\frac{1}{2} \int_{k\gamma}^{(k+1)\gamma} \text{FI}(\mu_t|\pi) dt + \frac{13\gamma\sigma^2}{4(1-(1-p)\beta)} \left[\frac{p}{b} + 2(1-p)(1-\beta)^2\right] + \frac{21}{2}\gamma L_\pi^2 d\phi(p, \beta) \\ &\quad + \frac{39}{4}\gamma\sigma_k^2 C_1^2 + \frac{39}{4}\gamma\varepsilon_{\sigma_k}^2 + \frac{39\gamma C_2^2(\alpha_k - 1)^2}{4\sigma_k^2}. \end{aligned}$$

Note that $\mathcal{L}_0 = \text{KL}(\mu_0|\pi) + \frac{\gamma(3+13\gamma^2 L_\pi^2 \phi(p, \beta))}{1-(1-p)\beta}e_0^2 \leq \text{KL}(\mu_0|\pi) + \frac{13\gamma e_0^2}{4(1-(1-p)\beta)} =: C_0$. Then, iterating the above inequality for $k = 0, 1, \dots, N-1$, multiplying both sides by $\frac{2}{N\gamma}$, and rearranging the terms, we obtain

$$\frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t|\pi) dt \leq \frac{2C_0}{N\gamma} + \frac{13\sigma^2}{2(1-(1-p)\beta)} \left[\frac{p}{b} + (1-p)(1-\beta)^2\right] + \frac{63\gamma L_\pi^2 d}{1-(1-p)\beta} + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2,$$

where $\bar{\sigma}^2$, $\bar{\varepsilon}_\sigma^2$, and $\bar{\alpha}^2$ are defined in (92). Note that we use $\phi(p, \beta) = 1 + \frac{2(1-p)}{1-(1-p)\beta} \leq 1 + \frac{2}{1-(1-p)\beta} \leq \frac{3}{1-(1-p)\beta}$. Since the definitions of γ , p , b , and σ_m are the same as in Theorem 3, we have

$$\text{FI}(\bar{\mu}_{N\gamma}|\pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t|\pi) dt \leq \mathcal{O} \left(\frac{\sigma_m^{2/3} L_\pi^{2/3} d^{2/3}}{N^{1/3}} \right) + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2.$$

We note that the annealing and noise schedules (12), and the SGM estimation error are the same as in Theorem 4a. Therefore, we set $\sigma_{\min} = \mathcal{O}(N^{-1/6})$ and follow the same steps in (88), (89), and (90) to obtain

$$\bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2 = \mathcal{O}\left(\frac{1}{N^{1/3}}\right).$$

This implies $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) = \mathcal{O}\left(\frac{\sigma_m^{2/3} L_\pi^2 d^{2/3}}{N^{1/3}}\right)$, that is, $\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \varepsilon$ is achieved with $\mathcal{O}\left(\frac{\sigma_m^2 L_\pi^2 d^2}{\varepsilon^3}\right)$ iterations and $pb + (1-p) = \mathcal{O}(1)$ gradient computations per iteration. $\square$

B.4.3 Proof of Corollary B.2 (ML-VRLD with SGM prior: squared TV distance convergence)

Corollary B.2 (ML-VRLD with SGM prior: squared TV distance convergence) *Let $\pi(\mathbf{x}|\mathbf{y}) \propto \ell(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ be the posterior with the likelihood $\ell(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ and the prior $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Suppose that the likelihood potential f satisfies Assumptions 1 and 2 with Lipschitz constant L_f , the prior potential h satisfies Assumption 1 and 5 with Lipschitz constant L_h , the target posterior π satisfies Assumption 3, and the SGM satisfies Assumption 6 with decreasing error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$ for $k \geq 1$. Define $L_\pi = L_f + L_h$. Let $(\mu_t)_{t \geq 0}$ denote the law of the interpolation generated by (11) with the estimator (8) and the schedules defined in (12). For any step size $\gamma \in \left(0, \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$, and for any $N \geq 1$, it holds that*

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0 C_{\text{PI}}}{N\gamma} + 52C_{\text{PI}}\sigma^2 \left[\frac{\frac{p}{b} + (1-p)(1-\beta)^2}{1 - (1-p)\beta} \right] + \frac{336C_{\text{PI}}\gamma L_\pi^2 d}{(1 - (1-p)\beta)^2} + 4C_{\text{PI}}(\bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2), \quad (100)$$

where

$$\bar{\sigma}^2 = \frac{39C_1^2}{2N} \sum_{k=1}^N \sigma_k^2, \quad \bar{\varepsilon}_\sigma^2 = \frac{39}{2N} \sum_{k=1}^N \varepsilon_{\sigma_k}^2, \quad \bar{\alpha}^2 = \frac{39C_2^2}{2N} \sum_{k=1}^N \frac{(\alpha_k - 1)^2}{\sigma_k^2}, \quad (101)$$

$$C_0 = \text{KL}(\mu_0 \|\pi) + \frac{13\gamma}{2(1 - (1-p)\beta)} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2], \quad (102)$$

and $C_{\text{PI}}, C_0, C_1, C_2$ are positive constants. Furthermore, let

$$\gamma = \frac{\sqrt{d}}{\sigma_m N^{3/4} \sqrt{L_\pi}}, \quad p = 1 - \beta = \frac{\sqrt{L_\pi d}}{\sigma_m N^{1/4}}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\},$$

and let the initial parameters satisfy $\sigma_0 \geq \sigma_{\min}$, $\varepsilon_{\sigma_0} > 0$, $\alpha_0 \geq 1$, and $\sigma_{\min} > 0$ is the minimum noise level. Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ achieves $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $\mathcal{O}\left(\frac{\sigma_m^4 C_{\text{PI}}^4 L_\pi^2 d^2}{\varepsilon^4}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. Applying Theorem 4 from Appendix B.4.1, we obtain

$$\text{FI}(\bar{\mu}_{N\gamma} \|\pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \|\pi) dt \leq \frac{2C_0}{N\gamma} + 13\sigma^2 \frac{\left(\frac{p}{b} + (1-p)(1-\beta)^2\right)}{(1 - (1-p)\beta)} + \frac{84\gamma L_\pi^2 d}{(1 - (1-p)\beta)^2} + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2.$$

Using Lemma 3 with Assumption 3 yields

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0 C_{\text{PI}}}{N\gamma} + 52C_{\text{PI}}\sigma^2 \frac{\left(\frac{p}{b} + (1-p)(1-\beta)^2\right)}{(1 - (1-p)\beta)} + \frac{336C_{\text{PI}}\gamma L_\pi^2 d}{(1 - (1-p)\beta)^2} + 4C_{\text{PI}}(\bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2)$$

Using the definitions of $\gamma, p, \beta, \sigma_k, \alpha_k$, and ε_{σ_k} from the corollary statement and following the steps in Appendix B.4.1, we prove that an iteration complexity of order $\mathcal{O}\left(\frac{C_{\text{PI}}^4 \sigma_m^4 L_\pi^2 d^2}{\varepsilon^4}\right)$ suffices to ensure $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $pb + (1-p) = \mathcal{O}(1)$ gradient computations per iteration. $\square$

B.4.4 Proof of Corollary B.3 (SL-VRLD with SGM prior: squared TV distance convergence)

Corollary B.3 (SL-VRLD with SGM prior: squared TV distance convergence) *Let $\pi(\mathbf{x}|\mathbf{y}) \propto \ell(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ be the posterior with the likelihood $\ell(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ and the prior $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Suppose that the likelihood potential f satisfies*

Assumptions 2 and 4 with Lipschitz constant L_f , the prior potential h satisfies Assumption 1 and 5 with Lipschitz constant L_h , the target posterior π satisfies Assumption 3, and the SGM satisfies Assumption 6 with decreasing error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$ for $k \geq 1$. Define $L_\pi = L_f + L_h$. Let $(\mu_t)_{t \geq 0}$ denote the law of the interpolation generated by (11) with the estimator (10) and the schedules defined in (12). For any step size $\gamma \in \left(0, \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}\right]$, where $\phi(p, \beta) = 1 + \frac{2(1-p)}{1-(1-p)\beta}$, and for any $N \geq 1$, it holds that

$$\|\mu_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0C_{\text{PI}}}{N\gamma} + 26C_{\text{PI}}\sigma^2 \left[\frac{\frac{p}{b} + 2(1-p)(1-\beta)^2}{1-(1-p)\beta} \right] + \frac{252C_{\text{PI}}\gamma L_\pi^2 d}{1-(1-p)\beta} + 4C_{\text{PI}}(\bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2), \quad (103)$$

where

$$\bar{\sigma}^2 = \frac{39C_1^2}{2N} \sum_{k=1}^N \sigma_k^2, \quad \bar{\varepsilon}_\sigma^2 = \frac{39}{2N} \sum_{k=1}^N \varepsilon_{\sigma_k}^2, \quad \bar{\alpha}^2 = \frac{39C_2^2}{2N} \sum_{k=1}^N \frac{(\alpha_k - 1)^2}{\sigma_k^2}, \quad (104)$$

$$C_0 = \text{KL}(\mu_0 \| \pi) + \frac{13\gamma}{4(1-(1-p)\beta)^2} \mathbb{E}[\|\nabla f(\mathbf{x}_0) - \mathbf{g}_0\|^2], \quad (105)$$

and $C_{\text{PI}}, C_0, C_1, C_2$ are positive constants. Furthermore, let

$$\gamma = \left(\frac{d}{\sigma_m^2 N^2 L_\pi^2} \right)^{1/3}, \quad p = 1 - \beta = \left(\frac{d^2 L_\pi^2}{\sigma_m^4 N} \right)^{1/3}, \quad b = \left\lceil \frac{1}{p} \right\rceil, \quad \text{where } \sigma_m = \max\{\sigma, 1\},$$

and let the initial parameters satisfy $\sigma_0 \geq \sigma_{\min}$, $\varepsilon_{\sigma_0} > 0$, $\alpha_0 \geq 1$, and $\sigma_{\min} > 0$ is the minimum noise level. Then, the time-averaged law $\bar{\mu}_{N\gamma} = \frac{1}{N\gamma} \int_0^{N\gamma} \mu_t dt$ achieves $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $\mathcal{O}\left(\frac{C_{\text{PI}}^3 \sigma_m^2 L_\pi^2 d^2}{\varepsilon^3}\right)$ iterations and $\mathcal{O}(1)$ gradient computations per iteration.

Proof. Applying Theorem 4b, we have

$$\text{FI}(\bar{\mu}_{N\gamma} \| \pi) \leq \frac{1}{N\gamma} \int_0^{N\gamma} \text{FI}(\mu_t \| \pi) dt \leq \frac{2C_0}{N\gamma} + \frac{13\sigma^2 \left(\frac{p}{b} + 2(1-p)(1-\beta)^2 \right)}{2(1-(1-p)\beta)} + \frac{63\gamma L_\pi^2 d}{1-(1-p)\beta} + \bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2.$$

Using Lemma 3 with Assumption 3 yields

$$\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \frac{8C_0C_{\text{PI}}}{N\gamma} + \frac{26C_{\text{PI}}\sigma^2 \left(\frac{p}{b} + 2(1-p)(1-\beta)^2 \right)}{(1-(1-p)\beta)} + \frac{252C_{\text{PI}}\gamma L_\pi^2 d}{1-(1-p)\beta} + 4C_{\text{PI}}(\bar{\sigma}^2 + \bar{\varepsilon}_\sigma^2 + \bar{\alpha}^2)$$

Using the definitions of $\gamma, p, \beta, \sigma_k, \alpha_k$, and ε_{σ_k} from the corollary statement and following the steps in Appendix B.4.2, we prove that an iteration complexity of order $\mathcal{O}\left(\frac{C_{\text{PI}}^3 \sigma_m^2 L_\pi^2 d^2}{\varepsilon^3}\right)$ suffices to ensure $\|\bar{\mu}_{N\gamma} - \pi\|_{\text{TV}}^2 \leq \varepsilon$ with $pb + (1-p) = \mathcal{O}(1)$ gradient computations per iteration. $\square$

B.4.5 Proof of Theorem 5a (ML-VRLD with SGM prior: weak convergence)

Theorem 5a (ML-VRLD with SGM prior: weak convergence) Let $\pi(\mathbf{x}|\mathbf{y}) \propto \ell(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ be the posterior with the likelihood $\ell(\mathbf{y}|\mathbf{x}) \propto e^{-f(\mathbf{x})}$ and the prior $p(\mathbf{x}) \propto e^{-h(\mathbf{x})}$. Suppose that the likelihood potential f satisfies Assumptions 1 and 2 with Lipschitz constant L_f , the prior potential h satisfies Assumption 1 and 5 with Lipschitz constant L_h , and the SGM satisfies Assumption 6 with decreasing error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$ for $k \geq 1$. Let $(\mu_t)_{t \geq 0}$ denote the law of the interpolation generated by (11) with the estimator (8) and the schedules defined in (12). Define the time-varying parameters as follows

$$\gamma_k = \frac{C_\gamma}{k^{3/2}}, \quad p_k = \frac{C_p}{k^{1/2}}, \quad b_k = \left\lceil \frac{1}{p_k} \right\rceil, \quad \forall k \geq 1,$$

where $0 < C_p < 1$ and $C_\gamma > 0$ are numerical constants. Then, the time-averaged law $\bar{\mu}_{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$, where $\tau_n = \sum_{k=1}^n \gamma_k$ converges weakly to π .

Proof. We use the same definitions of the time-varying parameters and the definition of the estimator $\mathbf{g}_k$ as in the proof of Theorem 4a. Therefore, the time-varying step sizes $(\gamma_k)_{k \geq 1}$ satisfy the step-size condition of Theorem 4a, that is

$$\gamma \in \left(0, \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}\right], \quad \text{where } \phi(p_k, \beta_k) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$$

for all $k \geq 1$. Thus, we can use the same upper bound in (87) (Theorem 4a; Appendix B.4.1) with time-varying parameters, and integrate both sides of the inequality over $[\tau_{n-1}, \tau_n]$ to get

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \|\pi) - \text{KL}(\mu_{\tau_{n-1}} \|\pi) &\leq -\frac{1}{2} \int_{\tau_{n-1}}^{\tau_n} \text{FI}(\mu_t \|\pi) dt - \frac{\gamma_n (6 + 26\gamma_n^2 L_\pi^2 \phi(p_n, \beta_n))}{1 - (1 - p_n)\beta_n} (e_n^2 - e_{n-1}^2) \\ &\quad + \frac{13\gamma_n \sigma^2}{2(1 - (1 - p_n)\beta_n)} [p_n^2 + (1 - p_n)(1 - \beta_n)^2] + \frac{21}{2} \gamma_n^2 L_\pi^2 d \phi(p_n, \beta_n) \\ &\quad + \frac{39}{4} \gamma_n \sigma_n^2 C_1^2 + \frac{39}{4} \gamma_n \varepsilon_{\sigma_n}^2 + \frac{39C_2^2 \gamma_n (\alpha_n - 1)^2}{4\sigma_n^2}, \end{aligned}$$

where we use $\frac{p_k}{b_k} \leq p_k^2$. Iterating the above bound, we obtain

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \|\pi) - \text{KL}(\mu_0 \|\pi) &\leq -\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \|\pi) dt - \underbrace{\sum_{k=1}^n \frac{\gamma_k (6 + 26\gamma_k^2 L_\pi^2 \phi(p_k, \beta_k))}{1 - (1 - p_k)\beta_k} (e_k^2 - e_{k-1}^2)}_{\text{(I)}} \\ &\quad + \underbrace{\frac{13\sigma^2}{2} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + (1 - p_k)(1 - \beta_k)^2}{1 - (1 - p_k)\beta_k} \right)}_{\text{(II)}} + \underbrace{\frac{21}{2} L_\pi^2 d \sum_{k=1}^n \gamma_k^2 \phi(p_k, \beta_k)}_{\text{(III)}} \\ &\quad + \underbrace{\frac{39C_1^2}{4} \sum_{k=1}^n \gamma_k \sigma_k^2}_{\text{(IV)}} + \underbrace{\frac{39}{4} \sum_{k=1}^n \gamma_k \varepsilon_{\sigma_k}^2}_{\text{(V)}} + \underbrace{\frac{39C_2^2}{4} \sum_{k=1}^n \frac{\gamma_k (\alpha_k - 1)^2}{\sigma_k^2}}_{\text{(VI)}} \end{aligned} \quad (106)$$

In the proof of Theorem 2 (Appendix B.3.3), we show that with the same parameters for $\gamma_k, p_k, \beta_k, b_k$, we have

$$\text{(I)} = -\sum_{k=1}^n \frac{\gamma_k (6 + 26\gamma_k^2 L_\pi^2 \phi(p_k, \beta_k))}{1 - (1 - p_k)\beta_k} (e_k^2 - e_{k-1}^2) \leq \underbrace{\frac{\gamma_1 (6 + 26\gamma_1^2 L_\pi^2 \phi(p_1, \beta_1))}{1 - (1 - p_1)\beta_1}}_{=: c_1} e_0^2 < \infty \quad (107)$$

$$\text{(II)} \leq \frac{13\sigma^2}{2} \sum_{k=1}^{\infty} \gamma_k \left(\frac{p_k^2 + (1 - p_k)(1 - \beta_k)^2}{1 - (1 - p_k)\beta_k} \right) =: S_2 < \infty \quad (108)$$

$$\text{(III)} \leq \frac{21L_\pi^2 d}{2} \sum_{k=1}^{\infty} \gamma_k^2 \phi(p_k, \beta_k) =: S_3 < \infty \quad (109)$$

We next show that the terms (IV), (V), and (VI) are uniformly bounded.

To bound the bias due to the noise schedule $(\sigma_k)_{k=1}^n$, we proceed as

$$\text{(IV)} = \frac{39C_1^2}{4} \sum_{k=1}^n \gamma_k \sigma_k^2 \leq \frac{39C_1^2 C_\gamma}{4} \sum_{k=1}^{\infty} \frac{\max\{\sigma_0^2 \rho_2^{2k}, \sigma_{\min}^2\}}{k^{3/2}} \stackrel{(*)}{\leq} \frac{39C_1^2 C_\gamma \sigma_0^2}{4} \sum_{k=1}^{\infty} k^{-3/2} =: A_1 < \infty, \quad (110)$$

where we use the facts that $\sigma_0 \geq \sigma_{\min}$ and $\rho_2 < 1$ to get $(*)$.

To bound the bias due to the SGM estimation errors $(\varepsilon_{\sigma_k})_{k=1}^n$, we use the fact that $\varepsilon_{\sigma_k} \leq \frac{C_\varepsilon}{k^{-1/2}}$ for some $C_\varepsilon > 0$ and proceed as

$$\text{(V)} = \frac{39}{4} \sum_{k=1}^n \gamma_k \varepsilon_{\sigma_k}^2 \leq \frac{39}{4} \sum_{k=1}^{\infty} \gamma_k \varepsilon_{\sigma_k}^2 \leq \frac{39C_\gamma C_\varepsilon^2}{4} \sum_{k=1}^{\infty} k^{-5/2} =: A_2 < \infty. \quad (111)$$

To bound the bias due to the annealing schedule $(\sigma_k)_{k=1}^n$, we follow

$$\begin{aligned} \text{(VI)} &= \frac{39C_2^2}{4} \sum_{k=1}^n \frac{\gamma_k (\alpha_k - 1)^2}{\sigma_k^2} = \frac{39C_2^2 C_\gamma}{4} \sum_{k=1}^{\infty} \frac{1}{k^{\frac{3}{2}}} \frac{\max\{\alpha_0 \rho_1^k - 1, 0\}^2}{\max\{\sigma_0^2 \rho_2^{2k}, \sigma_{\min}^2\}} \\ &\stackrel{(*)}{\leq} \frac{39C_2^2 C_\gamma (\alpha_0^2 - 1)}{4\sigma_{\min}^2} \sum_{k=1}^{\infty} k^{-3/2} =: A_3 < \infty, \end{aligned} \quad (112)$$

where we use the facts that $\alpha_0 \geq 1$, $\rho_1 < 1$, $\rho_2 < 1$, and $\sigma_0 \geq \sigma_{\min} > 0$ to obtain (*). Plugging the bounds (I)–(VI) into (106), using Jensen's inequality with the convexity of $\text{FI}(\cdot\|\pi)$, and rearranging the terms, we have

$$\text{FI}(\bar{\mu}_{\tau_n}\|\pi) \leq \frac{1}{\tau_n} \int_0^{\tau_n} \text{FI}(\mu_t\|\pi) dt \leq \frac{2\text{KL}(\mu_0\|\pi)}{\tau_n} + \frac{2(c_1 e_0^2 + S_2 + S_3 + A_1 + A_2 + A_3)}{\tau_n}, \quad (113)$$

where $\bar{\mu}_{\tau_n} := \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$ is the time-averaged law up to time $\tau_n = \sum_{k=1}^n \gamma_k$.

On the other hand, if $t \in [\tau_n, \tau_{n+1}]$, integrating (87) (Theorem 4a; Appendix B.4.1) from τ_n to t and removing the negative Fisher information term yields

$$\begin{aligned} \text{KL}(\mu_t\|\pi) &\leq \text{KL}(\mu_{\tau_n}\|\pi) - \frac{(t - \tau_n)(6 + 26\gamma_{n+1}^2 L_\pi^2 \phi(p_{n+1}, \beta_{n+1}))}{1 - (1 - p_{n+1})\beta_{n+1}} (e_{n+1}^2 - e_n^2) \\ &\quad + \frac{13(t - \tau_n)\sigma^2}{2(1 - (1 - p_{n+1})\beta_{n+1})} \left[\frac{p_{n+1}}{b_{n+1}} + (1 - p_{n+1})(1 - \beta_{n+1})^2 \right] \\ &\quad + \frac{21}{2}(t - \tau_n)\gamma_{n+1} L_\pi^2 d\phi(p_{n+1}, \beta_{n+1}) + \frac{39}{4}(t - \tau_n)\sigma_{n+1}^2 C_1^2 + \frac{39}{4}(t - \tau_n)\varepsilon_{\sigma_{n+1}}^2 \\ &\quad + \frac{39C_2^2(t - \tau_n)(\alpha_{n+1} - 1)^2}{4\sigma_{n+1}^2} \\ &\leq \text{KL}(\mu_{\tau_n}\|\pi) - \frac{(t - \tau_n)(6 + 26\gamma_{n+1}^2 L_\pi^2 \phi(p_{n+1}, \beta_{n+1}))}{1 - (1 - p_{n+1})\beta_{n+1}} (e_{n+1}^2 - e_n^2) + S_2 + S_3 + A_1 + A_2 + A_3 \\ &\leq \text{KL}(\mu_0\|\pi) - \sum_{k=1}^n c_k (e_k^2 - e_{k-1}^2) - c'_{n+1} (e_{n+1}^2 - e_n^2) + 2S_2 + 2S_3 + 2A_1 + 2A_2 + 2A_3, \end{aligned} \quad (114)$$

where

$$c'_{n+1} := \frac{(t - \tau_n)(6 + 26\gamma_{n+1}^2 L_\pi^2 \phi(p_{n+1}, \beta_{n+1}))}{1 - (1 - p_{n+1})\beta_{n+1}}.$$

As shown in (42) (Theorem 2; Appendix B.3.3), $(c_k)_{k=1}^n$ is a nonnegative and decreasing sequence. We note that $c_n > c_{n+1} \geq c'_{n+1}$. Using this property, we have

$$- \sum_{k=1}^n c_k (e_k^2 - e_{k-1}^2) - c'_{n+1} (e_{n+1}^2 - e_n^2) = c_1 e_0^2 + \sum_{k=1}^{n-1} (c_{k+1} - c_k) e_k^2 + (c'_{n+1} - c_n) e_n^2 - c'_{n+1} e_{n+1} \leq c_1 e_0^2.$$

Applying this upper bound to (114), we obtain

$$\text{KL}(\mu_t\|\pi) \leq \text{KL}(\mu_0\|\pi) + c_1 e_0^2 + 2S_2 + 2S_3 + 2A_1 + 2A_2 + 2A_3 < \infty.$$

Note that the initial mean estimation error $e_0^2 = \mathbb{E}[\|g_0 - \nabla f(x_0)\|^2]$ is bounded. This proves $\text{KL}(\mu_t\|\pi)$ is uniformly bounded for all $t \geq 0$. By the convexity of the KL divergence, it implies that $\{\text{KL}(\bar{\mu}_{\tau_n}\|\pi) \mid n \in \mathbb{N}\}$ is uniformly bounded as well. Since the sub-level sets of $\text{KL}(\cdot\|\pi)$ are weakly compact, $(\bar{\mu}_{\tau_n})_{n \in \mathbb{N}}$ is tight. To show that $\bar{\mu}_{\tau_n} \rightarrow \pi$ weakly, it is sufficient to show that every cluster point of $(\bar{\mu}_{\tau_n})_{n \in \mathbb{N}}$ is equal to π .

Consider a subsequence of $(\bar{\mu}_{\tau_n})_{n \in \mathbb{N}}$ converging to some limit $\bar{\mu}$. As $n \rightarrow \infty$, $\tau_n \rightarrow \infty$, and by (113) we also have $\text{FI}(\bar{\mu}_{\tau_n}\|\pi) \rightarrow 0$. Therefore, this is still true along the subsequence. By the weak lower semi-continuity of the Fisher information along the subsequence, $\text{FI}(\bar{\mu}_{\tau_n}\|\pi) = 0$. That implies; for $\psi := \frac{d\mu}{d\pi}$, we have $\sqrt{\psi} \in \text{dom}(\mathcal{E})$ and $\mathcal{E}(\sqrt{\psi}) = 0$, where $\mathcal{E}$ denotes the Dirichlet energy (i.e. the squared $L^2(\pi)$ -norm of the gradient; see Section 3 in Balasubramanian et al. [2022]). Since ∇f is Lipschitz by Assumption 1, then π has a continuous and strictly positive density on $\mathbb{R}^d$, so $\mathcal{E}(\sqrt{\psi}) = 0$ implies that ψ is a constant π with probability 1. Hence, $\bar{\mu} = \pi$ as $n \rightarrow \infty$. $\square$

B.4.6 Proof of Theorem 5b (SL-VRLD with SGM prior: weak convergence)

Theorem 5b (SL-VRLD with SGM prior: weak convergence) *Let $\pi(x|y) \propto \ell(y|x)p(x)$ be the posterior with the likelihood $\ell(y|x) \propto e^{-f(x)}$ and the prior $p(x) \propto e^{-h(x)}$. Suppose that the likelihood potential f satisfies Assumptions 2 and 4 with Lipschitz constant L_f , the prior potential h satisfies Assumption 1 and 5 with Lipschitz constant L_h , and the SGM*

satisfies Assumption 6 with decreasing error $\varepsilon_{\sigma_k} = \mathcal{O}(k^{-1/2})$ for $k \geq 1$. Let $(\mu_t)_{t \geq 0}$ denote the law of the interpolation generated by (11) with the estimator (10) and the schedules defined in (12). Define the time-varying parameters as follows

$$\gamma_k = \frac{C_\gamma}{k^{3/2}}, \quad p_k = \frac{C_p}{k^{1/2}}, \quad b_k = \left\lceil \frac{1}{p_k} \right\rceil, \quad \forall k \geq 1,$$

where $0 < C_p < 1$ and $C_\gamma > 0$ are numerical constants. Then, the time-averaged law $\bar{\mu}_{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$, where $\tau_n = \sum_{k=1}^n \gamma_k$ converges weakly to π .

Proof. We use the same definitions of the time-varying parameters and the definition of the estimator $\mathbf{g}_k$ as in the proof of Theorem 4b. Therefore, the time-varying step sizes $(\gamma_k)_{k \geq 1}$ satisfy the step-size condition of Theorem 4b, that is

$$\gamma \in \left(0, \frac{1}{L_\pi \sqrt{52\phi(p, \beta)}}\right], \quad \text{where} \quad \phi(p_k, \beta_k) = 1 + \frac{4(1-p)\beta^2}{(1-(1-p)\beta)^2}$$

for all $k \geq 1$. Thus, we can use the same upper bound in (99) (Theorem 4b; Appendix B.4.2), integrate both sides of the inequality over $[\tau_{n-1}, \tau_n]$ to get

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \parallel \pi) - \text{KL}(\mu_{\tau_{n-1}} \parallel \pi) &\leq -\frac{1}{2} \int_{\tau_{n-1}}^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt - \frac{\gamma_n (3 + 13\gamma_n^2 L_\pi^2 \phi(p_n, \beta_n))}{1 - (1-p_n)\beta_n} (e_n^2 - e_{n-1}^2) \\ &\quad + \frac{13\gamma_n \sigma^2}{4(1 - (1-p_n)\beta_n)} [p_n^2 + 2(1-p_n)(1-\beta_n)^2] + \frac{21}{2} \gamma_n^2 L_\pi^2 d \phi(p_n, \beta_n) \\ &\quad + \frac{39}{4} \gamma_n \sigma_n^2 C_1^2 + \frac{39}{4} \gamma_n \varepsilon_{\sigma_n}^2 + \frac{39C_2^2 \gamma_n (\alpha_n - 1)^2}{4\sigma_n^2}, \end{aligned}$$

where we use the fact that $b_n \geq 1/p_n$. Iterating the above bound, we obtain

$$\begin{aligned} \text{KL}(\mu_{\tau_n} \parallel \pi) - \text{KL}(\mu_0 \parallel \pi) &\leq -\frac{1}{2} \int_0^{\tau_n} \text{FI}(\mu_t \parallel \pi) dt - \underbrace{\sum_{k=1}^n \frac{\gamma_k (3 + 13\gamma_k^2 L_\pi^2 \phi(p_k, \beta_k))}{1 - (1-p_k)\beta_k} (e_k^2 - e_{k-1}^2)}_{\text{(I)}} \\ &\quad + \underbrace{\frac{13\sigma^2}{4} \sum_{k=1}^n \gamma_k \left(\frac{p_k^2 + 2(1-p_k)(1-\beta_k)^2}{1 - (1-p_k)\beta_k} \right)}_{\text{(II)}} + \underbrace{\frac{21}{2} L_\pi^2 d \sum_{k=1}^n \gamma_k^2 \phi(p_k, \beta_k)}_{\text{(III)}} \\ &\quad + \underbrace{\frac{39C_1^2}{4} \sum_{k=1}^n \gamma_k \sigma_k^2}_{\text{(IV)}} + \underbrace{\frac{39}{4} \sum_{k=1}^n \gamma_k \varepsilon_{\sigma_k}^2}_{\text{(V)}} + \underbrace{\frac{39C_2^2}{4} \sum_{k=1}^n \frac{\gamma_k (\alpha_k - 1)^2}{\sigma_k^2}}_{\text{(VI)}}. \end{aligned} \quad (115)$$

In the proof of Theorem B.1 (Appendix B.3.6), we show that with the same parameters for $\gamma_k, p_k, \beta_k, b_k$, we have

$$\text{(I)} = -\sum_{k=1}^n \frac{\gamma_k (3 + 13\gamma_k^2 L_\pi^2 \phi(p_k, \beta_k))}{1 - (1-p_k)\beta_k} (e_k^2 - e_{k-1}^2) \leq \underbrace{\frac{\gamma_1 (3 + 13\gamma_1^2 L_\pi^2 \phi(p_1, \beta_1))}{1 - (1-p_1)\beta_1}}_{=: c_1} e_0^2 < \infty \quad (116)$$

$$\text{(II)} \leq \frac{13\sigma^2}{4} \sum_{k=1}^{\infty} \gamma_k \left(\frac{p_k^2 + (1-p_k)(1-\beta_k)^2}{1 - (1-p_k)\beta_k} \right) =: S_2 < \infty \quad (117)$$

$$\text{(III)} \leq \frac{21L_\pi^2 d}{2} \sum_{k=1}^{\infty} \gamma_k^2 \phi(p_k, \beta_k) =: S_3 < \infty \quad (118)$$

Following the same steps in the proof of Theorem 5a (Appendix B.4.5), we have

$$\text{(IV)} = \frac{39C_1^2}{4} \sum_{k=1}^n \gamma_k \sigma_k^2 = \frac{39C_1^2 C_\gamma \sigma_0^2}{4} \sum_{k=1}^{\infty} k^{-3/2} =: A_1 < \infty, \quad (119)$$

$$(V) = \frac{39}{4} \sum_{k=1}^n \gamma_k \varepsilon_{\sigma_k}^2 \leq \frac{39C_\gamma C_\varepsilon^2}{4} \sum_{k=1}^{\infty} k^{-5/2} =: A_2 < \infty, \quad (120)$$

$$(VI) = \frac{39C_2^2}{4} \sum_{k=1}^{\infty} \frac{\gamma_k (\alpha_k - 1)^2}{\sigma_k^2} \leq \frac{39C_2^2 C_\gamma (\alpha_0^2 - 1)}{4\sigma_{\min}^2} \sum_{k=1}^{\infty} k^{-3/2} =: A_3 < \infty, \quad (121)$$

Applying these upper bounds to (115), dropping the negative KL term, and multiplying both sides by $2/\tau_n$, we obtain

$$\text{FI}(\bar{\mu}_{\tau_n} \|\pi) \leq \frac{1}{\tau_n} \int_0^{\tau_n} \text{FI}(\mu_t \|\pi) dt \leq \frac{2\text{KL}(\mu_0 \|\pi)}{\tau_n} + \frac{2(c_1 e_0^2 + S_2 + S_3 + A_1 + A_2 + A_3)}{\tau_n}, \quad (122)$$

where $\bar{\mu}_{\tau_n} := \frac{1}{\tau_n} \int_0^{\tau_n} \mu_t dt$ and $\tau_n := \sum_{k=1}^n \gamma_k$.

On the other hand, if $t \in [\tau_n, \tau_{n+1}]$, integrating (99) (Theorem 4b; Appendix B.4.2) from τ_n to t and removing the negative Fisher information term yields

$$\begin{aligned} \text{KL}(\mu_t \|\pi) &\leq \text{KL}(\mu_{\tau_n} \|\pi) - \frac{(t - \tau_n)(3 + 13\gamma_{n+1}^2 L_\pi^2 \phi(p_{n+1}, \beta_{n+1}))}{1 - (1 - p_{n+1})\beta_{n+1}} (e_{n+1}^2 - e_n^2) \\ &\quad + \frac{13(t - \tau_n)\sigma^2}{4(1 - (1 - p_{n+1})\beta_{n+1})} [p_{n+1}^2 + 2(1 - p_{n+1})(1 - \beta_{n+1})^2] \\ &\quad + \frac{21}{2}(t - \tau_n)\gamma_{n+1} L_\pi^2 d\phi(p_{n+1}, \beta_{n+1}) + \frac{39}{4}(t - \tau_n)\sigma_{n+1}^2 C_1^2 + \frac{39}{4}(t - \tau_n)\varepsilon_{\sigma_{n+1}}^2 \\ &\quad + \frac{39C_2^2(t - \tau_n)(\alpha_{n+1} - 1)^2}{4\sigma_{n+1}^2}. \\ &\leq \text{KL}(\mu_{\tau_n} \|\pi) - \frac{(t - \tau_n)(3 + 13\gamma_{n+1}^2 L_\pi^2 \phi(p_{n+1}, \beta_{n+1}))}{1 - (1 - p_{n+1})\beta_{n+1}} (e_{n+1}^2 - e_n^2) + S_2 + S_3 + A_1 + A_2 + A_3 \\ &\leq \text{KL}(\mu_0 \|\pi) - \sum_{k=1}^n c_k (e_k^2 - e_{k-1}^2) - c'_{n+1} (e_{n+1}^2 - e_n^2) + 2S_2 + 2S_3 + 2A_1 + 2A_2 + 2A_3, \end{aligned} \quad (123)$$

where

$$c'_{n+1} := \frac{(t - \tau_n)(3 + 13\gamma_{n+1}^2 L_\pi^2 \phi(p_{n+1}, \beta_{n+1}))}{1 - (1 - p_{n+1})\beta_{n+1}}.$$

As shown in (72) (Theorem B.1; Appendix B.3.6), $(c_k)_{k=1}^n$ is a nonnegative and decreasing sequence. We note that $c_n > c_{n+1} \geq c'_{n+1}$. Using this property, we have

$$-\sum_{k=1}^n c_k (e_k^2 - e_{k-1}^2) - c'_{n+1} (e_{n+1}^2 - e_n^2) = c_1 e_0^2 + \sum_{k=1}^{n-1} (c_{k+1} - c_k) e_k^2 + (c'_{n+1} - c_n) e_n^2 - c'_{n+1} e_{n+1} \leq c_1 e_0^2.$$

Applying this upper bound to (123), we obtain

$$\text{KL}(\mu_t \|\pi) \leq \text{KL}(\mu_0 \|\pi) + c_1 e_0^2 + 2S_2 + 2S_3 + 2A_1 + 2A_2 + 2A_3 < \infty.$$

Thus $\{\text{KL}(\mu_t \|\pi) \mid t \geq 0\}$ is bounded. Following the same argument as in the final part of the proof of Theorem 5 a (Appendix B.4.6), we conclude that $\bar{\mu}_{\tau_n}$ converges weakly to π as $n \rightarrow \infty$. $\square$

C ADDITIONAL EXPERIMENTAL DETAILS

We provide additional experimental details and results for the synthetic experiments, MRI reconstruction, and sparse-angle CT reconstruction tasks. Our code is available on the [project page](#).

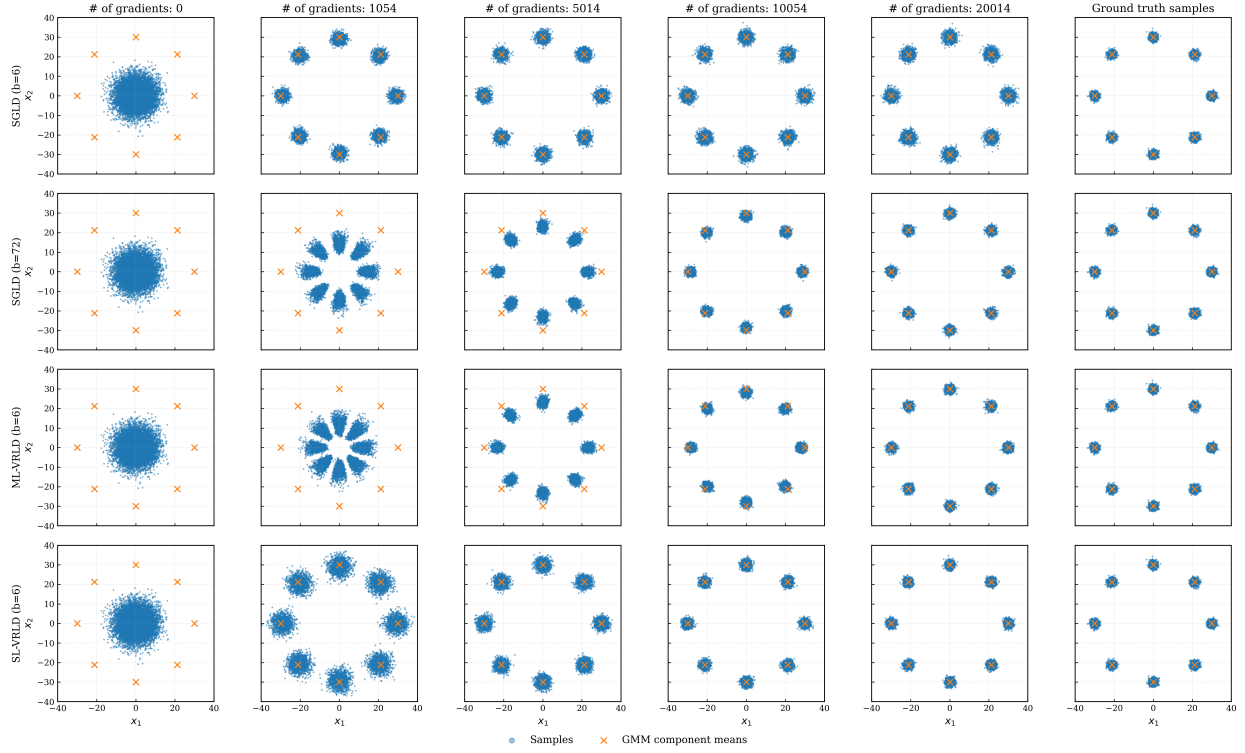


Figure 4: Visualization of samples generated by SGLD, ML-VRLD, and SL-VRLD on the 8-mode Gaussian mixture model at different numbers of gradient computations. Rows correspond to different methods and batch sizes, while columns show the evolution of the generated samples as the total number of gradient computations increases. The final column displays ground truth samples from the target GMM. SGLD with a fixed batch size ($b = 6$) remains trapped in a suboptimal distribution and fails to recover the target modes. Increasing the SGLD batch size to $b = 72$ enables convergence to the target distribution, but requires substantially more memory. In contrast, both ML-VRLD and SL-VRLD recover all target modes using the same fixed batch size ($b = 6$), demonstrating their ability to operate under fixed memory requirements. Moreover, SL-VRLD reaches the target distribution with fewer gradient computations, consistent with its improved theoretical convergence rate.

C.1 SYNTHETIC EXPERIMENTS

In this section, we give additional results and experimental details on our synthetic experiments, including computation of Fisher information, hyperparameters, and preprocessing steps.

Numerical Validation with GMM Sampling. We run SGLD, ML-VRLD and SL-VRLD estimators with the following iterations:

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma \mathbf{g}_k + \sqrt{2\gamma} \mathbf{Z}_k, \quad (124)$$

where $\mathbf{Z}_k \sim \mathcal{N}(0, I)$ and $\mathbf{g}_k$ is defined according to the corresponding estimator definitions. All algorithms are run for $N = 2000$ iterations to generate 10^4 samples, which are initialized over $[-50, 50]^2$ grid by sampling from a uniform distribution of $\mathcal{U}([-50, 50]^2)$. We use a step size of $\gamma = 0.02$ for SGLD and SL-VRLD, and $\gamma = 0.0015$ for ML-VRLD. We set the momentum parameter to $\beta = 0.9$ for ML-VRLD and $\beta = 0.999$ for SL-VRLD, and set $p = 0$ for both algorithm. We note that jointly using $\beta < 1$ and $p < 1$ does not improve the theoretical convergence rates. These parameters are combined primarily to provide a unified analysis of the underlying algorithms. Therefore, in our experiments, we set $p = 0$. We approximately compute the relative Fisher information at each iteration. Specifically, we use kernel density estimation (KDE) to estimate the sampling distribution from the generated samples of each algorithm and then evaluate the relative Fisher information with respect to the target GMM using ground truth samples. In our experiments, we assign equal probability $1/8$ to each mixture component and simulate stochastic gradients by adding Gaussian noise with standard deviation 30 to the gradient of the potential function.

In addition to the quantitative results in Fig. 1, Fig. 4 provides a qualitative comparison between the samples generated by each algorithm and ground truth samples drawn from the target GMM. SGLD with batch size $b = 6$ fails to accurately

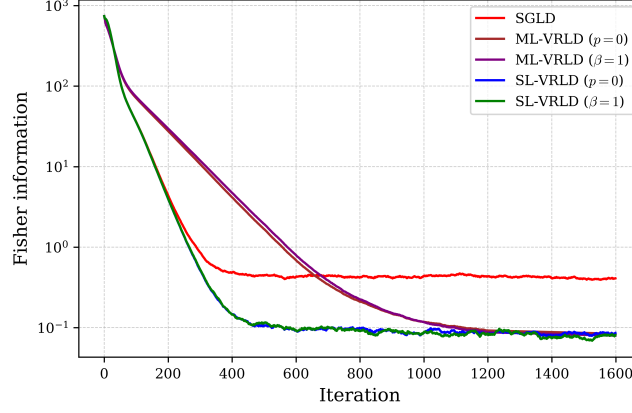


Figure 5: Results on a synthetic 2D inverse problem showing FI convergence for each method with batch size $b=6$. SGLD converges to a suboptimal solution, whereas ML-VRLD and SL-VRLD attain lower FI, consistent with our theory.

recover the target distribution, exhibiting a more dispersed sample distribution than the ground truth. This behavior is consistent with the $\mathcal{O}(Ld\sigma^2/\varepsilon)$ batch size requirement of SGLD, which necessitates larger batch sizes to attain higher accuracy. Indeed, increasing the SGLD batch size to $b = 72$ enables recovery of the target distribution. In contrast, both ML-VRLD and SL-VRLD closely match the ground truth distribution while operating with the same fixed batch size ($b = 6$), highlighting their ability to achieve convergence under fixed memory requirements. Moreover, SL-VRLD reaches the target distribution with fewer gradient computations than SGLD, consistent with their respective computational complexities of $\mathcal{O}(L^2d^2\sigma^2/\varepsilon^3)$ and $\mathcal{O}(L^3d^3\sigma^2/\varepsilon^3)$.

Numerical Validation with Compressed Sensing. We run SGLD, ML-VRLD and SL-VRLD estimators with the following iterations:

$$\mathbf{x}_{(k+1)\gamma} = \mathbf{x}_{k\gamma} - \gamma(\mathbf{g}_k + \alpha_k \nabla h_{\sigma_k}(\mathbf{x}_{k\gamma})) + \sqrt{2\gamma} \mathbf{Z}_k, \quad (125)$$

where $\mathbf{Z}_k \sim \mathcal{N}(0, I)$ and $\mathbf{g}_k$ is defined according to the estimator definition. We use the following annealing schedule defined in (12):

$$\alpha_k := \max\{\alpha_0 \rho_1^k, 1\} \quad \text{and} \quad \sigma_k := \max\{\sigma_0 \rho_2^k, \sigma_{\min}\}, \quad (126)$$

for $k \in \{0, \dots, N\}$, where $\alpha_0, \sigma_0 > 0$ are initial values, $\sigma_{\min} > 0$ is the minimum smoothing level, and $\rho_1, \rho_2 \in (0, 1)$ are decay rates. We recall that we consider a 2D Gaussian mixture prior for $p(\mathbf{x})$, thus $\nabla h_{\sigma_k}(\mathbf{x}_{k\gamma})$ is analytically tractable by the following formula

$$p_{\sigma_k}(\mathbf{x}) := \int_{\mathbb{R}^d} p(\mathbf{z}) \mathcal{N}(\mathbf{x}|\mathbf{z}, \sigma_k^2 I), \quad (127)$$

where $p_{\sigma_k}(\mathbf{x}) \propto e^{-h_{\sigma_k}(\mathbf{x})}$. In our experiments, we set $\sigma_0 = 10$, $\alpha_0 = 10$, $\rho_2 = 0.975$, $\sigma_{\min} = 0$, and define $\rho_1^k := \sigma_k^2$. To simulate SGM approximation error, we assume $\|\mathcal{S}_\theta(\mathbf{x}, \sigma_k) + \nabla h_{\sigma_k}(\mathbf{x})\| \leq \varepsilon_{\sigma_k}$ and perturb the ground-truth score $-\nabla h_{\sigma_k}(\mathbf{x})$ with additive Gaussian noise, i.e., $-\nabla h_{\sigma_k}(\mathbf{x}) + \varepsilon_{err}$, $\varepsilon_{err} \sim \mathcal{N}(0, I)$, followed by element-wise clipping of the noise to a maximum magnitude of $\varepsilon_{max} = 2.5$. We consider a linear inverse problem $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$ with AWGN $\mathbf{n} \sim \mathcal{N}(0, I)$. The measurement operator $\mathbf{A} \in \mathbb{R}^{1 \times 2}$ is constructed by sampling its entries independently from $\mathcal{U}(-1, 1)$, followed by normalization with respect to its Frobenius norm. We run each method (i.e. SGLD, ML-VRLD, and SL-VRLD) with 1000 sample points initialized according to a uniform distribution $\mathcal{U}([-50, 50]^2)$ over $[-50, 50]^2$ grid and run for $N = 1600$ iterations. Moreover, we add synthetic AWGN with standard deviation $\sigma = 10.0$ to the likelihood score gradient to simulate stochastic gradient estimates. To estimate the Fisher information at each iteration, we fit a two-component Gaussian mixture model (GMM) to the generated samples, thereby obtaining a parametric approximation of the underlying distribution of samples. This fitted density enables evaluation of the probability at any point on a predefined grid over $[-50, 50]^2$. Then, for each intermediate sample distribution, we compute the Fisher information for each unit area. The total sum over the grid gives the approximate relative Fisher information.

For SGLD, ML-VRLD ($p = 0$), and SL-VRLD ($p = 0$), we use a batch size of $b = 6$ and a step size of $\gamma = 0.01$. The momentum parameter is set to $\beta = 0.9$ for ML-VRLD ($p = 0$) and $\beta = 0.99$ for SL-VRLD ($p = 0$). For ML-VRLD ($\beta = 1$) and SL-VRLD ($\beta = 1$), we use large and small batch sizes of 8 and 5, respectively, with large batch size computation probability $p = 0.4$, and set the step size to $\gamma = 0.004$. This yields an expected batch size of 6. We intentionally use different

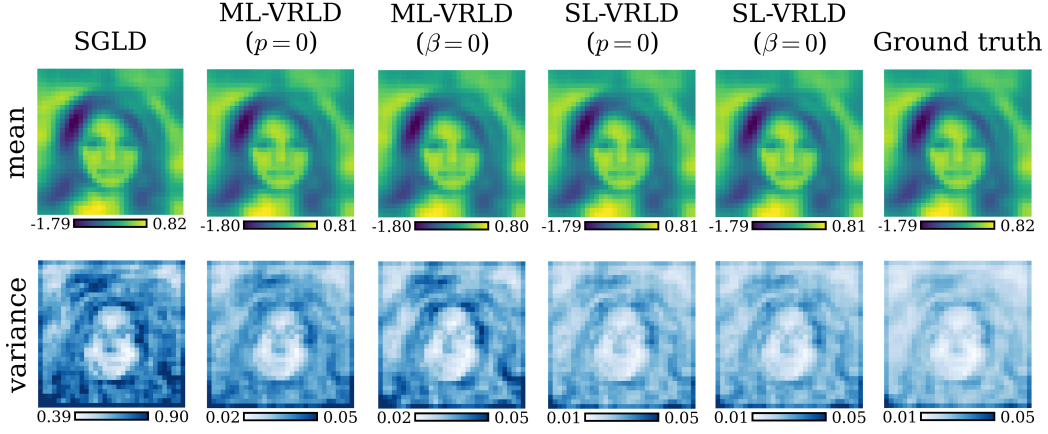


Figure 6: Results on a synthetic inverse problem used for statistical validation, where the posterior mean and variance are known in closed form. ML-VRLD and SL-VRLD yield more accurate variance estimates than SGLD under both the $\beta = 1$ and $p = 0$ configurations, whereas SGLD produces samples with substantially inflated variance.

step sizes for SL-VRLD and ML-VRLD, as ML-VRLD exhibits improved performance with a smaller step size, consistent with the theoretical results in Theorems 1 and 3. Each method is evaluated over 20 independent realizations of $\mathbf{A}$, with all other settings fixed. The mean performance across runs is reported in Fig. ??.

Statistical Validation. We validate that ML-VRLD and SL-VRLD can estimate the posterior statistics more accurately than SGLD under the same batch size. We construct a unimodal Gaussian prior by using the sample statistics of female images in CelebA [Liu et al., 2018]. Each image is downsampled to 32×32 and normalized to $[-1, 1]$. The measurement operator $\mathbf{A} \in \mathbb{R}^{115 \times 1024}$ is linear, with entries sampled independently from $\mathcal{N}(0, 0.01)$. The measurement noise in (4) is modeled as $\mathbf{n} \sim \mathcal{N}(0, 0.01)$. Similar to our numerical experiments, we use the exact smoothed score $\nabla h_{\sigma_k}(\mathbf{x})$ with the following annealing schedule parameters: $\alpha_0 = 1$, $\sigma_0 = 1$, and $\sigma_{\min} = 0.015$. For each method, 1000 initial samples are drawn uniformly from the hypercube $[-50, 50]^{1024}$, i.e., $\mathbf{x}_0 \sim \mathcal{U}([-50, 50]^{1024})$, and each algorithm is run for $N = 5000$ iterations. To simulate stochastic gradients within a mini-batch, we add AWGN with standard deviation $\sigma = 30.0$ to the likelihood score.

Although Fig. 2 reports results for the setting $p = 0$, we also conduct experiments with $\beta = 1$ and present those results in Fig. 6. For SGLD, ML-VRLD ($p = 0$), and SL-VRLD ($p = 0$), we use a batch size of $b = 6$ and a step size of $\gamma = 4 \times 10^{-4}$. For ML-VRLD and SL-VRLD with $\beta = 1$, we use large and small batch sizes $b_L = 8$ and $b_S = 5$, respectively, with large batch size computation probability $p = 0.4$, yielding an expected batch size of $\mathbb{E}_p[b] = 6$. The step size is set to $\gamma = 10^{-4}$. As shown in Fig. 6, SGLD produces samples with significantly higher variance than the ground truth variance. ML-VRLD and SL-VRLD substantially reduce this variance, with SL-VRLD achieving the closest match to the ground-truth posterior variance under both the $\beta = 1$ and $p = 0$ configurations, highlighting the effectiveness of the variance-reduction mechanism.

C.2 ACCELERATED PARALLEL MRI RECONSTRUCTION

We consider a multi-coil MRI reconstruction task with a radial subsampling pattern. The inverse problem is defined as

$$\mathbf{y} = \sum_{i=1}^K \mathbf{P} \mathbf{F} \mathbf{S}_i \mathbf{x} + \mathbf{n}, \quad (128)$$

where $\mathbf{A}_i := \mathbf{P} \mathbf{F} \mathbf{S}_i$ denotes the measurement operator corresponding to the i -th coil. Here, $\mathbf{P}$ is a diagonal subsampling matrix, $\mathbf{F}$ denotes the Fourier transform matrix, and $\mathbf{S}_i$ represents the receiver coil sensitivity map. We set $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I})$ such that the input signal-to-noise ratio (SNR) corresponds to 40 dB. Then the likelihood is $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{A}\mathbf{x}, \sigma_n^2 \mathbf{I})$, so the score of the likelihood at k^{th} iteration is

$$\nabla f(\mathbf{x}_{k\gamma}) = -\frac{1}{\sigma_n^2} \sum_{i=1}^K \mathbf{A}_i^H \left(\mathbf{y} - \sum_{j=1}^K \mathbf{A}_j \mathbf{x}_{k\gamma} \right), \quad (129)$$

Table 3: Peak GPU memory consumption (GB) per iteration for generating a single sample for each method. Lower values indicate better memory efficiency. All results were obtained from a single A100 NVIDIA GPU.

Method	MRI		CT	
	Accel. = $8\times$	Accel. = $12\times$	360 views	180 views
TV	0.38	0.38	1.6	0.81
RED	0.71	0.71	1.96	1.16
PnP-FISTA	0.71	0.71	1.96	1.16
DPS	2.53	2.53	3.76	2.96
PnP-DM	0.71	0.71	1.96	1.16
SGLD	0.71	0.71	1.68	1.15
ML-VRLD	0.71	0.71	1.68	1.15
SL-VRLD	0.71	0.71	1.68	1.15

where $\mathbf{A}^H$ denotes the Hermitian adjoint (conjugate transpose) of $\mathbf{A}$. In our experiments, for all methods, we sample a subset of coils $\xi_k \subset \{1, \dots, K\}$ with $|\xi_k| = b$. Then, an unbiased mini-batch stochastic gradient estimate is given by

$$\nabla f(\mathbf{x}_{k\gamma}, \xi_k) = -\frac{K}{b\sigma_n^2} \sum_{i \in \xi_k} \mathbf{A}_i^H (\mathbf{y}_i - \mathbf{A}_i \mathbf{x}_{k\gamma}), \quad (130)$$

where $\mathbf{y}_i$ denotes the measurements from the i -th coil, and $\xi_k \subset \{1, \dots, K\}$ is a uniformly sampled subset of size b , i.e.,

$$\mathbb{P}(\xi_k = S) = \frac{1}{\binom{K}{b}}, \quad \forall S \subset \{1, \dots, K\}, |S| = b. \quad (131)$$

We conduct our experiments with two different acceleration factors $R \in \{8, 12\}$, representing highly undersampled regimes used in fast and volumetric MRI acquisitions. We synthesize the multiple coils by using `sigpy.mri.birdcage_maps` function from `SigPy` [Ong and Lustig, 2019]. We compare ML-VRLD and SL-VRLD against SGLD, used as a stochastic gradient baseline, as well as a diverse set of competing approaches spanning diffusion-based, denoiser-based, and classical regularization frameworks. Specifically, we include DPS, PnP-DM, PnP-FISTA, RED, and TV as representative state-of-the-art baselines. We use brain MRI data from the fastMRI dataset [Zbontar et al., 2019] to train the denoiser and diffusion models used in the baseline methods, and to construct the test set. The training and test sets consist of 70,684 and 100 slices, respectively, where each slice has a shape of 256×256 . For ML-VRLD, SL-VRLD, and SGLD methods, we use the annealing schedule defined in (12) with the following hyperparameters: $\alpha_0 = 10^4$, $\sigma_0 = 348$, $\rho_2 = 0.99$, $\sigma_{\min} = 0.01$, and define $\rho_1^k := \sigma_k^2$. These hyperparameters are selected based on dataset-specific characteristics, following the guidelines provided in [Song and Ermon, 2020]. We adopt these parameters from [Song and Ermon, 2020, Sun et al., 2024]. In our experiments, we fix $p = 0$ for both ML-VRLD and SL-VRLD, and tune the step size γ and momentum parameter β using Optuna, based on performance evaluated on a validation set of 5 samples. Specifically, for an acceleration factor of 8, we set the step sizes for SGLD, ML-VRLD, and SL-VRLD to $\gamma = 5 \times 10^{-6}$, $\gamma = 7.787 \times 10^{-6}$, and $\gamma = 1.83 \times 10^{-5}$, respectively. The corresponding momentum parameters for ML-VRLD and SL-VRLD are $\beta = 0.318$ and $\beta = 0.155$. For an acceleration factor of 12, the step sizes are $\gamma = 5 \times 10^{-6}$, $\gamma = 7.899 \times 10^{-6}$, and $\gamma = 3.452 \times 10^{-5}$ for SGLD, ML-VRLD, and SL-VRLD, respectively, with corresponding momentum parameters $\beta = 0.208$ and $\beta = 0.922$ for ML-VRLD and SL-VRLD. For each posterior sampling method, we generate 5 samples by using batch size of $b = 32$ coils at each iteration and report the mean sample results. Furthermore, we implement the denoiser-based methods and TV regularization using `DeepInverse` [Tachella et al., 2025]. For the training and evaluation of DPS and PnP-DM, we use `InverseBench` [Zheng et al., 2025]. Visual and quantitative results are presented in Fig. 3 and Table 2, respectively.

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C.3 SPARSE-ANGLE CT RECONSTRUCTION

We study sparse-angle CT reconstruction, where the goal is to recover high-quality images from a restricted set of projection views. The limited angular coverage renders the problem ill-posed due to incomplete sampling. We simulate parallel-beam CT

measurements using 360 and 180 projection views, each with 258 detector elements. The forward operator $\mathbf{A}$ is implemented via `deepinv.physics.Tomography` from the `DeepInverse` package. Additional details regarding the acquisition geometry and the implementation of the measurement operator are provided in the `DeepInverse` documentation. To state the inverse problem formally, let $\mathbf{x} \in \mathbb{R}^d$ denote the unknown CT image and let $\mathbf{R} \in \mathbb{R}^{(KM) \times N}$ denote the full discrete Radon (parallel-beam) projection operator over M projection views and K detector elements per view. We assume $\mathbf{R}$ is stacked by views

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \vdots \\ \mathbf{R}_M \end{bmatrix}, \quad \mathbf{R}_m \in \mathbb{R}^{K \times N}. \quad (132)$$

Thus the full sinogram is $\mathbf{R}\mathbf{x} \in \mathbb{R}^{KM}$. To define the sparse-angle sampling operator, let $\xi_k \subset \{1, \dots, M\}$ denote the index set of measured views at the k -th iteration, where $|\xi_k| = b \ll M$, and b denotes the batch size. The measured vies are selected according to a uniform distribution given as

$$\mathbb{P}(\xi_k = S) = \frac{1}{\binom{M}{b}} \quad \text{for every } S \subset \{1, \dots, M\}, |S| = b, \quad (133)$$

and the view-selection matrix $\mathbf{P}_{\xi} \in \{0, 1\}^{b \times M}$ is defined by

$$(\mathbf{P}_{\xi_k})_{i,m} = \begin{cases} 1, & m = \omega_i, \\ 0, & \text{otherwise,} \end{cases} \quad \text{where } \xi_k = \{\omega_1, \dots, \omega_b\}. \quad (134)$$

Then the sampling operator for projection views is

$$\mathbf{S}_{\xi_k} := \mathbf{P}_{\xi_k} \otimes \mathbf{I}_K \in \{0, 1\}^{(Kb) \times (KM)}, \quad (135)$$

where $\otimes$ denotes the Kronecker product and $\mathbf{I}_K$ is the $K \times K$ identity matrix. Then the sparse-angle measurement operator is

$$\mathbf{A} := \mathbf{S}_{\xi_k} \mathbf{R} \in \mathbb{R}^{(Kb) \times d}. \quad (136)$$

We consider the linear inverse problem in (4) with AWGN $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I})$. Thus the likelihood is given as $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{A}\mathbf{x}, \sigma_n^2 \mathbf{I})$ and the likelihood score at step k is given as

$$\nabla f(\mathbf{x}_{k\gamma}) = -\frac{1}{\sigma_n^2} \mathbf{A}^T (\mathbf{y} - \mathbf{A}\mathbf{x}). \quad (137)$$

In our experiments, we use mini-batch stochastic gradient estimates so we consider the following decompositions of $\mathbf{A}$ and $\mathbf{y}$:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ \vdots \\ \mathbf{A}_M \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_M \end{bmatrix}, \quad (138)$$

where $\mathbf{A}_i \in \mathbb{R}^{K \times d}$ and $\mathbf{y}_i \in \mathbb{R}^K$. Then the likelihood score becomes a sum

$$\nabla f(\mathbf{x}_{k\gamma}) = -\frac{1}{\sigma_n^2} \sum_{i=1}^M \mathbf{A}_i^T (\mathbf{y}_i - \mathbf{A}_i \mathbf{x}). \quad (139)$$

Hence, given a mini-batch views $\xi_k \subset \{1, \dots, M\}$ with $|\xi_k| = b$, sampled uniformly at random, we obtain an unbiased stochastic gradient estiamte

$$\nabla f(\mathbf{x}_{k\gamma}, \xi_k) = -\frac{M}{b\sigma_n^2} \sum_{i \in \xi_k} \mathbf{A}_i^T (\mathbf{y}_i - \mathbf{A}_i \mathbf{x}). \quad (140)$$

In our experiments, we use this formulation for mini-batch stochastic likelihood score estimates. Moreover, we adopt the same baselines as in the MRI experiments, except that zero-filled reconstruction is replaced by filtered back-projection (FBP) with a Ramp filter, which is more suitable for CT reconstruction. For all methods, we consider two batch sizes: 60 views out of 360 and 30 views out of 180 projection views, respectively. We use abdominal CT scans downscaled to 256×256 from

the 2DeteCT dataset [Kiss et al., 2023] for our training and test sets. The training set consists of 4,500 slices and is used to train: (i) the denoiser for RED and PnP-FISTA, (ii) the diffusion model for DPS and PnP-DM, and (iii) the SGM for SGLD, ML-VRLD, and SL-VRLD. All methods are evaluated on 100 slices and mean of 5 samples are reported for each posterior sampling algorithm. We use the same annealing schedule and parameters in MRI reconstruction experiments. For ML-VRLD and SL-VRLD, we use the configuration with $p = 0$ and optimize the step size γ and momentum parameter β using Optuna over 5 samples as a validation set. Specifically, for total number of projection views 360, we set the step sizes for SGLD, ML-VRLD, and SL-VRLD, to $\gamma = 5.76 \times 10^{-6}$, $\gamma = 1.11 \times 10^{-5}$, and $\gamma = 1.35 \times 10^{-5}$, respectively. The corresponding momentum parameters for ML-VRLD and SL-VRLD are $\beta = 0.583$ and $\beta = 0.774$. For the experiments with total number of projection views 180, we set the step sizes for SGLD, ML-VRLD, and SL-VRLD, to $\gamma = 5.16 \times 10^{-6}$, $\gamma = 2.56 \times 10^{-5}$, and $\gamma = 1.7 \times 10^{-5}$, respectively. The corresponding momentum parameters are for ML-VRLD and SL-VRLD are $\beta = 0.41$ and $\beta = 0.988$. We use the same implementations of baseline methods in MRI experiments. Visual and quantitative results are presented in Fig. 3 and Table 2. The implementations of SGLD, ML-VRLD, and SL-VRLD are provided in the supplementary material, along with representative test cases to facilitate reproducibility.